Finite Information Cuts in Markov Prediction: Precision, Depth, and Robustness

Comprehensive research draft revision 5 2 November 2026

Abstract

We study probability prediction at the actual terminal context of an unknown homogeneous second-order source through a complete finite information cut. For fixed-locality one-summary representations, the grid and attention-weight ratio have a sharp joint risk profile, with a binary mixing cutoff different from that of every fixed larger alphabet. Bounded primitive depth supplies another resource: a dyadic compander has a sharp mixing-dependent depth threshold at square-root grid resolution. Growing nonlinear encoders can also use multiple coarse coordinates to represent one finer scalar, so a fixed-family $1/Q$ floor is not universal at growing depth. We establish IID capacity and depth sandwiches, exhibit a pre-grid failure of ordinary folding, and construct robust separated-curve representations with paid finite evaluators. Companding a full triple histogram before robust packing returns the upper bound to general actual-terminal Markov prediction under arbitrary initialization. It attains statistical-order risk with RK^3 cut coordinates and internal resolution $Q^R \gtrsim \sqrt{M}$, using a paid rational output rather than rounding back to the coarse grid. The lower capacity exponent uses all RK^3 coordinates, leaving a dimension gap. A separate IID theorem treats independent real noise amplitude through effective precision $\min\{Q, 1/\delta\}$. Every claim specifies its interface, side information, arithmetic certificate and resource charges. These are scoped mathematical constructions and lower bounds, not training, total-computation optimality, or unrestricted Transformer claims.

Status and scope. This is an AI-assisted comprehensive research draft, with independent LLM mathematical reviews and finite diagnostic checks. It is not externally peer reviewed, formally verified, or an audit of an emitted network. No authorship or institutional affiliation is assigned. Scientific scope is frozen at 08:09:06 UTC on 2 November 2026. Earlier statistical inputs are explicitly identified rather than silently treated as newly proved; the new depth and robustness arguments are developed in full below.

1 Introduction

A normalized attention vector can retain a conditional empirical distribution exactly in real arithmetic while losing statistically important information after coordinate rounding. The difficulty is not simply the number of bits in a prediction. A rare context may contribute only a small fraction of the global attention mass. Dividing by that fraction after a finite information cut amplifies the rounding error; more post-cut computation cannot reconstruct distinctions that the complete message no longer contains.

This manuscript studies that phenomenon in an explicit stochastic experiment. The source table is unknown, transitions may vanish, initialization need not be stationary, and the loss is measured at the actual final pair of the same trajectory used to construct the summary. These choices exclude an independent-query simplification. They also make the occupation count of the terminal context a

central random quantity. A mixing promise reduces the numerical penalty, but neither the encoder nor the constructive decoder needs the unknown stationary law or a runtime mixing-parameter estimate.

The first resource axis is grid resolution. An original finite normalized-statistic interface with $9K + 1$ retained scalars has a complete all-resolution risk profile. For larger alphabets that profile remains valid after optimizing the encoder within a specified fixed-depth causal family evaluated by a certified finite arithmetic procedure. The latter assertion requires more than an analogy with bounded attention: it requires a causal influence estimate, a complete-message concentration and covering argument, and an actual finite-arithmetic upper construction belonging to the family.

The second axis is attention-weight range. The representation can also be changed. A complete K^3 triple histogram, with a positive integer amplification of the queried pair before quantization, attains sampling-order risk at square-root grid resolution. A rescaled rare-window proof and a weight-envelope-uniform IID floor give a complete joint weight/grid frontier in the fixed-dimensional, fixed-locality, one-global-summary family. Increasing the weight ratio cannot eliminate the separate $1/Q$ floor. Statistical-order prediction requires both $Q \gtrsim \sqrt{M}$ and a mixing-dependent lower bound on the product κQ.

The alphabet changes that mixing dependence. A binary active Markov segment yields a lower with cutoff $\max\{\sqrt{\beta}, M^{-1}\}$. A stationary weighted-return analysis first matches it; a path-sensitive first-hit and occupation argument then extends the same paid decoder to every initialization, including off-support starts. At $\beta = 1/M$, binary sources need only constant weight ratio in a square-root grid band, whereas each fixed $K \geq 3$ needs order $\log M$. At zero mixing promise both require logarithmic ratio. An exact-label histogram inclusion establishes the binary refinement separately for the bounded causal finite family.

The third axis is actual bounded-primitive depth. A fixed-width dyadic compander with zero attention logits achieves statistical-order risk at square-root grid resolution with $O(\log \log M)$ primitive stages, uniformly in the source's initialization and mixing promise. A depth-independent rescaled late-window tube proves necessity of this order for $K \geq 3$ at $\beta = 0$. Both statements use a genuinely uniform growing-depth arithmetic certificate; neither hides increasing complexity inside a per-layer graph or a fixed architecture constant.

The central distinction is between fixing a representation and redesigning it within a paid primitive budget. A complete cut permits strong lower bounds even for arbitrary post-cut decoders; changing the pre-cut geometry can invalidate a floor specific to the old family. Section 4 states the enlarged conclusions and remaining gaps before the technical proofs.

1.1 Position relative to existing constructions and lower bounds

The counting representation is established prior art. Behrens, Biggio and Zdeborová give a weighted inventory with equal-token and unequal-token weights e and 1, selected mass $ek/[L + (e - 1)k]$, and an inversion formula, in an infinite-precision model [2, Appendix B.3]. Yehudai et al. study full histogram banks and query-count inversion [13, Sections 4.1–4.2]. Grouping a local tuple as one categorical feature and normalizing a selected sub-block is not a new counting method. The questions here concern the complete post-summary grid, terminal-context occupation, and a lower bound uniform in a growing weight ratio.

Conditional empirical Markov prediction by shallow attention also predates this synthesis. Rajaraman et al. construct constant-depth conditional k-gram representations and state logarithmic-bit approximations for their attention-only constructions; their separate layer-normalized construction

does not carry the same finite-error analysis [8, Theorems 4.4 and 5.1; Remarks 4.2 and 5.2]. Real-arithmetic representability and a paid all-operations implementation are therefore distinct claims, not competing descriptions of one theorem.

Bounded-score dilution is likewise a known mechanism. Hahn's sensitivity analysis must be read with the 2021 correction retaining attention-weight variation [4, 5]. Chiang and Cholak show why small continuous sensitivity is not itself information loss: exact recognition can survive shrinking margins, and zero-epsilon normalization or length-scaled logits can evade bounded-Lipschitz assumptions [3]. Our decoder is unrestricted after the cut, so its impossibility result must follow from the complete finite message, not from a Lipschitz readout hypothesis.

The statistical tools have close predecessors. Han, Jana and Wu use waiting-state sources, a common prior and harmonic averaging in stationary full-history KL prediction; their spectral-gap results also show a mixing-dependent disappearance of a prediction logarithm [6]. Those are not same-rate TV results for a quantized interface. Xu and Raginsky give finite-message Bayes lower bounds and the half-logarithmic communication threshold for Bernoulli estimation [12, Theorem 3 and Example 6]. A generic count of Q^D labels, however, is too weak for our D-coordinate bank. The specific obligation is to prove that, with high probability, its *complete* message occupies only $O(1 + Qj/M)$ labels on a rare window of length j.

Nonlinear scalar quantization is also established. Adler, Tang and Polyanskiy study probability companders, including power transforms x^s and $s = 1/2$, followed by inverse reconstruction and normalization under KL distortion [1, Sections I–III, Theorem 5]. Our depth construction does not introduce companding or square-root behavior near zero; it supplies an explicit bounded-primitive dyadic geometry, one-sided arithmetic and actual-terminal conditional-row analysis. Ou, Schenkel and Bölcskei construct a depth–weight-precision conversion for ReLU networks [7, Proposition 4.1]. That coefficient-precision experiment differs from a quantized observed-summary cut with all internal bits paid. The cited statements and immediate explanations were inspected; their full appendix proofs are not claimed to have been audited here.

Accordingly, the contribution asserted here is a source- and interface-specific synthesis of established mechanisms with explicit additional obligations: complete-message accounting, causal depth, dependence-safe terminal occupation, finite arithmetic, and quantitative growing-range constants. The literature comparison is targeted, not a certificate of novelty or an exhaustive survey.

Repeated ReLU tents are established depth constructions [10, Section 3.3]. Uss et al. encode a high-resolution scalar in two low-precision curve coordinates and reconstruct it by postprocessing [11]; the inspected arXiv version's Sections 2.1–2.4 discuss curve length, discontinuity and noise assumptions. Taylor, Mili and Zaghloul develop scalar stretching/folding, separated tubes and nearest-curve decoding for analog error correction under AWGN [9]. The inspected scope is their abstract, introduction, Sections 2.1–2.3 and the nearest-curve discussion in Section 3.1, not their full experiments or optimization. Accordingly geometric dimensional expansion and noise suppression are not new concepts claimed here. The distinction between these works and the present bounded-adversarial statistical/finite-cut model is an inference from that limited reading, not evidence that no closer result exists.

2 Experiment and information interfaces

Fix an alphabet $\mathcal{A}$ of size $K \geq 2$, $M \geq 64$, and a dyadic resolution $Q = 2^h$ with integer $h \geq 4$. Observe $X_1, \ldots, X_{M+2}$ from an unknown homogeneous second-order table $\theta_{ab}(c)$. Set

$$Y_j = (X_{j+1}, X_{j+2}), \quad 0 \leq j \leq M, \qquad P((a,b),(b,c)) = \theta_{ab}(c).$$

The target is θ_{Y_M}, the next-symbol law at the observed terminal pair. A predictor $\widehat{p}$ has risk

$$\mathbb{E}_{\theta,\mu}\,\mathrm{TV}(\widehat{p}, \theta_{Y_M}), \qquad \mathrm{TV}(p,q) = \frac{1}{2}\sum_{c\in\mathcal{A}} |p_c - q_c|, \tag{1}$$

where μ is the initial pair law. All expectations are unconditional on the actual trajectory; Y_M is not an independent query.

Definition 2.1 (All-state certificate)**.** The class $\mathcal{D}_2(\beta)$, $0 \leq \beta \leq 1$, consists of tables admitting a stationary pair law Π such that

$$\Pi P = \Pi, \qquad P^2(s,t) \geq \beta\Pi(t) \quad \text{for every pair } s,t. \tag{2}$$

The certificate is existential and hidden. If it is nonunique, the stationary experiment includes its choice. In particular, the condition includes starting states outside $\operatorname{supp}\Pi$. Zero transition probabilities are allowed. In the stationary experiment initialization equals a qualifying hidden Π; in the arbitrary-start experiment μ is unrestricted.

For an encoder family $\mathcal{E}$, define $\mathcal{R}_{\mathcal{E},\mathrm{all}}$ as the infimum over encoder/decoder pairs independent of the source law, of the worst risk (1) over $\mathcal{D}_2(\beta)$ and all μ; use $\mathcal{R}_{\mathcal{E},\mathrm{stat}}$ for the stationary restriction. Public K, M, Q, β may be known when choosing a pair. Source-independent common randomness is permitted, with every uniform architecture bound holding seedwise. The constructive predictors below are deterministic and do not read β at a supplied budget.

2.1 Experiments that must not be conflated

The fixed-prefix original interface, the one-block ratio family and the fixed-depth causal family below are separate comparisons. Their information lower bounds already permit arbitrary real simplex decoders. Their specified finite upper programs, including the early compander, additionally round predictions to a Q-simplex grid. Later growing-depth packing changes the encoder and treats Q only as an internal cut; its paid finite decoder can emit finer dyadic or rational probabilities. This is not a relaxation invented for a previously grid-constrained lower bound.

The clean grid, a canonical grid-to-grid corruption *after* quantization, one bounded real perturbation *before* clipping/quantization, and faults in repeated internal arithmetic are different channels. Only the declared single pre-grid channel is covered by the new robust constructions. The independent-amplitude theorem permits real attacks, not merely attacks on a fine arithmetic lattice. Reliable finite computation elsewhere is assumed and paid.

All growing-depth lower bounds fix width, scalar output dimension, heads, locality, stage primitive count, coefficient/state/Lipschitz bounds, mask density and the exact positional boundary budget. Each stage has bounded work; a growing affine/ReLU graph cannot be hidden inside one layer. Historical processing is causal, and the synthetic initial terminal state contains no history statistic beyond its fixed boundary symbols. All formal histories obey the ideal contracts within every frozen branch. A uniform error certificate E/Q with depth-independent E connects the actual finite evaluator to that ideal family. Unless explicitly stated otherwise, constants are for fixed alphabet and fixed exposed dimension, not jointly uniform in them.

2.2 The complete cut and distinct resource coordinates

The cut carries a vector $Z_d = Q^{-1}\lfloor QF_d \rfloor$, finitely many exact boundary symbols S, and public metadata. Once the cut is made, the decoder may compute arbitrarily and output any real simplex vector. It may not read the history, clean numerator or denominator, an unquantized residual, another token's output, or a duplicate pre-cut state. This restriction is mathematical access control, not an uncharged physical isolation device.

Grid precision h is not the complete arithmetic word length. A coordinate in $[0,1]$ on the closed grid has $Q+1$ labels and requires $h+1$ unsigned bits. A fixed bounded signed range requires $h+O(1)$ bits. Accumulators must additionally pay $\log M$, signs and full products. The scalar dimension D, number of physical heads, attention depth, total retained bits, and depth of an expanded Boolean/ReLU circuit are different quantities. A lower bound for one such coordinate is not a lower bound for all the others.

2.3 Baseline bounded encoder models

The *original interface* retains the complete $9K+1$ scalar bank of four statistical heads, together with its declared exact query, endpoint and flag fields. Two additional paid singleton heads acquire local predecessor features. Its actual finite exponential evaluator, not ideal softmax, is part of the fixed encoder. The allowed decoder is optimized; the particular conservative decoder used for the upper bound is described in Section D.

For the *one-summary local family*, fix integers $D, r \geq 1$ and constants $0 < a \leq b < \infty$ independently of M, β, Q. It returns D ratios

$$F_d = \frac{\sum_{i=3}^{M+2} w_{di}(L_i, q(S))\, v_{di}(L_i, q(S))}{\sum_{i=3}^{M+2} w_{di}(L_i, q(S))}, \qquad a \leq w_{di} \leq b, \quad |v_{di}| \leq 1. \tag{3}$$

Here L_i is a trailing window of fixed radius r, and S contains the first and last r symbols. The source-independent tables may vary with position and public parameters. Every coordinate pools all M keys. Query-selective masks, nonlocal precomputed features and interior exact side statistics are excluded. D counts scalar outputs; several coordinates may share a head and denominator. Arbitrary real tables are allowed for the information lower; the upper uses a specified finite construction.

The *finite causal family* has a fixed architecture budget: depth, head count, dimensions, local radius, affine/ReLU operations per position, uniform state and coefficient bounds, score/value Lipschitz constants, and dense-mask fraction. History states depend only on causal local windows and public finite dyadic tables. All historical parameters are independent of future symbols. The exact boundary field, first and last three symbols in the basic version, may enter only the appended terminal initial query and the decoder. Public masks $\mathcal{K}_t \subseteq \{1, \ldots, t\}$ are independent of source symbols and boundary branches and satisfy $|\mathcal{K}_t| \geq ct$ for fixed $c > 0$.

Queries, keys and values are affine maps, scores are bounded dot products with bounded public dyadic scale/bias, and positionwise updates/readout are fixed finite affine/ReLU directed acyclic graphs. Ideal states have a declared uniform range on all formal histories. Every public numerical datum is a supplied finite dyadic whose actual description length is charged. Layer normalization and unspecified real nonlinear evaluators are excluded from this finite subclass. Section C specifies the integer evaluator used to implement each ideal encoder.

The density budget must permit $c \leq 1/3$. The architecture budget is assumed to accommodate one global block, four statistical heads, radius three, width $12K+4$, output dimension $D \geq 9K+1$,

affine row-sum bounds at least two, score bound at least two, state bound at least one and a sufficient score Lipschitz bound, for example $4K$. Larger fixed depths can add identity updates. Capacity does not certify arbitrary parameters: each selected network must obey the primitive bounds. Section I states the smaller binary histogram capacity separately. Section J allows the number of stages to grow while fixing every per-stage primitive budget and making terminal initialization and arithmetic certification uniform. These are explicitly different comparison classes.

3 Bounded-score risk and precision frontiers

Write

$$\begin{aligned} \beta_0 &= \max\{\beta, M^{-1}\}, \qquad & W(M,\beta,Q) &= \frac{1+\ln(1+Q/(M\beta_0))}{Q}, \\ H(M,\beta,Q) &= \max\{M^{-1/2}, W(M,\beta,Q)\}. \end{aligned} \tag{4}$$

Constants below may depend on the fixed alphabet and architecture, but not on M, β, Q. They are not sharp joint-K constants.

Theorem 3.1 (Bounded finite information cut)**.** *Assume $K \geq 3$. For the original fixed interface,*

$$2^{-336} H \leq \mathcal{R}_{\mathrm{orig,stat}} \leq \mathcal{R}_{\mathrm{orig,all}} \leq \min\{1, C_K H\}, \qquad C_K = 2^{16}K^{5/2} + 2^{14}K^3 + 5320K^5. \tag{5}$$

For the finite causal family of Section 2, evaluated at target precision $h+3$ by the integer compiler of Section C and then rounded at resolution Q, there is $c_{\mathrm{arch}} > 0$ such that

$$c_{\mathrm{arch}} H \leq \mathcal{R}_{\mathrm{causal,stat}} \leq \mathcal{R}_{\mathrm{causal,all}} \leq \min\{1, C_K H\}. \tag{6}$$

The same profile holds for (3) *when $D \geq 9K+1$, $r \geq 3$, $0 < a \leq 1$ and $6 \leq b < \infty$, with a lower constant depending on D, r, a, b.*

The underlying lower, finite arithmetic and upper steps are presented below; the dependency scope is in Section Q. The three upper bounds do not identify the evaluated programs. In particular, the causal upper replaces the original head evaluator. Its deterministic accuracy envelope suffices; neither equality of old and new grid labels nor equality of selector decisions is claimed.

Corollary 3.2 (Grid threshold)**.** *Under Theorem 3.1, at fixed $K \geq 3$ and architecture, the resolution required for statistical-order risk is, up to multiplicative constants,*

$$Q_* = \sqrt{M}\left[1 + \ln\left(1 + \frac{1}{\sqrt{M}\max\{\beta, M^{-1}\}}\right)\right]. \tag{7}$$

A dyadic budget within a constant factor of Q_* attains a sufficiently large fixed multiple of $M^{-1/2}$. Attaining any specified fixed multiple requires $Q \gtrsim Q_*$, with the lower constant also depending on that multiplier. Thus

$$h_* = \tfrac{1}{2}\log_2 M + \log_2\left[1 + \ln\left(1 + \frac{1}{\sqrt{M}\max\{\beta, M^{-1}\}}\right)\right] + O(1).$$

Proof. Let $s = \sqrt{M}$, $t = M\beta_0$ and $f = 1+\ln(1+s/t)$. For $Q = sf$, the inequality $1+sf/t \leq f(1+s/t)$ gives $1 + \ln(1+Q/t) \leq f + \ln f \leq 2f$, so $W \leq 2/s$. Increasing Q decreases W; dyadic rounding

upward costs at most a factor two. Conversely suppose $H \le A/s$ with fixed $A > 0$, enlarged to $A \ge 1$. Then $Q \ge s/A$ from $W \ge 1/Q$. By concavity, $\ln(1 + x/A) \ge A^{-1} \ln(1 + x)$ for $A \ge 1$. Therefore $1 + \ln(1 + Q/t) \ge f/A$, and $W \le A/s$ forces $Q \ge sf/A^2$. Apply the theorem's lower constant to a desired risk multiplier. □

For $\beta = 0$ or $\beta = O(M^{-1})$, the threshold is $\sqrt{M} \log M$ and $h_* = \frac{1}{2} \log_2 M + \log_2 \log M + O(1)$. If $\beta\sqrt{M}$ is bounded below, it is $\Theta(\sqrt{M})$. The expression covers intermediate, length-dependent mixing promises without choosing a new decoder. Selecting a smaller budget from a known class promise is not learning an unknown β.

4 Depth, coordinate capacity, and robust prediction

The fixed-family frontier does not impose a universal $1/Q$ floor after the encoder's primitive depth is allowed to grow. Three complementary results explain what replaces that assertion. Full statements and proofs are given in the technical appendices, with the experiment held explicit throughout.

Mixing determines how much companding depth is needed. At a fixed square-root grid band, let

$$e_K = \begin{cases} \max\{\sqrt{\beta}, M^{-1}\}, & K = 2, \\ \max\{\beta, M^{-1}\}, & K \ge 3, \end{cases} \qquad f_K = 1 + \ln(1 + Q/(Me_K)).$$

Theorem K.1 proves least feasible 1+primitive depth of order $1 + \ln f_K$, for stationary and arbitrary initialization, at sufficiently large fixed statistical-risk multiplier. Every fixed positive multiplier has the stated necessary lower. The encoder uses zero attention logits and a bounded-width dyadic compander; its increasing gain is not an increasing attention-weight ratio. At $\beta = 1/M$, the required order is constant for binary sources but double-logarithmic for each fixed larger alphabet. At $\beta = 0$ both have the double logarithm. Internal word precision, counts, descriptions and all-token work still grow and are paid.

Growing depth can use the whole finite-label capacity. For IID Bernoulli prediction through J bounded coordinates, Theorem L.1 gives risk order $\max\{M^{-1/2}, Q^{-J}\}$ once the appropriate depth is allowed. Partial folding yields statistical order when Q^J is a fixed positive multiple of $\sqrt{M}$, with sufficient depth $O_J(1 + \log_+(\sqrt{M}/Q))$. Theorem L.2 supplies a lower $c/(QC^d)$ using a deterministic reveal of the largest influences and the resulting nonuniform posterior. It requires exact side information to consist of fixed public-position source bits, not an arbitrary nonlocal finite statistic. Necessary and sufficient numerical multipliers and exponential bases are not identified. The decoder reconstructs a finer finite scalar from the coarse labels; it receives no clean mean, count, residual state or history reread.

Clean folding and robust geometry are different representations. A common pre-grid reset of the first fold rail creates a reflection collision; with the actual terminal pair retained, Proposition M.1 still gives an order-$1/Q$ attacked risk for a family whose clean risk is order $1/Q^2$ or smaller. This is a failure of that specific encoder, not a universal robust lower. Redesigned separated snakes give robust IID error $O_{J,c}(M^{-1/2} + Q^{-J})$ under one pre-grid perturbation of size c/Q. A dyadic plateau construction has an exact-lattice evaluator; a no-pause rational construction gives the statistical-order depth sandwich. Lemma M.5 explicitly extends the earlier dyadic coefficient-description contract and pays conversion; it does not call rational stages lattice exact.

The upper returns to the general actual-terminal Markov problem. Theorem N.1 compands every coordinate of a complete K^3 triple histogram and packs each in R noisy coordinates. With $D = RK^3$ cut coordinates, it attains $O_{K,R,c}(M^{-1/2})$ risk under arbitrary initialization and any c/Q attack when $Q^R \gtrsim_{R,c} \sqrt{M}$. Corollary N.2 gives sufficient primitive depth

$$O_{K,R,c}(1 + \log\log(M+2) + \log_+(\sqrt{M}/Q)).$$

The proof uses a symmetric inverse estimate, add-one normalization and unconditional occupation of the actual terminal pair. Robust reconstruction may introduce mass into zero-count cells; it is not treated as downward or zero-pure. The output is a paid exact rational probability vector. Rounding it back to the coarse internal grid would add an error of order $1/Q$ and generally destroy this benefit. The universal alphabet lower uses all $D = RK^3$ coordinates, whereas the sufficient exponent is R. This dimension-capacity gap remains open. At the named smallest sufficient scale $Q = \Theta(M^{1/(2R)})$, the common-family influence lower and this upper match primitive depth order $\log M$; they do not establish an all-resolution sharp Markov frontier.

Independent noise sets an effective precision ceiling. For the separate IID experiment with one real pre-grid amplitude $0 \le \delta \le 1/4$, Theorem O.1 uses $\tau = \max\{Q^{-1}, \delta\}$ and $Q_{\text{eff}} = 1/\tau$. Risk is at least a fixed multiple of each of $M^{-1/2}$, τ^J and τ/C^d. At enough depth, robust risk order is $\max\{M^{-1/2}, \tau^J\}$, with a statistical-order depth sandwich in $\log_+(\tau\sqrt{M})$. Constructive selection requires a finite public dyadic noise-scale certificate, or paid rational noise input and comparisons; an arbitrary real amplitude is not a computational oracle. Fixed positive noise and fixed J prevent consistency even if Q, d grow. This IID upper is not asserted for the general Markov class.

Reading and evidence. Appendices A–I give the fixed-family information, arithmetic and actual-terminal occupation arguments. Appendices J–K treat companding and mixing-dependent depth. Appendices L–O develop capacity, robust geometry and the return to general Markov prediction. Complete resource ledgers distinguish all-token arithmetic, masks, coefficients, generation, addresses, full products, decoder conversion, coexisting banks and PURE expansion. The statistical add-one and selector inputs inherited from the earlier research are stated explicitly in Appendix D; their full original proof bodies remain part of the accompanying source dossier, rather than being presented as newly proved facts here. All newly integrated depth/packing/robustness arguments are developed in the present text.

Technical proofs and resource accounts

A Complete messages on stationary rare windows

The lower-bound proof has a source part and a representation part. The source is shared across local and causal encoders. The representation analysis must control *every* retained coordinate and side field simultaneously.

A.1 A legal stationary source with a common prior

Use three symbols W, C_0, C_1. For $\beta \le 1/16$, let

$$\alpha = \frac{1}{2M}, \qquad \eta = \beta_0 - \alpha, \qquad p \in [1/4, 3/4], \qquad p_0 = 1 - p,\ p_1 = p.$$

Define a first-order kernel, embedded in the second-order model, by

$$\begin{aligned} T(W,W) &= 1-\alpha, & T(W,C_i) &= \alpha p_i, \\ T(C_i,W) &= \eta, & T(C_i,C_j) &= (1-\eta)p_j, \\ \pi(W) &= \eta/\beta_0, & \pi(C_i) &= \alpha p_i/\beta_0. \end{aligned} \tag{8}$$

Unused-symbol rows equal π. Direct substitution gives $\pi T = \pi$ and $T(x,y) \ge \beta_0 \pi(y)$ for all symbols. Set $\theta_{uv}(x) = T(v,x)$ and $\Pi(u,v) = \pi(u)T(u,v)$. Then

$$P^2((u,v),(x,y)) = T(v,x)T(x,y) \ge \beta_0 \Pi(x,y) \ge \beta \Pi(x,y),$$

including off-support starting states. Initialize at this hidden stationary pair law.

For $Q \le M/4$, put

$$J_* = \left\lfloor \min\{(4\beta_0)^{-1}, M^2/(4Q^2)\} \right\rfloor. \tag{9}$$

Here $4 \le J_* \le M/4$. Let E_j be the word event consisting of $M + 2 - j$ waiting symbols followed by j active symbols, $2 \le j \le J_*$. Its probability is independent of p and obeys

$$e_j = \Pi(W,W)\alpha(1-\alpha)^{M-j}(1-\eta)^{j-1} \ge \frac{1}{32M}. \tag{10}$$

Conditional on E_j, the active suffix is IID Bernoulli(p) *before* conditioning on endpoint information. The events are disjoint. One uniform prior on $p \in [1/4, 3/4]$ is used across all j, and remains uniform conditional on each E_j. We may grant j to the decoder without weakening a lower bound.

The actual next-symbol target on E_j is

$$\eta\delta_W + (1-\eta)((1-p)\delta_{C_0} + p\delta_{C_1}). \tag{11}$$

The no-exit event in the observed suffix does not remove the next-step exit mass η. This distinction is essential for the stated loss.

A.2 Local ratios and finite boundary branches

For the local model (3), freeze a possible boundary value s and its query while sampling all j active bits. There are at most $H_b = 3^{2r}$ such values on the witness. Write the numerator and denominator sums as A_{ds} and B_{ds} and define

$$m_{ds}(p) = \frac{\mathbb{E}_p A_{ds}}{\mathbb{E}_p B_{ds}}.$$

This is a ratio of expectations, not an expectation of a ratio. Both denominators are at least aM. Replacing one coin changes at most r summands, giving replacement bounds $2br$ and variances at most $2b^2r^2j$. Chebyshev and a union bound over all $2DH_b$ sums imply, with probability at least $9/10$, simultaneous deviations at most $T_b\sqrt{j}$, where $T_b = br\sqrt{40DH_b}$. Subtracting ratios then gives

$$|F_{ds} - m_{ds}(p)| \le \frac{2T_b\sqrt{j}}{aM}. \tag{12}$$

For a function of independent Bernoulli bits the exact polynomial identity

$$\frac{d}{dp}\mathbb{E}_p f = \sum_i \mathbb{E}_p[f(B_i = 1) - f(B_i = 0)] \tag{13}$$

implies $\|m_s'(p)\|_\infty \le 4brj/(aM)$. Thus the whole bank lies near a short one-parameter curve, uniformly over arbitrary public position-dependent tables.

Choose a parameter mesh so that adjacent curve values differ by at most $1/Q$. Because $Q\sqrt{j}/M \le 1/2$, every mesh point's tube intersects only a constant number of grid cells per coordinate. Counting all coordinates and the actual boundary label gives a fixed, p-independent set of at most

$$A_b(1 + Qj/M), \qquad A_b = H_b(2 + 4br/a)(\lceil 8T_b/a + 4\rceil + 1)^D \tag{14}$$

complete good messages. The good event may depend on p; the containing alphabet may not. Freezing all possible boundary values first and selecting the actual one afterwards avoids a false assertion of conditional IID.

Lemma A.1 (Good-message Bayes bound)**.** *Let p be uniform on an interval of length ℓ. Suppose for every p a good event has probability at least $9/10$, and every complete good message is in one fixed set of at most R labels. Any possibly randomized scalar decoder has mean absolute error at least $\ell/(16R)$.*

Proof. Fix the source-independent decoder seed. Radius-$\ell/(8R)$ neighborhoods of its at most R good-message outputs cover at most one quarter of the prior interval. The joint probability of being good and outside those neighborhoods is at least $9/10 - 1/4$. On that event the error is at least $\ell/(8R)$; multiplying gives a bound stronger than the claim. Average over the seed. No posterior-uniformity assertion after the good event is used. □

By (11), $\mathrm{TV}(v, \theta) \ge (1 - \eta)|v(C_1)/(1 - \eta) - p|$. Applying the lemma, (10) and (14) yields

$$\begin{aligned}\mathcal{R} &\ge \frac{1}{2048A_bM}\sum_{j=2}^{J_*}\frac{1}{1 + Qj/M} \\ &\ge \frac{1}{8192A_bQ}\ln\left(1 + \frac{1}{8}\min\left\{\frac{Q}{M\beta_0}, \frac{M}{Q}\right\}\right).\end{aligned} \tag{15}$$

The second cutoff M/Q is real: it comes from the tube thickness in (9). It cannot simply be dropped.

A.3 IID floor and completion of the resolution domain

On stationary binary IID sources, freezing the boundary gives an $O(1)$-length expectation curve and stochastic thickness $O(M^{-1/2})$. For $Q \le \sqrt{M}$, the same proof produces at most $A_I Q$ good complete messages and hence $\mathcal{R} \ge c_I/Q$. For $Q > \sqrt{M}$, and also outside the useful rare-window range, take Bernoulli parameters

$$p_\pm = \tfrac{1}{2} \pm \frac{1}{16\sqrt{M+2}}.$$

The history KL divergence is at most $1/8$; Pinsker and two-point testing give risk at least $1/(128\sqrt{M+2}) \ge M^{-1/2}/256$, even with full history. These IID sources satisfy (2) for every $\beta \le 1$. Therefore separate lower witnesses supply a statistical floor, a $1/Q$ floor and (15); their maximum, not a sum of their losses, is available.

Here is the all-Q completion used in both bounded-family theorems. If $\beta > 1/16$, then $W \le Q^{-1} + 16/M \le Q^{-1} + 2M^{-1/2}$. If $Q > M/4$ or $Q > M^{3/4}$, $\ln(1+x) \le 4x^{1/4}$ gives $W \le 5M^{-1/2}$. In the remaining domain put $x = Q/(M\beta_0)$ and $y = M/Q$. Since $x \le Q$ and $Q \le M^{3/4}$, $x > y$ implies $x \le y^3$. Thus

$$\ln(1+x) \le 3\ln(1+\min\{x,y\}) \le 24\ln(1+\min\{x,y\}/8).$$

The three lower bounds therefore control H throughout the legal domain. This calculation neither extrapolates a thin-tube estimate nor imports an original-interface-only lower into a larger family.

B Causal depth and the correct influence envelope

Recursive causal features require a different representation argument from the local ratio calculation. Hold the appended initial query fixed within a boundary branch. Let the ideal state at row t be h_t, and compare two formal histories with discrepancies $\delta_t = \|h_t - h'_t\|$. Assume a head has scores in $[-B, B]$, values bounded by V, score Lipschitz constant S, value Lipschitz constant V_L, and positionwise update Lipschitz constant F_L for the sum norm. Write H_a for the maximum heads per layer.

Along the line segment between two logit vectors, every weight is at most C/t, where $C = e^{2B}/c$ for masks of density c. The softmax differential is

$$d\alpha_i = \alpha_i \left(ds_i - \sum_k \alpha_k ds_k \right), \qquad \sum_i |d\alpha_i| \le 2 \sum_i \alpha_i |ds_i|.$$

Separating value and weight changes and integrating gives the conservative bound

$$\|o_t - o'_t\| \le 2VCS\delta_t + \frac{C(V_L + 2VS)}{t} \sum_{i \le t} \delta_i.$$

The weight-change term is retained. Summing the heads and applying the update yields

$$\delta'_t \le a\delta_t + \frac{b}{t} \sum_{i \le t} \delta_i, \quad a = F_L(1 + 2H_a VCS), \quad b = F_L H_a C(V_L + 2VS). \tag{16}$$

No interpolation of hidden states or global Euclidean Lipschitzness of a dot product is needed; the primitive bounds apply on the declared reachable domain.

Proposition B.1 (Triangular influence)**.** *Let $A_{ti} = \mathbf{1}\{i \le t\}/t$, $B_* = aI + bA$, and let the terminal row be $T = M + 3$. For depth L, readout Lipschitz constant G and initial symbol diameter Δ, a replacement at $i < T$ changes the terminal vector by at most*

$$d_i = G\Delta(B_*^L)_{Ti} \le \frac{G\Delta[(a+b)^L - a^L]}{T}[1 + \ln(T/i)]^{L-1}. \tag{17}$$

Moreover $\sum_{i<T} d_i \le C_0$ and $\sum_{i<T} d_i^2 \le \Lambda/M$ for architecture constants, while $d_i \le \Gamma/M$ for $i \ge T/2$.

Proof. For $k \ge 1$,

$$(A^k)_{ti} = \frac{1}{t} \sum_{i \le u_1 \le \cdots \le u_{k-1} \le t} \frac{1}{u_1 \cdots u_{k-1}} \le \frac{1}{t}\left(\sum_{u=i}^{t} \frac{1}{u}\right)^{k-1} \le \frac{[1 + \ln(t/i)]^{k-1}}{t}.$$

The empty product covers $k = 1$. Binomial expansion of B_*^L proves (17). The row sum of B_*^L is $(a+b)^L$, so the sum bound applies to its actual matrix entries, not merely their larger envelopes. For $m = 2L - 2$,

$$I_m = \int_0^1 [1 + \ln(1/x)]^m \, dx = \sum_{k=0}^{m} \binom{m}{k} k! < \infty.$$

A decreasing-function Riemann sum gives $\Lambda = (G\Delta[(a+b)^L - a^L])^2 I_m$. Restricting $i \ge T/2$ gives $\Gamma = G\Delta[(a+b)^L - a^L](1 + \ln 2)^{L-1}$. Radius-$r$ preprocessing multiplies the sum and late-influence constants by at most r and the square-sum constant by at most r^2, since each source symbol affects at most r initial states. □

A uniform $O(1/T)$ bound for every earlier token is false. With two zero-score scalar causal averaging layers, the map is A^2 and

$$(A^2)_{T1} = \frac{1}{T} \sum_{u=1}^{T} \frac{1}{u} = \frac{H_T}{T}. \tag{18}$$

This is an admissible bounded one-head example. The theorem uses late-suffix influence and square summability, not the false uniform bound.

For a j-coin late suffix, Efron–Stein and (13), now applied to the entire nonlinear terminal output, give

$$\mathrm{Var}(F_d) \le \frac{\Gamma^2 j}{2M^2}, \qquad \|(\mathbb{E}_p F)'\|_\infty \le \frac{\Gamma j}{M}.$$

For a fully IID history they give variance at most $\Lambda/(2M)$ and curve speed at most C_0. Thus the same tube count, common prior and domain completion apply, with new architecture constants. Unlike the local proof, the curve here is an expectation of the whole network output; no ratio-of-expectations substitution is made.

B.1 Boundary information and temporal causality

The proof can grant a stronger finite-side-information model: freeze each of finitely many boundary labels and allow branch-dependent tables and parameters. The uniform bounds must hold on *all formal histories in each frozen branch*, including histories inconsistent with that branch's boundary

value. Simultaneous branchwise concentration followed by selection of the actual label is valid, and its finite cardinality is counted.

That enlarged model need not be end-to-end causal. For example, assigning every initial history state the future bit X_{M+2} through the branch label preserves causal attention edges but broadcasts future information to row one. The finite causal theorem excludes this behavior: historical tables, parameters and masks are independent of future boundary symbols, and the full boundary field enters only the appended terminal initial query. It is a subclass of the stronger lower-bound model. This scope restriction is not an extra statistical assumption or a claim that lower-triangular masks alone imply temporal causality.

C A constructive finite arithmetic bridge

The influence argument concerns an ideal bounded architecture. Intermediate rounding is discontinuous and need not satisfy (16). The bridge instead proves a uniform end-to-end approximation from explicit signed arithmetic.

Theorem C.1 (Finite evaluator)**.** *For the finite-data affine/ReLU subclass in Section 2, an integer evaluator with fractional precision $P = 4h + C_{\mathrm{arch}}$ satisfies*

$$\|F_{\mathrm{fin}}(X) - F_{\mathrm{ideal}}(X)\|_\infty \le 2^{-h}$$

for every length and every formal history. Full online words additionally include $\log n$ accumulation guards and range/sign allowances. All supplied data, conversion, evaluation, movement and storage costs are charged. This evaluator excludes arbitrary unevaluated Lipschitz maps and layer normalization.

Constructive proof. Set $D_P = 2^P$, $\varepsilon = D_P^{-1}$ and represent a signed real by integer X with value X/D_P. Rounding is downward, including negative inputs. Convert every supplied dyadic parameter by exact signed division. For an affine coordinate, accumulate the complete signed integer

$$N_j = D_P B_j + \sum_k A_{jk} X_k, \qquad Y_j = \lfloor N_j / D_P \rfloor .$$

All products and the entire sum are retained before the single floor; ReLU is exact integer $\max(0, Y_j)$. With ideal input range U, discrepancy $\delta \le 1$, and fan-in d, one affine step has error at most

$$\|A\|_\infty \delta + d\varepsilon(U + \delta) + 2\varepsilon.$$

Finite recursion yields computable constants for each fixed affine/ReLU block.

Interval bounds on the rounded-state cube of radius $R+1$ give an architecture-only integer $a \ge 1$ bounding both ideal and finite logits in $[-a, a]$. Shift by a, which cancels in softmax. For the exact grid value $x = s_{\mathrm{fin}} + a \in [0, A]$, $A = 2a$, let $S = D_P x$, $j_0 = \lceil 8A^2 \rceil + 2$ and $J = j_0 + P$. Compute

$$T_0 = D_P, \qquad T_j = \left\lfloor \frac{T_{j-1} S}{j D_P} \right\rfloor, \qquad W = \sum_{j=0}^{J} T_j. \tag{19}$$

The integer $H'_a = 3^A$ bounds e^x. Each term-rounding error contributes at most $H'_a \varepsilon$ to the sum of all later terms. For $j \ge 8A^2$, $A^j/j! \le 2^{-j}$ by bounding the last half of the factorial. Hence

$$1 \le W/D_P \le e^x, \qquad 0 \le e^x - W/D_P \le (H'_a J + 1)\varepsilon. \tag{20}$$

No exponential or logarithm is evaluated as an oracle.

For $m \geq 1$ keys and signed value integers V_i, form

$$Z = \sum_i W_i \geq mD_P, \qquad N = \sum_i W_i V_i, \qquad O = \lfloor N/Z \rfloor, \qquad o_{\text{fin}} = O/D_P. \tag{21}$$

There is no extra division by D_P inside O. The numerator uses full signed products with $\log m$ sum guards, so cancellation causes no invalid truncation. If finite and ideal values differ by at most η_v and shifted weights by at most η_w, subtracting the two ratios gives

$$|o_{\text{fin}} - o_{\text{ideal}}| \leq \varepsilon + \eta_v + 2V\eta_w.$$

Both weight sums are at least m in real units, which removes dependence on key count. Combining this with the affine score bounds and (20) gives an error recursion

$$\delta_{\ell+1} \leq a_* \delta_\ell + b_*(P+1)2^{-P}$$

while $\delta_\ell \leq 1$. The readout and initialization obey the same finite recursion. Choose a computable $C_* \geq 1$ dominating all resulting constants and select C_{arch} by finite search so that

$$C_*(C_{\text{arch}} + 5)2^{-C_{\text{arch}}} \leq 1.$$

For $P = 4h + C_{\text{arch}}$, this ensures $C_*(P+1)2^{-P} \leq 2^{-h} \leq 1$. Induction therefore keeps the actual states in the neighborhood used to derive the estimates, closing rather than assuming the bootstrap. □

If the finite output is within E/Q of an ideal output, enlarging each good-message tube by E/Q transfers the lower proof. With the compiler targeted at $h + 3$, $E = 1/8$; one may safely use $E = 1$ in constants. An arbitrarily small error can cross a grid boundary, so this statement does not imply equality of rounded labels. It also does not say that the expanded finite arithmetic is a fixed-depth ordinary scalar Transformer.

D Terminal occupation and the constructive upper

This section identifies the inherited statistical inputs and the additional precision analysis. Let

$$J = \sum_{j=0}^{M-1} \mathbf{1}\{Y_j = Y_M\}, \qquad C_c = \sum_{j=0}^{M-1} \mathbf{1}\{Y_j = Y_M, X_{j+3} = c\}, \qquad H_c = \sum_{t=3}^{M+2} \mathbf{1}\{X_t = c\}.$$

These are proof variables, not decoder ports. Define the full-history add-one row $q_c = (C_c+1)/(J+K)$ and pooled row $b_c = (H_c + 1)/(M + K)$.

D.1 Inherited full-history statistical statements

For precision about dependencies, put $n_0 = M + 1$, $\delta = 1 - \beta$ and

$$i = \min\{1, \sqrt{K/n_0}\}, \qquad a_0 = \min\{1, \sqrt{K^3/n_0}\},$$
$$g = \min\{1, \sqrt{K^5/n_0}\}, \qquad h_0 = \min\{1, \sqrt{K^4/n_0}\},$$
$$F_0 = \max\{i, \min\{\delta, a_0\}\}, \qquad D_0 = \delta^{n_0} g, \quad T_0' = \delta^{M/2} h_0.$$

The inherited full-history statistical scales are

$$\mathcal{L} = \max\{F_0, D_0, T_0'\}, \qquad \mathcal{H}_{\mathrm{orig}} = \max\{a_0, D_0, T_0'\} \le g. \tag{22}$$

Its *ORIGINAL* assertion is $\sup \mathbb{E}\,\mathrm{TV}(q, \theta_{Y_M}) \le 2^{13}\mathcal{H}_{\mathrm{orig}}$. Its common, mixing-blind selector pools when

$$\mathrm{TV}(q, b) \le \rho(J), \qquad \rho(J) = \min\{g, 64\sqrt{K/(J+K)}\},$$

and has risk at most $2^{16}\mathcal{L}$. The underlying event-radius and rejection-restricted estimates, not just that scalar risk inequality, are used by the finite decoder. These statistical facts are inherited from the full-history statistical, event-radius and rejection proof chain in the accompanying dossier; they are not independently reproved in this article. In particular, no unconditional next-sample IID claim is substituted for them. Both scales are at most $K^{5/2}/\sqrt{M}$.

D.2 From four bounded heads to a conservative decoder

The compressed bank uses four score/value patterns at the terminal query (a, b):

Head	Ideal score using supplied dyadics	Values
P_1	$L_2\mathbf{1}\{\mathrm{lag2} = a\} + L_2\mathbf{1}\{\mathrm{lag1} = b\}$	token, lag2
P_2	$L_2\mathbf{1}\{\mathrm{lag2} = a\} + L_3\mathbf{1}\{\mathrm{lag1} = b\}$	token, lag2
T	$L_2\mathbf{1}\{\mathrm{lag2} = b\}$	lag1, initial flag vector
0	0	lag2, lag1, flag vector, flag

Here L_2, L_3 are finite downward approximations to $\ln 2, \ln 3$ and the flag vector at key three is e_{X_2}, zero elsewhere. The table has $2K + 2K + 2K + (3K + 1) = 9K + 1$ outputs. All features are local bits; equalities are dot products of one-hots. Scores are at most $\ln 6 < 2$, and values lie in $[0, 1]$. The public mask is keys $3, \ldots, \min(t, M + 2)$ at rows $t \ge 3$, with self fallback at rows one and two. Its size is at least $t/3$ including the appended row. Only the terminal query uses future boundary information relative to historical positions.

The exact finite original weights lie in $[1, 6]$. Their ratio formula verifies membership in the local family. For the causal-family upper, apply Theorem C.1 at target $h + 3$ instead. The public log approximations are generated at

$$p_0 = 4h + 4\,\lceil \log_2(M + K + 4) \rceil + 64.$$

Their error perturbs each ideal head coordinate by at most $1/(8Q)$. Compiler error is at most $1/(8Q)$, and final downward rounding costs less than $1/Q$. Thus the new actual bank lies within $5/(4Q) < 2/Q$ of the true-log identity bank. Positive Taylor weights and bit-valued projections also keep it on the closed $[0, 1]$ grid.

The deterministic identity-and-error ledger in the original finite-interface identities and conservative decoder proof then recovers grid frequencies $a_c, y_c \in [0, 1]$ satisfying

$$|a_c - C_c/M| \le 76/Q, \qquad |y_c - H_c/M| \le 6/Q. \tag{23}$$

All exact endpoint corrections and finite denominators are included in that ledger. This is the precise reusable interface; it does not require the old exponential recurrence once the head-error and range conditions are established.

Normalize positive weights $Ma_c + 1$ and $My_c + 1$ on the ordered Q-simplex grid, giving $\widehat{q}, \widehat{b}$. With $E_0 = 76KM/Q$, $A = J + K$ and $k = K/Q$, (23) implies

$$\mathrm{TV}(\widehat{q}, q) \le \min\{1, E_0/A\} + k, \qquad \mathrm{TV}(\widehat{b}, b) \le 7k.$$

Let $\widehat{A} = \sum_c (Ma_c + 1)$, $A_- = \max\{K, \widehat{A} - E_0\}$, and

$$e_{\mathrm{obs}} = \min\{1, E_0/A_-\} + 8k, \qquad \rho_- = \min\{g, 64\sqrt{K/A_-}\}.$$

The finite rule pools when $\mathrm{TV}(\widehat{q}, \widehat{b}) \le \min\{1, \rho_- + e_{\mathrm{obs}}\}$. The apparent square roots can be eliminated by the exact integer comparisons in the conservative finite-decoder proof; they are not runtime real oracles. Since $A_- \le A$, any history pooled by the exact selector is also pooled by the finite rule. Equality of decisions is unnecessary; the one-sided rejection containment is the relevant invariant.

Using the inherited event-radius and rejection estimates, the conservative finite-decoder proof establishes the pre-integration reduction

$$\mathrm{risk} \le \min\left\{1, 2^{16}\mathcal{L} + 3\mathbb{E}\min\left\{1, \frac{E_0}{J+K}\right\} + \mathbb{P}(J + K < 4E_0) + \frac{9K}{Q}\right\}. \tag{24}$$

This is a substantive inherited interface, not something inferred backwards from a final risk bound. The causal compiler satisfies its deterministic premises, so the same downstream map and reduction apply to the changed encoder.

D.3 Dependence-safe occupation estimates

For any homogeneous chain on d states and arbitrary initialization, let J_m count earlier visits to its actual state at time m. Then

$$\mathbb{P}(J_m < w) \le \min\{1, d^2 w/(m+1)\}, \qquad w \in \{1, 2, \ldots\}. \tag{25}$$

Indeed the largest such probability over initial states is nonincreasing in m by terminal-suffix domination and the Markov property. On each deterministic path only the first w occurrences of a state have fewer than w earlier visits. Summing over time and then over the d possible starting states bounds the sum of these maxima by $d^2 w$. Monotonicity proves (25). This is the inherited universal occupation argument, reproduced here because it is also used for the changed histogram encoder.

For $d = K^2$, layer cake at the exact integer threshold gives

$$\mathbb{E}\min\{1, A/(J_m + k)\} \le \Psi(K^4 A/(m+1)), \quad k \ge 1, \tag{26}$$

where $\Psi(0) = 0$, $\Psi(x) = x(1 + \ln(1/x))$ on $(0, 1)$ and $\Psi(x) = 1$ on $[1, \infty)$. Specifically $J_m < A/t - k$ is either empty or $J_m < \lceil A/t - k \rceil$, and that ceiling is at most A/t. There is no independence assumption about the terminal query.

Under stationarity and $\beta > 0$, the inherited stationary occupation analysis proves the sharper bound

$$\mathbb{E}_\Pi \min\{1, A/(J_m + k)\} \le \frac{dA}{m}\left[17 + \ln\left(1 + \frac{480}{\beta A}\right)\right], \quad d = |\operatorname{supp}\Pi| \le K^2. \tag{27}$$

The proof reverses the chain conditional on its actual endpoint s. The reversed kernel on positive stationary support satisfies the same two-step minorization. If $p = \Pi(s)$ and τ_w is its wth positive

return, the stopped occupation-vector argument gives $\mathbb{E}_s\tau_w = w/p$ and $\mathbb{P}_s(J_m < w) \le w/(mp)$. A tilted singular-operator comparison, valid without reversibility, gives

$$\mathbb{P}_s(J_m < mp/16) \le 5e^{-\beta mp/96}.$$

Integrating the minimum of this tail and $A/(mpt)$, then averaging over s, yields (27). The factor p cancels only after the conditional estimate has been proved. The full operator and return proof is retained in the accompanying stationary-occupation proof, rather than hidden behind an informal mixing-concentration citation.

D.4 Exact residual suffix laws

A TV burn error alone would leave a nonvanishing additive term as $Q \to \infty$. The all-Q upper instead uses an exact path mixture. Let $u = \lfloor M/2 \rfloor$, $m = M - u$, $r = \lfloor M/4 \rfloor$ and, for $0 < \beta < 1$,

$$R = \frac{P^2 - \beta\mathbf{\Pi}}{1-\beta}, \qquad \mathbf{\Pi}(s,t) = \Pi(t), \qquad \epsilon = (1-\beta)^r.$$

All-state minorization makes R stochastic even off support. Since $R\mathbf{\Pi} = \mathbf{\Pi}R = \mathbf{\Pi}$, for every initial μ,

$$\mu P^u = (1-\epsilon)\Pi + \epsilon\nu, \qquad \nu = \mu R^r P^{u-2r}. \tag{28}$$

Apply the same original-P path-extension kernel to both components: the *entire* suffix law is a mixture of a stationary P trajectory and an ordinary P trajectory from ν. The residual suffix does not evolve under R. At $\beta = 1$ the stationary component is exact and the residual is omitted.

Full-history visits dominate suffix visits pathwise. Apply (27) to the stationary component and (26) to the residual component in (24). The result from arbitrary-start occupation analysis is

$$\text{risk} \le \min\left\{1, 2^{16}\mathcal{L} + \frac{2^{14}K^3}{Q}\left[1 + \ln\left(1 + \frac{Q}{\beta KM}\right)\right] + 7\epsilon\Psi(152K^5/Q)\right\}.$$

For $\beta M = t \ge 1$, $\epsilon \le ee^{-t/4}$ and

$$\epsilon(1 + \ln Q) \le 1 + \ln(1 + Q/t) + 4.$$

This bounds the residual by $5320K^5W$. If $\beta < 1/M$, the universal occupation bound directly yields the same envelope. Hence

$$\text{risk} \le \min\{1, 2^{16}\mathcal{L} + (2^{14}K^3 + 5320K^5)W\} \le \min\{1, C_K H\}.$$

The suffix, reverse chain and residual kernel are analysis objects. The predictor does not compute them or receive another history port. Together with the lower proof and stationary inclusion this completes the theorem's stated modular argument.

E Weighted inventories and the effective score budget

Bounded-range families have a genuine representation restriction. We now change the finite evaluator and retain more coordinates. The complete triple-bank construction and rescaled-window lower below do not reinterpret the old exponential program.

Two paid singleton-copy heads acquire the previous two one-hots. Paid local Boolean gates form pair keys $p_t(a,b)$ and query-independent one-hot triple values $v_t(a,b,c)$. A dense statistical head has key/query dimension K^2 and retains all K^3 coordinates. Its exact pair dot product selects the integer weight κ on matches and 1 otherwise, where $\kappa \geq 1$ is public. Its effective natural logits are $0, \ln \kappa$; no logarithm or exponential is executed. All M terminal keys have positive weight.

At every row compute exact weighted numerators and denominator, then retain only

$$z_{abc} = \lfloor QA_{abc}/D \rfloor .$$

The decoder selects the K coordinates for the exact terminal pair and normalizes their sum. If all selected floors vanish, it uses the uniform vector. Ordered simplex rounding gives its final Q-grid output. The clean A_{abc}, D, counts and unquantized block mass are not decoder inputs.

E.1 Deterministic error including the zero-floor branch

Let J and C_c denote the earlier occurrences and successors of the terminal pair. For $J > 0$, the exact selected block has mass

$$r = \frac{\kappa J}{M + (\kappa - 1)J} \tag{29}$$

and conditional composition C/J. Nonmatches occupy other coordinates rather than contaminating this composition. If nonnegative y_c sum to r and $f_c = Q^{-1} \lfloor Qy_c \rfloor$, their lost mass is less than K/Q. When $\sum f_c > 0$, subtracting normalized vectors gives $\mathrm{TV}(f/\sum f, y/r) \leq K/(Qr)$. If all floors vanish, $r < K/Q$, so the same capped bound covers any simplex fallback.

Comparison to add-one smoothing and final ordered rounding therefore gives, on every history including $J = 0$,

$$\mathrm{TV}(T_\kappa, q) \leq \frac{K}{J+K} + \min\left\{1, \frac{K(K+1)M}{\kappa Q(J+K)}\right\} + \frac{2K-1}{Q}. \tag{30}$$

The inequality uses $1/r = (\kappa-1)/\kappa + M/(\kappa J)$ and $J + K \leq (K+1)J$ when $J \geq 1$. The first term equals one at $J = 0$. No positive mass surviving the grid is assumed.

Theorem E.1 (Amplified triple bank)**.** *Let $D_K = K^5(K+1)$. For arbitrary initialization and every mixing promise,*

$$\mathrm{risk}(T_\kappa) \leq \min\left\{1, 2^{13}\mathcal{H}_{\mathrm{orig}} + \Psi\left(\frac{K^5}{M+1}\right) + \Psi\left(\frac{D_K}{\kappa Q}\right) + \frac{2K-1}{Q}\right\}. \tag{31}$$

With the integer, mixing-blind choices

$$L_M = \mathrm{bitlength}(M+1), \quad \kappa = L_M, \quad h = \max\{4, \lceil L_M/2 \rceil\}, \quad Q = 2^h,$$

the risk is at most

$$\min\left\{1, \frac{(2^{13}+2)K^{5/2} + 3D_K + 2K - 1}{\sqrt{M+1}}\right\}. \tag{32}$$

Thus the effective logit range is $\ln\ln(M+1) + O(1)$ at $Q \asymp \sqrt{M}$.

Proof. Apply (26) to the first two terms in (30), and use the inherited full-history statistical risk bound. This proves (31). At the integer schedule, $\sqrt{M+1} \leq Q \leq 2\sqrt{M+1}$ and $\ln(1+Q) \leq L_M$. For $D \geq 1, z > 0$,

$$\Psi(D/z) \leq \frac{D[1 + \ln(1+z)]}{z}, \qquad \Psi(x) \leq 2\sqrt{x}.$$

Taking $z = L_M Q$ gives $\Psi(D_K/(L_M Q)) \leq 3D_K/Q$, which proves (32). Budget generation needs only bitlength, integer ceiling and a power of two. □

E.2 A rescaled tube with uniform weight-ratio constants

Fix $K \geq 3$ and a one-block local family with at most D coordinates, trailing radius r, actual weights in $[1, \kappa]$, values in $[-1, 1]$, and only the boundary side information of (3). Here $\kappa \geq 1$ may be real; D, r are fixed independently of M, β, Q, κ. The lower permits arbitrary real public positional tables and source-independent shared randomness. It does not include recursively global features or query-selected masks. Smaller banks can be conceptually zero-padded for this information argument without changing a physical implementation.

The earlier complete triple-bank construction substitution in local-family analysis gives the valid lower $\Omega(\ln M/(\kappa^{D+1}\sqrt{M}))$ in weak mixing at $Q \asymp \sqrt{M}$. It loses a power of κ through the number of grid cells across a thickened tube. The following refinement changes the suffix cutoff instead of treating that old constant as uniform.

Proposition E.2 (Rescaled rare-window lower)**.** *Put $z = \kappa Q$, $H_b = 3^{2r}$, $T_b = r\sqrt{40DH_b}$ and*

$$N_b = \lceil 8T_b + 4 \rceil + 1, \qquad A_b^* = H_b(2 + 4r)N_b^D, \qquad c_b = (8192A_b^*)^{-1}.$$

For $0 \leq \beta \leq 1/16$ and $z \leq M/4$, the stationary minimax risk, optimized over this entire family, satisfies

$$\mathcal{R}_{\kappa,\mathrm{stat}} \geq \frac{c_b}{z} \ln\left(1 + \frac{1}{8}\min\left\{\frac{z}{M\beta_0}, \frac{M}{z}\right\}\right). \tag{33}$$

The constant depends only on D, r, not on κ, Q, M, β, K or the tables. This is not a matching all-Q two-resource risk formula.

Proof. Use the stationary witness (8), the same uniform prior and the events E_j, but replace (9) by

$$J_* = \left\lfloor \min\{(4\beta_0)^{-1}, M^2/(4z^2)\} \right\rfloor.$$

Both terms are at least four, so $J_* \geq 4$, J_* is at least half the unfloored minimum, and $z\sqrt{J_*}/M \leq 1/2$. Freeze every boundary query before sampling the suffix. A coin changes at most r numerator or denominator terms, by at most $2\kappa r$ in total. The simultaneous sum deviations are therefore at most $\kappa T_b\sqrt{j}$ with probability at least $9/10$. Denominators are at least M, giving

$$|F_{ds} - m_{ds}(p)| \leq 2\kappa T_b\sqrt{j}/M, \qquad \|m_s'(p)\|_\infty \leq 4\kappa r j/M.$$

After scaling by Q, the tube thickness is at most T_b, independent of κ. A parameter mesh with gaps at most $M/(4rzj)$ has at most $(2 + 4r)(1 + zj/M)$ points and one grid unit of approximation error. Rounding and the boundary label give at most $A_b^*(1 + zj/M)$ complete good messages. Lemma A.1 and (10) now yield

$$\mathcal{R}_{\kappa,\mathrm{stat}} \geq \frac{1}{2048A_b^*M} \sum_{j=2}^{J_*} (1 + zj/M)^{-1}.$$

The harmonic comparison from Section A, with z in place of Q, proves (33). The prior is unchanged across events and independent seeds. The constant is uniform because κ shortens the admissible window and enters the mesh length through z; it no longer enlarges the grid-box cross-section. □

E.3 The least feasible budget in a square-root grid band

Fix $0 < c_- \le c_+ < \infty$, a legal dyadic Q with $c_-\sqrt{M} \le Q \le c_+\sqrt{M}$, and a target multiplier $C > 0$. For $a \in \{\mathrm{stat}, \mathrm{all}\}$ define

$$\kappa_{\min,a}(M, \beta, Q; C) = \inf\{\kappa \in [1, \infty) : \mathcal{R}_{\kappa,a} \le C/\sqrt{M}\}, \tag{34}$$

with $\inf \varnothing = +\infty$. The envelope budget κ is real and the lower allows real tables; a feasible upper below uses actual integer weights. This definition assumes neither attainment of the infimum nor a noninteger implementation.

The unchanged complete triple-bank construction upper is more precise than (31). Put

$$t = M\beta_0, \quad G(z, t) = 1 + \ln(1 + z/t), \quad C'_K = 48K^3(K + 1) + 10D_K.$$

Then its same paid triple bank obeys

$$\mathrm{risk}(T_\kappa) \le \min\left\{1, 2^{13}\mathcal{H}_{\mathrm{orig}} + \Psi\left(\frac{K^5}{M+1}\right) + C'_K\frac{G(\kappa Q, t)}{\kappa Q} + \frac{2K-1}{Q}\right\}. \tag{35}$$

Theorem E.3 (Sharp weight-ratio order in the fixed band)**.** *Let $K \ge 3$, $D \ge K^3$ and $r \ge 3$ be fixed. Set*

$$f = 1 + \ln(1 + Q/(M\beta_0)), \qquad C_{\mathrm{suff}} = (2^{13} + 2)K^{5/2} + (2C'_K + 2K - 1)/c_-.$$

For every fixed $C \ge C_{\mathrm{suff}}$, uniformly in $M \ge 64$, $\beta \in [0, 1]$ and all legal Q in the stated band,

$$c_* f \le \kappa_{\min,\mathrm{stat}} \le \kappa_{\min,\mathrm{all}} \le \lceil f \rceil \le 2f, \tag{36}$$

where $c_* > 0$ depends on D, r, c_+, C. The necessary bound holds for every fixed $C > 0$; feasibility at arbitrarily tiny multipliers is not claimed.

Proof. For sufficiency choose the public integer $\kappa = \lceil f \rceil$. Since $1 + \kappa Q/t \le \kappa(1 + Q/t)$,

$$G(\kappa Q, t)/\kappa \le (f + \ln \kappa)/\kappa \le 2.$$

Use $\mathcal{H}_{\mathrm{orig}} \le K^{5/2}/\sqrt{M+1}$, $\Psi(x) \le 2\sqrt{x}$ and $Q \ge c_-\sqrt{M}$ in (35). This proves arbitrary-start feasibility with C_{suff}. Its all-zero-block and $J = 0$ branches remain covered by (30).

For necessity write $x = Q/(M\beta_0)$, so $x \le Q$ and $Q^2/M \le c_+^2$. Choose a finite $F \ge 2(1 + \ln 8)$, depending only on c_+, such that for every $s \ge F$,

$$\frac{e^{s-1} - 1}{8c_+^2 s} \ge e^{s/2} - 1.$$

Such a choice exists by exponential domination. Exhaust the cases. If $\beta > 1/16$, then $x \le 2c_+$ and $\kappa \ge 1 \ge f/(1 + 2c_+)$. If $z > M/4$, then $\kappa > \sqrt{M}/(4c_+)$, whereas $f \le (1/8 + c_+)\sqrt{M}$; this gives a fixed multiple of f without extrapolating the tube. In the remaining domain (33) applies. If $f \le F$, use $\kappa \ge f/F$; if $\kappa \ge f$, there is nothing to prove.

Otherwise $f > F$ and $1 \le \kappa < f$. The two logarithmic branches satisfy

$$\ln(1 + \kappa x/8) \ge f - 1 - \ln 8 \ge f/2, \qquad \ln(1 + M/(8\kappa Q)) \ge f/2.$$

For the second, use $M/(\kappa Q) \geq x/(\kappa c_+^2) > x/(f c_+^2)$ and $x = e^{f-1} - 1$. Thus (33) and risk at most $C/\sqrt{M}$ force $\kappa \geq c_b f/(2Cc_+)$. Taking the minimum of the five positive constants

$$1, \quad (1+2c_+)^{-1}, \quad [4c_+(1/8+c_+)]^{-1}, \quad F^{-1}, \quad c_b/(2Cc_+)$$

proves uniform necessity. It applies to every feasible real budget, hence its infimum. Stationary inclusion gives the middle inequality in (36). □

When $\beta \leq 1/M$, $f \asymp \ln M$ on this band, so $\kappa_{\min,a} = \Theta(\ln M)$ and $\ln \kappa_{\min,a} = \ln\ln M + O(1)$. For bounded f, the meaningful uniform range comparison is $1 + \ln \kappa_{\min,a} = \Theta(1 + \ln f)$, not a positive lower on $\ln \kappa_{\min,a}$. The integer upper changes no program or cost ledger. Specifying $\lceil f \rceil$ from a known class promise is not learning β or evaluating a free transcendental primitive: an implemented selection rule must pay its supplied-parameter and arithmetic costs. The mixing-blind bitlength schedule in Theorem E.1 remains available. Bands containing no legal dyadic Q have no instance at that M.

E.4 Uniform attention and representation costs

At $\kappa = 1$, the construction is ordinary uniform attention over the full triple histogram, followed by conditional block normalization. It is a strong elementary baseline, not a new counting mechanism.

Corollary E.4 (Uniform full-histogram synthesis)**.** *At fixed $K \geq 3$, its best post-grid decoder has stationary and arbitrary-start risk of order $H(M, \beta, Q)$, uniformly over the legal β, Q.*

Proof. For the upper, set $\kappa = 1$ in (35): the first two terms are $O_K(M^{-1/2})$, since $\Psi(x) \leq 2\sqrt{x}$, and the remaining terms are $O_K(W)$. For the lower, this exact K^3 bank belongs to (3) with $a = b = 1, r = 3$; its decoder risk obeys the local-family $1/Q$, statistical and rare-window lower bounds, combined in Section A.3 into $\Omega_K(H)$. This is a synthesis of the complete triple-bank construction and local-family analysis, not a separately audited source unit. □

The full bank pays K^3 coordinates instead of the compressed bank's $9K + 1$, so fixed-K rate equality does not imply equal dimension or costs. The earlier growing-range successor-only construction with weights $1, M, M^2$ still gives a valid sufficient point with effective range $2 \ln M$. The old triple-bank lower also remains true; the rescaled proof sharpens it rather than refuting it. The fixed-band theorem above has no exact threshold constant and no implication for arbitrary deep or growing architectures. Section F extends the one-block analysis to all legal joint weight/grid regimes using an additional uniform IID argument.

F The joint weight-ratio and grid frontier

The fixed-band threshold of Section E does not by itself determine the risk when both resources vary. In particular, the earlier IID tube constant depended on the weight envelope. The next argument proves that even an arbitrarily large finite weight ratio cannot remove the final-grid penalty. Together with the rescaled rare-window lower and the unchanged triple-bank upper, this closes the joint frontier in the one-summary-block family. It does not enlarge the bounded causal family of Theorem 3.1.

F.1 A grid floor uniform in the weight envelope

Proposition F.1 (Uniform IID grid floor)**.** *Fix $D, r \geq 1$ in the local-ratio model* (3)*, with actual weights in $[1, \kappa]$. For all finite real $\kappa, Q \geq 1$ and $M \geq 64$,*

$$\mathcal{R}_{\kappa,\mathrm{stat}} \geq c_I(D, r)/Q$$

even on stationary binary IID sources. The constant is independent of the weight envelope, alphabet size and mixing promise. The decoder may still return arbitrary real probabilities.

Proof. Use IID Bernoulli(p) observations on two designated symbols, and define every second-order row, including unsupported rows, to emit that same law. Then $P^2(s, t) = \Pi(t)$ on every state, so every $\beta \in [0, 1]$ is allowed and the actual terminal target is always p. Fix and grant an encoder seed. The following prior and constants do not depend on that seed.

Write

$$H_0 = 3^{2r}, \quad F_0 = DH_0, \quad q_0 = 4^{-r}, \quad L_0 = rF_0 + 2r,$$
$$J_0 = (2rL_0 + 1)/q_0, \qquad V_0 = 4r^2 J_0/q_0,$$

and choose an integer

$$C_0 \geq \max\{1,\ 400F_0(2r-1)/q_0^2,\ 3200F_0V_0\}.$$

Put $m = \lceil C_0 Q^2 \rceil$. If $M < m$, the full-history two-point bound of Section A.3 gives risk at least $1/(128\sqrt{M+2}) \geq 1/(128\sqrt{C_0+3}\,Q)$. This branch already handles a locality as large as the history.

Suppose $M \geq m$ and give p the uniform prior on $I = [1/4, 3/4]$. Freeze every possible boundary branch s before sampling. For each coordinate, branch and position, let a_{dsi} be the largest weight over the attainable binary patterns in its trailing window. Select the m largest such maxima for each (d, s), with public tie-breaking. Reveal all bits in those windows and all boundary bits. The revealed position set U is fixed independently of the sample and p; for $N = |U|$,

$$m \leq N \leq rF_0 m + 2r \leq L_0 m.$$

This is a proof-only oracle, not a new encoder feature. Write $B = X_U$.

A chosen maximizing pattern has probability at least q_0. Each selected window overlaps at most $2r - 2$ others, so the variance of the count of realized maximizing patterns is at most $(2r-1)m$. Chebyshev and a union over the F_0 branches give a B-measurable event E, with probability at least .99 for every p, on which at least $q_0 m/2$ selected patterns realize their maxima in every branch. If a_* is the largest unselected maximum in one branch, its now-known selected weight mass is at least $(q_0 m/2)a_*$.

Condition on $B \in E$ and fixed p. For a numerator A and denominator D_n, let $\mu_A = \mathbb{E}[A \mid B]$, $\mu = \mathbb{E}[D_n \mid B]$ and $h = \mu_A/\mu$. At most rN unselected windows intersect U, and their total maxima are bounded by $rL_0 m a_*$. Do not assign probability q_0 to their old maximizing patterns: those patterns may have become impossible after the reveal. Each disjoint window, however, retains its IID law and has expected weight at least $q_0 a_i$. Charging the intersecting windows to the selected mass gives

$$\sum_{\text{unselected}} a_i \leq J_0\mu, \qquad a_* \leq \frac{2\mu}{q_0 m}.$$

Efron–Stein on the remaining independent bits, with at most r affected summands per bit, therefore yields

$$\mathrm{Var}(A \mid B),\ \mathrm{Var}(D_n \mid B) \leq 2r^2 \sum a_i^2 \leq V_0\mu^2/m.$$

A simultaneous Chebyshev event of conditional probability at least .99 bounds both deviations by $\mu/(4Q)$. Exact ratio subtraction then gives

$$|A/D_n - h| \le 2/(3Q) < 1/Q.$$

For every remaining local pattern the absolute logarithmic derivative of its Bernoulli probability is at most $4r$. Positivity of weights gives $|\mu'_A|, |\mu'| \le 4r\mu$, hence $|h'| \le 8r$, independently of κ.

The oracle posterior is not uniform. If B contains t_B ones and $y = t_B/N$, its density on I is proportional to $p^{t_B}(1-p)^{N-t_B}$. Put

$$I_B = I \cap [y - 3/\sqrt{N}, y + 3/\sqrt{N}].$$

Hoeffding followed by Markov shows that, except for a set of B of probability at most $200e^{-18}$, this interval has posterior mass at least .99. Call this event T. On T its posterior density is at most $128\sqrt{N}$: for the clamped mode p_0, the log density ℓ satisfies $|\ell'(p_0)| \le 16\sqrt{N}$ and $|\ell''| \le 16N$. Integrating for distance $1/(64\sqrt{N})$ in a direction remaining inside I gives a normalizer at least $e^{\ell(p_0)}/(128\sqrt{N})$. This also handles a mode on a truncation boundary. Consequently the density is at most

$$P_0 Q, \qquad P_0 = 128\sqrt{L_0(C_0+1)}.$$

For $B \in E \cap T$, an economical mesh of I_B with spacing at most $1/(8rQ)$ has at most $2 + 48r$ points, because $N \ge Q^2$. The derivative bound, random tube and downward grid together put every good complete bank in at most

$$R_0 = (2 + 48r)9^D$$

labels. There is no missing boundary factor: B fixes the actual side field. The joint posterior probability of lying in I_B and being in the tube is at least .98. For a fixed decoder seed, neighborhoods of radius $\varepsilon = (8R_0P_0Q)^{-1}$ about all R_0 possible scalar outputs have posterior mass at most $1/4$. The conditional expected error is thus at least $1/(16R_0P_0Q)$. Since $\mathbb{P}(E \cap T) > .98$, averaging over B and both seeds proves the result with

$$c_I = \min\{(128\sqrt{C_0+3})^{-1}, (32R_0P_0)^{-1}\}.$$

The fixed reveal and its posterior are both essential. Treating the reveal as free model information, or retaining a uniform prior after it, would not prove this statement. □

F.2 All joint regimes and both initializations

For $t \ge 1$ define

$$\mathcal{W}(z,t) = \frac{1 + \ln(1 + z/t)}{z}, \qquad \mathcal{F}(M,\beta,\kappa,Q) = \max\left\{M^{-1/2},\, Q^{-1},\, \mathcal{W}(\kappa Q, M\beta_0)\right\}.$$

Here κ is a weight envelope, not an additional exact message. The following theorem permits arbitrary real tables inside that envelope; its upper witness uses only integer weights.

Theorem F.2 (Complete one-block frontier)**.** *Fix $K \ge 3$, $D \ge K^3$ and $r \ge 3$. In the one-block local positive-ratio family, for every integer $\kappa \ge 1$, $M \ge 64$, $\beta \in [0,1]$ and legal dyadic $Q \ge 16$,*

$$c\,\mathcal{F} \le \mathcal{R}_{\kappa,\mathrm{stat}} \le \mathcal{R}_{\kappa,\mathrm{all}} \le \mathrm{risk}_{\mathrm{all}}(T_\kappa) \le \min\{1, C_K^{\mathrm{joint}}\mathcal{F}\}. \tag{37}$$

One may take

$$c = \min\{(256 \cdot 5)^{-1}, c_I/2, c_b/48\}, \qquad C_K^{\text{joint}} = (2^{13}+2)K^{5/2} + C'_K + 2K - 1.$$

The lower constants depend only on the fixed D, r, not on κ. This is not a theorem for arbitrary deep or growing architectures.

Proof. The statistical floor from Section A.3 gives $\mathcal{R} \geq M^{-1/2}/256$, and Proposition F.1 gives c_I/Q. Proposition E.2 supplies $c_b L$ only for $\beta \leq 1/16$ and $z = \kappa Q \leq M/4$, where

$$L = z^{-1} \ln(1 + \min\{z/t, M/z\}/8), \qquad t = M\beta_0.$$

These bounds may use different witnesses; take their maximum, not a sum of losses.

The completion is elementary but must be uniform in z. The function $\mathcal{W}$ decreases in z and t. If $\beta > 1/16$, $\mathcal{W} \leq Q^{-1} + 16/M \leq Q^{-1} + 2M^{-1/2}$. If $z > M/4$, then

$$\mathcal{W}(z,t) \leq 4[1 + \ln(1 + M/4)]/M < 5M^{-1/2}.$$

If $z > M^{3/4}$, then

$$\mathcal{W}(z,t)\sqrt{M} \leq \frac{1 + \ln(1+z)}{z^{1/3}} \leq 1 + \ln 2 + 3/e < 3.$$

On the remaining domain, set $x = z/t$, $y = M/z$. Then $x \leq z \leq y^3$ and $y \geq 4$. Splitting at $x = y$ and using log concavity gives

$$\ln(1+x) \leq 24 \ln(1 + \min\{x, y\}/8), \qquad \mathcal{W} \leq Q^{-1} + 24L.$$

The separate floors and the truncated lower therefore imply the displayed $c\mathcal{F}$ everywhere, including equality at the branch boundaries.

For the upper, apply (35) and $\mathcal{H}_{\text{orig}} \leq K^{5/2}/\sqrt{M+1}$, $\Psi(u) \leq 2\sqrt{u}$. The complete triple bank is in this family because $D \geq K^3$, $r \geq 3$, all keys have positive weights, and its query uses only the permitted terminal pair. The zero-block fallback, $J = 0$ case and post-grid normalization remain unchanged. Stationary inclusion gives the middle inequality. All full-history statistical input and whole-suffix mixture dependencies remain exactly those identified in Section D. □

Corollary F.3 (The two simultaneous statistical-order budgets)**.** *At fixed K, D, r, statistical-order risk requires and, up to constant factors, is achieved by*

$$Q \gtrsim \sqrt{M}, \qquad \kappa Q \gtrsim Z_*, \quad Z_* = \sqrt{M}\left[1 + \ln\left(1 + \frac{1}{\sqrt{M}\beta_0}\right)\right]. \tag{38}$$

Necessity holds for every prescribed fixed target multiplier; sufficiency asserts an explicit sufficiently large multiplier. The quantity Z_* equals the earlier numerical threshold Q_*, but is now a product *threshold, not a replacement for the separate grid requirement.*

Proof. Put $a = \sqrt{M}$, $u = a/(M\beta_0)$, $f_* = 1 + \ln(1+u)$ and $v = \kappa Q/a$. If risk is at most C_0/a, let $B = \max\{1, C_0/c\}$. Theorem F.2 gives $Q \geq a/B$ and $[1 + \ln(1+vu)]/v \leq B$. For $v \geq 1$ the numerator is at least f_*, so $v \geq f_*/B$. For $v < 1$, concavity gives $1 + \ln(1+vu) \geq v f_*$, hence $f_* \leq B$; the numerator is also at least one, so $v \geq 1/B \geq f_*/B^2$. Thus $\kappa Q \geq Z_*/B^2$ in every case.

Conversely, $1 + f_* u \leq f_*(1+u)$ implies $\mathcal{W}(af_*, M\beta_0) \leq 2/a$. If $Q \geq a$ and $\kappa Q \geq Z_*$, the unchanged program has risk at most $2C_K^{\text{joint}}/a$. Fixed multiplicative relaxations of either budget change only this target constant. Legal choices always exist: take $Q = 2^{\max\{4, \lceil \log_2 \sqrt{M} \rceil\}}$ and $\kappa = \max\{1, \lceil Z_*/Q \rceil\}$. □

At a fixed square-root grid band, (38) recovers the sharp weight-ratio conclusion of Theorem E.3. At $\kappa = 1$ it recovers the ordinary full-histogram baseline. Sending κ to a very large finite value cannot compensate for $Q \ll \sqrt{M}$. These are representation constraints, not an optimum at fixed total word length, memory, runtime or expanded circuit depth.

G Binary sources: active windows and weighted returns

The three-symbol waiting construction does not describe the entire alphabet dependence. We now set $K = 2$, keeping exactly the same one-block interface. The upper requires $D \geq 8, r \geq 3$, integer $\kappa \geq 1$ and legal dyadic $Q \geq 16$. The lower permits any fixed $D, r \geq 1$ and real $\kappa, Q \geq 1$. Its active segment is Markov, not IID, and the useful stationary mixing scale is different.

G.1 A binary stationary rare-window witness

Put

$$e_0 = \max\{\sqrt{\beta}, M^{-1}\}, \qquad z = \kappa Q.$$

Proposition G.1 (Binary truncated lower)**.** *For $M \geq 64$, $0 \leq \beta \leq 1/256$ and $z \leq M/4$,*

$$\mathcal{R}^{(2)}_{\kappa,\mathrm{stat}} \geq \frac{c_{\mathrm{bin}}}{z} \ln\left(1 + \frac{1}{8}\min\left\{\frac{z}{Me_0}, \frac{M}{z}\right\}\right), \tag{39}$$

where $c_{\mathrm{bin}} > 0$ depends only on D, r.

Proof. Order the pair states as $W = 00, A = 01, B = 10, C = 11$. Let $a = 1/(2M)$, $p \in [1/4, 3/4]$, and use the actual homogeneous kernel and stationary law

$$P = \begin{pmatrix} 1-a & a & 0 & 0 \\ 0 & 0 & 1-p & p \\ e_0 & 1-e_0 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix}, \qquad \Pi = x(e_0/a, 1, 1, 2p), \quad x = (e_0/a + 2 + 2p)^{-1}.$$

Direct multiplication gives stationarity and $\Pi_W \geq 4/11$. The two-step entries, together with $e_0^2 \geq \beta$ and $e_0 \geq 4\beta$, give the all-state certificate. For example its restrictive entry is $P^2(B, A) = e_0 a \geq \beta x$; the first row's active entries are at least $a/4$, and the other nonexceptional active entries are at least $1/8$. The proof uses the actual p-dependent hidden Π, not a fixed waiting-state start.

Set $J_* = \lfloor \min\{(4e_0)^{-1}, M^2/(4z^2)\} \rfloor \geq 4$. For $4 \leq j \leq J_*$, consider histories with $M + 2 - j$ initial zeros, then a first 1, and an active suffix with no pair 00. On the active states A, B, C use the reference chain

$$R = \begin{pmatrix} 0 & 1-p & p \\ 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \end{pmatrix},$$

started at A. Relative to its $j-1$-transition law, the actual history density is

$$\Pi_W (1-a)^{M-j} a (1-e_0)^{N_B} \geq (32M)^{-1},$$

where $N_B \leq j-1$. The events for different j are disjoint. Their probabilities need not be independent of p; pointwise domination, integrated against one common uniform prior, replaces the IID-event argument.

Every row of R^2 puts mass at least $1/8$ on B. Coupling in two-step blocks therefore gives expected differing-state count at most $2\sum_{b\ge 0}(7/8)^b = 16$. For any positional sum F of local terms bounded by κ, two continuations after a common prefix differ in expected F by at most $34\kappa r$. Doob martingale increments and single-transition differentiation then give

$$\operatorname{Var}_p F \le (34\kappa r)^2 j, \qquad |(\mathbb{E}_p F)'| \le 34\kappa r j.$$

These estimates cover overlapping windows and arbitrary positional tables; no independent-coin argument is used. Also $R^3(s, A) \ge 1/8$ for every s, so the terminal state is A with probability at least $1/8$ for all $j \ge 4$.

Freeze all $H_2 = 2^{2r}$ boundary branches. With

$$T_2 = 34r\sqrt{64DH_2}, \quad N_2 = \lceil 2T_2 + 4\rceil + 1, \quad A_2 = H_2(2 + 68r)N_2^D,$$

simultaneous Chebyshev gives a good event of probability at least $31/32$, ratio deviation at most $2\kappa T_2\sqrt{j}/M$, and curve derivative at most $68\kappa r j/M$. The rescaled cutoff makes the grid-tube radius at most T_2. A mesh therefore contains all complete good messages in at most $A_2(1 + zj/M)$ labels.

Only terminal A has actual next-symbol parameter p. The good-and-A event has probability at least $3/32$ for every p. Radius $1/(128R_j)$ neighborhoods about $R_j = A_2(1 + zj/M)$ decoder outputs cover prior mass at most $1/32$. Their complement within the good-and-A event gives mean loss at least $1/(2048R_j)$. This is a joint-prior calculation, not a uniform posterior after conditioning on the endpoint.

After pathwise domination and summation,

$$\mathcal{R}^{(2)}_{\kappa,\mathrm{stat}} \ge \frac{1}{65536A_2M}\sum_{j=4}^{J_*}(1 + zj/M)^{-1}.$$

For $u = z/M \le 1/4$, the sum is at least $\ln(1 + uJ_*)/(8u)$, including $J_* = 4$. Since $uJ_* \ge \min\{z/(Me_0), M/z\}/8$, this proves (39) with $c_{\mathrm{bin}} = (524288A_2)^{-1}$. □

For this topology the necessity of a square-root mixing scale is visible in $P^2(B, A) \ge \beta\Pi_A$, which is equivalent to $e_0^2 + e_0a(2 + 2p) \ge \beta$. Taking both exit and launch probabilities of order M^{-1} only allows $\beta = O(M^{-2})$. This identity alone is not a universal binary obstruction; a matching upper needs a separate argument.

G.2 Stationary weighted occupation and the actual floor decoder

Let a stationary binary source have

$$\Pi_{00} = y, \quad \Pi_{01} = \Pi_{10} = x, \quad \Pi_{11} = w$$

and exit parameters $a = \theta_{00}(1)$, $b = \theta_{01}(1)$, $e = \theta_{10}(0)$, $t_1 = \theta_{11}(0)$. Define analysis weights $q_{00} = a$, $q_{11} = t_1$, and $q_{01} = q_{10} = 1$. The crossing weights are clipping constants, not assertions that the crossing rows have unit exit probability.

Lemma G.2 (Binary weighted reciprocal occupation)**.** *For $\beta > 0$, $A > 0$ and integer $k \ge 1$,*

$$\mathbb{E}\min\{q_{Y_M}, A/(J + k)\} \le \frac{4A}{M}\left[17 + \ln\left(1 + \frac{96}{\sqrt{\beta}A}\right)\right]. \tag{40}$$

Zero weights contribute zero before any division.

Proof. Stationarity gives $ya = xe$ and $wt_1 = xb$. If $x = 0$, positive β permits only a singleton diagonal support; its weight is zero, so the assertion follows directly. Henceforth assume $x > 0$. Positive-β minorization makes the supported chain irreducible, and its two-step entries and balance now imply $e^2 \geq \beta y$ and $b^2 \geq \beta w$. First-step equations for the largest supported hitting mean H_s give, for example,

$$H_{00} \leq \frac{b}{et_1} + \frac{2}{e} + \frac{1}{t_1} + 1, \qquad H_{01} \leq \frac{2}{a} + \frac{1}{t_1} + 2.$$

Multiplication by $\sqrt{\beta}\,\Pi_s q_s$, balance and the two square inequalities show

$$H_s \leq \frac{5}{\sqrt{\beta}\,\Pi_s q_s}.$$

The other targets follow by exchanging symbols. Absent states are omitted using the smaller supported first-step system, not evaluated by $0/0$.

The stationary reverse pair chain becomes the forward chain after swapping pair coordinates. Thus the same bound applies when conditioning on the actual terminal state s and looking backward. Kac gives the mean positive return $1/\Pi_s$. For $\overline{H} = \max\{H_s, 1/\Pi_s\}$, $\overline{H} \leq 6/(\sqrt{\beta}\,\Pi_s q_s)$. The elementary hitting-time moment induction yields

$$\mathbb{E}\tau^m \leq m!\overline{H}^{m-1}\mathbb{E}\tau.$$

Indeed $T^m \leq m\sum_{j<T}(T-j)^{m-1}$, and on survival the remaining law is a supported hitting law. With $\lambda = \sqrt{\beta}\,\Pi_s q_s/24$, this gives $\mathbb{E}e^{\lambda\tau} \leq e^{2\lambda/\Pi_s}$. Strong Markov at returns and a separate small-$M\Pi_s$ cap imply

$$\mathbb{P}_s(J < M\Pi_s/16) \leq 2e^{-\sqrt{\beta}\,M\Pi_s q_s/48}.$$

This is a weighted return estimate, not a claim of a global $\sqrt{\beta}$ spectral gap.

Put $\mu = M\Pi_s$ and $\alpha = A/\mu$. The inherited Kac–Markov count bound is $\mathbb{P}_s(J < j) \leq j/\mu$. In layer cake for $\min\{q_s, A/(J+k)\}$, layers below 16α cost at most 16α; higher layers are bounded by $\min\{\alpha/u, 2e^{-\sqrt{\beta}\mu q_s/48}\}$. Using $ve^{-\sqrt{\beta}v/48} \leq 48/\sqrt{\beta}$ gives the conditional bound

$$\frac{A}{M\Pi_s}\left[17 + \ln\left(1 + \frac{96}{\sqrt{\beta}A}\right)\right].$$

Only now multiply by Π_s and sum over at most four supported states. □

Proposition G.3 (Unchanged binary program, stationary upper)**.** *For the same paid eight-coordinate triple bank and decoder, every legal κ, Q, M and $\beta \in [0,1]$ satisfy*

$$\mathrm{risk}_{\mathrm{stat}}(T_\kappa) \leq \min\left\{1, \frac{2^{17}}{\sqrt{M}} + \frac{3}{Q} + 544\,\mathcal{W}(\kappa Q, Me_0)\right\}. \tag{41}$$

The proof here is stationary. Section H establishes the arbitrary-start extension by a different path/count argument.

Proof. Let S_0 be the inherited actual-terminal add-one risk and $B_0 = \mathbb{E}[2/(J+2)]$. Then

$$S_0 \leq 2^{13}\sqrt{32}/\sqrt{M+1}, \qquad B_0 \leq 2\sqrt{32}/\sqrt{M+1}.$$

For $J > 0$, the empirical exit fraction $\widehat{p}$ differs from the add-one fraction by at most $2/(J+2)$. Thus its expected error from the actual exit probability is at most $S_0 + B_0$.

Let $\rho = \kappa J/[M + (\kappa - 1)J]$ be the selected mass and n_0, n_1 its two integer floors. On $N = n_0 + n_1 \geq 1$, write $n_0 = Q\rho(1 - \widehat{p}) - f_0$, $n_1 = Q\rho\widehat{p} - f_1$, with $f_i \in [0, 1)$. The normalized exit $v = n_1/N$ obeys

$$v - \widehat{p} = \frac{\widehat{p}f_0 - (1 - \widehat{p})f_1}{N}, \qquad |v - \widehat{p}| \leq \min\{\widehat{p}, 1 - \widehat{p}, 2/(Q\rho)\}.$$

On diagonal rows, clipping at the true exit probability and its Lipschitz dependence charge one empirical error. On crossing rows use the direct bound with $q_s = 1$. Since $2/(Q\rho) \leq 2/Q + 6M/[z(J + 2)]$ for $J \geq 1$, both cases give

$$|v - \theta_{\text{exit}}| \leq \min\{q_s, A/(J + 2)\} + 2/Q + 2|\widehat{p} - \theta_{\text{exit}}|, \qquad A = 6M/z.$$

The exact weights and true exits here are analysis variables, not added ports.

If the entire selected block floors to zero, $Q\rho < 2$ and hence $J < 4M/z$; this also includes $J = 0$. The unchanged uniform fallback is charged by universal occupation:

$$\mathbb{P}(N = 0) \leq 16\,\lceil 4M/z \rceil\,/(M + 1) \leq 64/z + 16/M.$$

Final ordered output rounding costs at most $1/Q$. Therefore the total risk is at most

$$2S_0 + 2B_0 + \mathbb{E}\min\{q_{Y_M}, A/(J + 2)\} + 64/z + 16/M + 3/Q.$$

For $\beta \geq M^{-2}$, substitute $A = 6M/z$ in (40); the weighted term is at most $480[1+\ln(1+z/(M\sqrt{\beta}))]/z$. Together with $64/z$ and the statistical bounds this is (41). For $\beta < M^{-2}$, including zero, use the unchanged beta-free triple-bank bound and $\Psi(96/z) \leq 96[1 + \ln(1 + z)]/z$ instead. No positive-β return argument is used at zero. □

G.3 The complete stationary binary frontier and alphabet contrast

Define

$$\mathcal{F}_2(M, \beta, \kappa, Q) = \max\{M^{-1/2}, Q^{-1}, \mathcal{W}(\kappa Q, Me_0)\}, \qquad e_0 = \max\{\sqrt{\beta}, M^{-1}\}.$$

Theorem G.4 (Stationary binary frontier)**.** *Fix $K = 2$, $D \geq 8$ and $r \geq 3$. For every $M \geq 64$, $\beta \in [0, 1]$, integer $\kappa \geq 1$ and legal dyadic $Q \geq 16$,*

$$c_2\mathcal{F}_2 \leq \mathcal{R}^{(2)}_{\kappa,\text{stat}} \leq \text{risk}_{\text{stat}}(T_\kappa) \leq \min\{1, 131619\,\mathcal{F}_2\}, \tag{42}$$

where $c_2 = \min\{(256 \cdot 5)^{-1}, c_I/2, c_{\text{bin}}/48\}$ depends only on D, r. It is a stationary theorem only.

Proof. The uniform IID and statistical floors are already binary witnesses. For the logarithmic term repeat the exhaustive completion in Theorem F.2, now with $t = Me_0$ and Proposition G.1. Specifically, when $\beta > 1/256$, $e_0 > 1/16$ and $\mathcal{W} \leq Q^{-1} + 16/M \leq Q^{-1} + 2M^{-1/2}$. The large-$z$ cases $z > M/4$ and $z > M^{3/4}$ depend only on $t \geq 1$ and give the same $5M^{-1/2}$ bound. On the complement the binary lower is within its stated domain; $x = z/t \leq z \leq (M/z)^3$ gives

$$\mathcal{W}(z, t) \leq Q^{-1} + 24z^{-1}\ln(1 + \min\{z/t, M/z\}/8).$$

Thus the three separate lower witnesses imply $c_2\mathcal{F}_2$ with no uncovered boundary case. The upper follows from (41), since $2^{17} + 3 + 544 = 131619$. The same eight-coordinate program supplies admissibility; none of its costs or ports changes. □

The exact stationary statistical-order conditions are consequently

$$Q \gtrsim \sqrt{M}, \qquad \kappa Q \gtrsim Z_{*,2}, \quad Z_{*,2} = \sqrt{M}\left[1 + \ln\left(1 + \frac{1}{\sqrt{M}e_0}\right)\right]. \tag{43}$$

The inversion and constructive allocation proof of Corollary F.3 applies verbatim with $t = Me_0$: necessity holds for every fixed target multiplier, and sufficiency gives an explicitly large one. This transfers an elementary scalar argument, not the three-symbol lower.

For clarity about the least budget, in a fixed legal band $c_-\sqrt{M} \le Q \le c_+\sqrt{M}$ define

$$\kappa^{(2),\mathrm{int}}_{\mathrm{min,stat}} = \min\{k \in \mathbb{N} : k \ge 1,\ \mathcal{R}^{(2)}_{k,\mathrm{stat}} \le C_0/\sqrt{M}\},$$

with value $+\infty$ if the set is empty, and put $f_{Q,2} = 1 + \ln(1 + Q/(Me_0))$. The lower proof holds also for every finite real envelope. Since $\mathcal{W}(\kappa Q, Me_0) \ge f_{Q,2}/(\kappa Q)$, every feasible real budget satisfies

$$\kappa \ge \frac{c_2}{C_0 c_+} f_{Q,2}.$$

For the actual integer choice $\kappa = \lceil f_{Q,2} \rceil \le 2f_{Q,2}$, $\mathcal{W}(\kappa Q, Me_0) \le 2/Q$. Hence the unchanged binary program has risk at most

$$\frac{2^{17} + 1091/c_-}{\sqrt{M}}.$$

For C_0 at least this constant, the feasible integer set is nonempty, its least element exists, and $\kappa^{(2),\mathrm{int}}_{\mathrm{min,stat}} = \Theta(f_{Q,2})$. This also bounds the real-budget infimum between the same lower and integer feasible upper; it does not assert a fractional-weight implementation.

Corollary G.5 (A stationary alphabet distinction)**.** *At $\beta = 1/M$ and a fixed square-root grid band, the least feasible weight ratio is $\Theta(1)$ for $K = 2$, but $\Theta(\log M)$ for every fixed $K \ge 3$, at sufficiently large fixed target-risk multipliers. Both statements use the same all-state certificate, actual-terminal stationary loss and one-summary-block interface.*

Proof. For $K = 2$, $e_0 = M^{-1/2}$, so $f_{Q,2} = 1 + \ln(1 + Q/\sqrt{M})$ is bounded by band-dependent constants. For $K \ge 3$, $M\beta_0 = 1$ and $1 + \ln(1 + Q) = \Theta(\log M)$; Theorems E.3 and F.2 supply the comparison. For a chosen fixed $K \ge 3$, one may use a common fixed budget $D \ge K^3$, $r \ge 3$ for both alphabets. Constants still depend on that architecture; no dimension-uniform or universal Transformer claim follows. □

At $\beta = 0$, both alphabets instead require logarithmic weight ratio in this grid band; the binary witness gives the same weak-mixing behavior throughout $\beta \le M^{-2}$. Thus the distinction is not that binary sources avoid rare contexts altogether. The stationary proof above by itself does not establish arbitrary-start refinement: its reversal step uses stationarity, and the older beta-free upper is insufficient. The next section closes that gap without changing the implemented decoder, and extends the alphabet distinction to both initializations.

H Initialization through counts, not endpoint mixing

The stationary binary proof conditions on the terminal state and reverses a stationary path. Neither operation is available under an arbitrary initial law. A direct replacement by rapid weighted endpoint

mixing is false, even with full stationary support. The successful argument instead retains the first hitting time, the actual terminal local time and the source's true row-exit weights.

Throughout this section write $W = 00, A = 01, B = 10, C = 11$ and use row parameters

$$a = \theta_W(1), \quad b = \theta_A(1), \quad e = \theta_B(0), \quad t_1 = \theta_C(0).$$

For arbitrary starts the appropriate analysis weights are

$$q = (a, b, e, t_1),$$

the probabilities of a designated actual successor on each row. In particular the crossing weights are now b, e, not the unit caps used in the stationary return lemma. All these quantities are proof variables, never decoder inputs.

H.1 A negative control for endpoint-only reasoning

Proposition H.1 (Failure of weighted endpoint transfer)**.** *There are full-support binary sources with genuine all-state coefficient $\beta_* > 0$ for which neither*

$$\sum_s q_s|(\mu P^n)_s - \Pi_s|, \qquad \sum_s |(\mu P^n)_s - \Pi_s|(P^n q)(s)$$

admits a uniform bound $Cn^{-\alpha} + C\exp(-cn\sqrt{\beta_*})$ for any fixed $\alpha, C, c > 0$, even when q is the row-minority vector. This weighted endpoint counterexample obstructs a proof method; it is not a precision-risk lower bound.

Proof. Take $0 < \eta \le \varepsilon \le 1/32$ and

$$a = \varepsilon, \quad b = \eta, \quad e = 1/2, \quad t_1 = \varepsilon\eta, \qquad \Pi = \frac{(1/2, \varepsilon, \varepsilon, 1)}{3/2 + 2\varepsilon}.$$

Multiplying the binary pair kernel shows that the exact largest all-state coefficient is $\beta_* = \eta^2(3/2+2\varepsilon)$; the binding entry is $P^2(A, B)/\Pi_B$. Every other entry dominates it because $\eta \le \varepsilon$. The minority vector is $(\varepsilon, \eta, 1/2, \varepsilon\eta)$.

Start at W and reflect an $A \to C$ transition back to B. In the reflected W, A, B chain, both crossing-state probabilities are at most 2ε at every time: the recursion is $r_A(k) = \varepsilon r_W(k-1) + r_A(k-2)/2$. A first-separation coupling to the original chain consequently gives

$$P^n(W, W) \ge 1 - 4\varepsilon - 2n\varepsilon\eta.$$

At $n = \lfloor 1/(16\varepsilon\eta) \rfloor$, this is at least $3/4$, whereas $\Pi_W \le 1/3$. Keeping only state W bounds the two displayed residuals below by $5\varepsilon/12$ and $5\varepsilon/16$, respectively. For any fixed $\alpha > 0$, choose fixed $k \ge 1$ with $\alpha(k+1) > 1$ and let $\eta = \varepsilon^k$. Then $n \asymp \varepsilon^{-k-1}$ and $n\sqrt{\beta_*} \asymp \varepsilon^{-1}$, so the proposed right side is $o(\varepsilon)$. □

Long initial residence can itself supply many observations of the likely terminal row. Both residuals discard this information. Their failure therefore does not separate stationary and arbitrary-start prediction risks; it explains why the following proof must preserve counts.

H.2 A same-target reciprocal identity

Lemma H.2 (Schur return identity)**.** *For any homogeneous chain on d states, any initial/target state s and $n \geq 1$, let $J_s(n) = \sum_{j<n} \mathbf{1}\{Y_j = s\}$. Then*

$$\mathbb{E}_s \frac{\mathbf{1}\{Y_n = s\}}{J_s(n)} = \frac{\operatorname{tr}(P^n) - \operatorname{tr}(P^n_{-s})}{n} \leq \frac{d}{n}. \tag{44}$$

The fraction is evaluated only on $Y_n = s$, where its denominator is positive. No irreducibility, stationarity or aperiodicity is assumed.

Proof. Let $F_s(z)$ be the first-positive-return probability generating function, allowing a defective law. First-step decomposition and Schur complementation give the formal identities

$$F_s(z) = zP_{ss} + z^2 P_{s,-s}(I - zP_{-s})^{-1} P_{-s,s}, \qquad 1 - F_s(z) = \frac{\det(I - zP)}{\det(I - zP_{-s})}.$$

Both determinants have constant term one. Expand their formal logarithms:

$$[z^n]\{-\ln(1 - F_s(z))\} = \frac{\operatorname{tr}(P^n) - \operatorname{tr}(P^n_{-s})}{n}.$$

On the other hand, $-\ln(1-F_s) = \sum_{k\geq 1} F_s^k/k$. Strong Markov makes its coefficient $\sum_k \mathbb{P}_s(\tau_k = n)/k$. At the kth positive return there are exactly k earlier visits, including time zero. This is the left side of (44). Nonnegativity follows from the return expansion, and the upper bound follows from $\operatorname{tr}(P^n) \leq d$ and nonnegativity of the killed trace. □

This identity controls a return to the initial target. For a different terminal target it is used after a first-hit decomposition, rather than as a bound on an independently chosen row.

H.3 Absorbing diagonals and alternating geometric runs

Suppose W is absorbing, so $a = 0$; no restrictions are placed on the other rows or initial distribution. A true path ending in A, B, C never visited W. Its complete-path probability is at most that under the reflected chain

$$R = \begin{pmatrix} 0 & 1-b & b \\ 1 & 0 & 0 \\ 0 & t_1 & 1-t_1 \end{pmatrix}$$

on A, B, C, since each departure from B contributes an additional factor $1 - e \leq 1$ in the true law. This domination preserves the terminal state and its entire prior count. It is not a conditional IID statement.

For $0 < b, t_1 < 1$, an A-run has G_b visits to A and duration $2G_b - 1$, while a C-run has G_{t_1} visits and duration $G_{t_1} + 1$, where the geometric variables are supported on positive integers. Their generating functions and the cycle function are

$$D_A(z) = \frac{bz}{1 - (1-b)z^2}, \quad D_C(z) = \frac{t_1 z^2}{1 - (1-t_1)z}, \quad F(z) = D_A(z) D_C(z).$$

Put $L(z) = -\ln(1 - F(z)) = \sum_{n\geq 1} L_n z^n$. Then

$$0 \leq L_n \leq \min\{b, t_1, 3/n\}. \tag{45}$$

Indeed $L_n = \sum_{k\ge 1} \mathbb{P}(S_k = n)/k$. Dropping $1/k$ gives a cycle-completion probability; completing at C requires a final $A \to C$ transition, giving the bound b. Reversing the order of the independent run durations gives the same cycle law and the bound t_1. Finally

$$\det(I - zR) = [1 - (1-b)z^2][1 - (1-t_1)z][1 - F(z)]$$

and the trace/log identity give $nL_n = \operatorname{tr}(R^n) - (1-t_1)^n - 2\mathbf{1}_{\{n \text{ even}\}}(1-b)^{n/2} \le 3$.

Here is the count-sensitive reduction behind the use of these coefficients. For

$$g_q(r,N) = \binom{N-1}{r-1} q^r (1-q)^{N-r}, \quad N \ge r \ge 2,$$

the exact identity

$$\frac{g_q(r,N)}{q(N-1)} = \frac{g_q(r-1,N-1)}{r-1}$$

removes one completed run and one visit. To a path ending at A, virtually append $A \to C$, multiplying its probability by b; its completed A-visit count is $J_A + 1$. To a path ending at C, append $C \to B \to A$, multiplying by t_1, with completed count $J_C + 1$. Applying the identity with the larger denominator $1 + q(N-1)$ reduces the weighted terminal sums to coefficients of L, $D_A L$, $D_C L$ and one-cycle exceptions. The removed durations are exactly one for an A-run and two for a C-run. No previous visit is reset.

For completeness, splitting the remaining duration at half the horizon gives

$$[z^n]F \le 8/n, \quad [z^M]D_C L \le 15/M, \quad [z^M]D_A L \le 41/M.$$

On the short-duration part use $L_{M-\ell} \le 6/M$. On the long-duration part use respectively $L_{M-\ell} \le t_1, b$, and geometric density bounds $t_1 e^{-t_1 M/4}$ and $b e^{-bM/8}$. The resulting maxima $M t_1^2 e^{-t_1 M/4} < 9/M$ and $M b^2 e^{-bM/8} < 35/M$ give the constants. The run reductions thus imply, for either target $s = A, C$ and every initial active state,

$$\mathbb{E}_R\left[\mathbf{1}\{Y_M = s\}\frac{q_s}{1 + q_s J_s}\right] \le 64/M, \qquad q_A = b, \quad q_C = t_1, \quad M \ge 8.$$

An initial B shifts the horizon by one; use that shift only for $M \ge 9$, and the unit integrand bound at $M = 8$. Finite-horizon continuity extends the estimates to $b, t_1 \in [0,1]$. Since

$$\min\{q, A_0/(j+2)\} \le 2\max\{A_0, 1\}\frac{q}{1+qj},$$

the total A, C contribution is at most $256\max\{A_0, 1\}/M$.

The true B contribution uses its exit weight e. An exit from a terminal B into absorbing W is the first visit to W, so

$$e\mathbb{P}(Y_M = B) \le \mathbb{P}(J_{Y_{M+1}}(M+1) = 0) \le 16/(M+2).$$

The absorbing row has weight zero. Complete-path domination therefore proves

$$\mathbb{E}\min\{q_{Y_M}, A_0/(J+2)\} \le 256\max\{A_0, 1\}/M + 16/(M+2).$$

Combined with the historywise floor reduction below, this gives the uniform all-start bound

$$\operatorname{risk}(T_\kappa) \le \min\{1, 2^{17} M^{-1/2} + 3/Q + 1600/(\kappa Q)\}. \tag{46}$$

The earlier reflected-core argument obtains the same no-log order when b, t_1 stay in a fixed compact subinterval. The run argument is what removes that extra assumption. Neither argument changes the finite predictor.

H.4 Full support: diagonal starts and first-hit convolutions

Assume now $\beta > 0$ and full stationary support. All four exits are positive. Write the stationary masses as (y, x, x, w); then $ya = xe$, $wt_1 = xb$ and $x = at_1/(et_1 + 2at_1 + ab)$. The full all-state certificate supplies

$$ae \geq \beta x, \qquad bt_1 \geq \beta x, \qquad at_1 \geq \beta x.$$

The third inequality uses $P^2(W, C) = ab \geq \beta w$ and positive b.

The diagonal-to-successor-crossing estimate is

$$\mathbb{E}_W[\mathbf{1}\{Y_M = A\} \min\{1, A_0/(J_A + 2)\}] \leq \frac{40A_0}{M}\left[1 + \ln\left(1 + \frac{1}{d_0 A_0}\right)\right], \quad d_0 = \max\{\sqrt{\beta}, M^{-1}\}. \tag{47}$$

Its key algebra is the certificate consequence

$$\min\{1/e, (1 + Mt_1)/b\} \leq 12(1 + Ma)/\sqrt{\beta}, \qquad \beta M^2 \geq 1.$$

To see why it is uniform, set $\alpha = Ma, E = Me, B = Mb, T = Mt_1$ and $D = \alpha B + 2\alpha T + ET$. The three certificate inequalities become $\beta M^2 \leq D, ED/T, BD/\alpha$. For $H = \max\{E, B/(1 + T)\}$ and $X = (1 + \alpha)H$, splitting at $T = 1$ and $\alpha = 1$ gives $\beta M^2 \leq 6X^2 + 6X$; hence $X \geq M\sqrt{\beta}/12$.

Before an excursion from A hits W, its expected truncated number of A visits is at most $\min\{1/e, (1 + Mt_1)/b\}$. The first bound counts the geometric tests at B; the second counts outgoing $A \to C$ entries and uses that they are at most the number of C exits plus one. Include the outgoing transition after a possible visit at the observation endpoint. In the full chain there are at most Ma further $W \to A$ entrances in expectation. Thus the global visit count V_A obeys

$$\mathbb{E}_A V_A \leq 12(1 + Ma)^2/\sqrt{\beta}.$$

For every path, summing the terminal functional over all horizons counts successive visits once:

$$\sum_{n=0}^{M} \mathbf{1}\{Y_n = A\} \min\{1, A_0/(J_A(n) + 2)\} \leq A_0[1 + \ln(1 + V_A/A_0)].$$

Jensen applies to this *global* count, not to a sum of reset excursion counts.

From initial W, the first hit of A is geometric(a). Split its convolution with the same-target functional at $M/2$. Early hits cost at most $8A_0/M$ by Lemma H.2. Late hits have point mass at most $ae^{-aM/4}$. Multiplying the preceding cumulative-count bound and using $xe^{-x/4} \leq 2$, $x^2e^{-x/4} \leq 9$ proves (47). If $\beta M^2 < 1$, use the universal occupation bound directly; it gives the same cutoff and avoids every zero divisor.

For the other crossing target, every initial-W, terminal-B path has a last A visit, at time k, and $J_B(M) = J_A(k)$. The intervening phase has distribution

$$U_C(1) = 1 - b, \qquad U_C(\ell) = bt_1(1 - t_1)^{\ell-2}, \quad \ell \geq 2.$$

Consequently its terminal count functional is exactly the convolution of (47)'s functional at time k with $U_C(M - k)$. For $k \geq M/2$ use (47); for $k < M/2$ use the long-hold density $bt_1e^{-t_1M/4}$ and the global bound $b\mathbb{E}_W V_A \leq 1 + Mt_1$. The term $b\ln(1/b)$ stays multiplied by b and is bounded by $1/\exp(1)$. This proves the opposite-crossing estimate with constant 100 in place of 40. Horizons $8 \leq M < 16$ use the direct cap, and the weak-β branch is separated before dividing by b. Symbol exchange supplies both crossing targets from initial C.

H.5 Opposite diagonals: a density envelope and a hitting potential

The remaining component is initial W, terminal C, with true exit weight t_1. Let T_C be the first hit of C, $\rho = b + e - be$, and

$$m = \mathbb{E}_W T_C = \frac{\rho + a(2-b)}{ab}, \qquad \lambda = 1/m.$$

The first-step equations give a largest supported hitting mean H with $m \le H \le m+1 \le 2m$. Blockwise Markov and strong Markov therefore yield $\mathbb{P}_i(T_C > n) \le 2e^{-\lambda n/16}$. This survival estimate alone is insufficient; the required first-hit *density* has the stronger envelope

$$d_n = \mathbb{P}_W(T_C = n) \le 12\lambda e^{-\lambda n/16}. \tag{48}$$

Indeed the killed probabilities satisfy $A_k = aW_{k-1} + (1-e)(1-b)A_{k-2}$, giving

$$d_n = ab \sum_{j=0}^{\lfloor (n-2)/2 \rfloor} (1-\rho)^j W_{n-2-2j}.$$

Since $\lambda \le b \le \rho$, the survival envelope bounds this by $4(ab/\rho)e^{-\lambda n/16}$; independently it is at most $3be^{-\lambda n/16}$. The identity $\min\{b, ab/\rho\}/\lambda = \min\{1, a/\rho\}(\rho/a + 2 - b) \le 3$ proves (48).

Let $p = \Pi_C$ and $h_i = \mathbb{E}_i T_C$, $h_C = 0$. The hitting potential satisfies

$$Ph - h = -1 + \mathbf{1}_{\{C\}}/p.$$

At C this is Kac's positive-return identity; elsewhere it is the first-step equation. Telescoping from initial C yields, for $V_C = \sum_{n=0}^{M} \mathbf{1}\{Y_n = C\}$,

$$\mathbb{E}_C V_C = p[M + 1 + \mathbb{E}_C h(Y_{M+1})] \le p(M + 1 + H).$$

The weighted hitting bound from Lemma G.2's proof gives $\sqrt{\beta}\, p t_1 H \le 5$, hence

$$\mathbb{E}_C(t_1 V_C) \le \frac{10}{\sqrt{\beta}}(1 + M\lambda).$$

The weight cannot be dropped.

Put $f(n) = \mathbb{E}_C[\mathbf{1}\{Y_n = C\} \min\{t_1, A_0/(J_C(n)+2)\}]$. Lemma H.2 gives $f(n) \le 4A_0/n$ for $n \ge 1$, while chronological visit summation and Jensen give

$$\sum_{n=0}^{M} f(n) \le A_0 \left[1 + \ln\left(1 + \frac{10(1+M\lambda)}{\sqrt{\beta} A_0}\right)\right].$$

There are no target visits before T_C, so the desired quantity equals $\sum_{r=1}^{M} d_r f(M-r)$, including $f(0)$ when the first hit is terminal. Early hits cost $8A_0/M$; late hits use (48) and the cumulative bound. With $x = M\lambda$, the maxima $xe^{-x/32} \le 12$ and $x^2 e^{-x/32} \le 555$ give

$$\mathbb{E}_W[\mathbf{1}\{Y_M = C\} \min\{t_1, A_0/(J_C+2)\}] \le \frac{8192 A_0}{M}\left[1 + \ln\left(1 + \frac{1}{\sqrt{\beta} A_0}\right)\right]. \tag{49}$$

This opposite-diagonal estimate needs no condition $\beta M^2 \ge 1$. That fact matters when applying it to a shorter suffix.

H.6 All initial states, all supports, and the unchanged decoder

For $\beta > 0$, stationarity gives $\Pi = (y, x, x, w)$. If $x > 0$, the all-state entries $ea \geq \beta x$ and $bt_1 \geq \beta x$ force all four exits positive, and balance gives $y, w > 0$. If $x = 0$, supported diagonals are absorbing; two supported absorbing diagonals contradict positive minorization. Thus the stationary support is either full or a single absorbing diagonal. The latter is already covered by (46), including initial states outside that support. At $\beta = 0$ no support classification is required.

For full support put

$$V_M = \sum_s \mathbf{1}\{Y_M = s\} \min\{q_s, A_0/(J_s(M) + 2)\}, \qquad G = 1 + \ln(1 + 1/(\sqrt{\beta} A_0)).$$

From either diagonal start, the same-target, successor-crossing, other-crossing and opposite-diagonal bounds have constants $4, 40, 100, 8192$. Hence, at every horizon $m \geq 8$,

$$\mathbb{E}_{\text{diag}} V_m \leq 8336 A_0 G/m.$$

The crossing bounds use $\max\{\sqrt{\beta}, 1/m\}$, whose larger value only decreases the logarithm; (49) has no strong-certificate horizon restriction. This proves the bound even for the shorter suffixes below.

For a crossing start, let $h_c = b + e - be$. Before the first diagonal hit the chain alternates between A, B. If $h_c \leq M^{-1/2}$, the crossing endpoint weights contribute at most $b + e$. Last-entry geometric sums give $a\mathbb{P}(Y_M = W) \leq e$ and $t_1\mathbb{P}(Y_M = C) \leq b$; a zero exit contributes zero before any division. Since $b + e \leq 2h_c$, $\mathbb{E}V_M \leq 4M^{-1/2}$. If $h_c > M^{-1/2}$, the probability of no diagonal hit by $\lfloor M/2 \rfloor$ is at most $e^{-\sqrt{M}/8} \leq 3M^{-1/2}$. On its complement, strong Markov gives an ordinary suffix starting at a diagonal, of length at least $\lceil M/2 \rceil$. The full terminal count dominates its suffix count. Consequently, for all initial laws,

$$\mathbb{E}V_M \leq 4M^{-1/2} + \frac{16672 A_0}{M} G, \qquad \beta M^2 \geq 1. \tag{50}$$

This is a path/count argument, not the false weighted endpoint transfer of Proposition H.1.

The binary floor identity in Proposition G.3 is historywise. For any chosen true exit probability q_s, including $q_A = b, q_B = e$, its positive-block bound is

$$|v - \widehat{p}| \leq \min\{\widehat{p}, 1 - \widehat{p}, 2/(Q\rho)\} \leq \min\{q_s, 2/(Q\rho)\} + |\widehat{p} - q_s|.$$

The arbitrary-start add-one comparator and universal occupation bound therefore give exactly

$$R(T_\kappa) \leq 2S_0 + 2B_0 + \mathbb{E}V_M + 64/z + 16/M + 3/Q, \qquad A_0 = 6M/z. \tag{51}$$

The separate empty-floor charge includes $J = 0$ and does not change the uniform fallback. No stationary return estimate is being imported under a nonstationary law.

For full support and $\beta M^2 \geq 1$, substitution in (50) gives numerical coefficient $6 \cdot 16672 + 64 = 100096 < 2^{17}$ on $\mathcal{W}(z, Me_0)$, where $e_0 = \max\{\sqrt{\beta}, M^{-1}\}$. The statistical coefficient $(2^{14} + 4)\sqrt{32} + 6$ is also below 2^{17}. Singleton support uses (46). For $0 \leq \beta < M^{-2}$, use the unchanged beta-free triple-bank upper, including $\Psi(96/z) \leq 96[1 + \ln(1 + z)]/z$; here $Me_0 = 1$. Thus every support, initialization and cutoff boundary is accounted for.

Theorem H.3 (Binary one-block frontier for both initializations)**.** *In the family of Theorem G.4, with the same legal budgets,*

$$c_2\mathcal{F}_2 \leq \mathcal{R}^{(2)}_{\kappa,\text{stat}} \leq \mathcal{R}^{(2)}_{\kappa,\text{all}} \leq \text{risk}_{\text{all}}(T_\kappa) \leq \min\left\{1, \frac{2^{17}}{\sqrt{M}} + \frac{3}{Q} + 2^{17}\mathcal{W}(\kappa Q, Me_0)\right\}. \tag{52}$$

In particular both minimax risks are of order $\mathcal{F}_2$, with common upper constant $2^{18} + 3 = 262147$.

Proof. The preceding support and decoder argument proves the upper. Theorem G.4's stationary lower applies to the same encoder family, and stationary laws are an admissible subset of arbitrary initializations. No new lower witness or all-regime extrapolation is needed. □

Both simultaneous budgets in (43) now characterize statistical-order risk for either initialization. In a fixed square-root grid band, every feasible real weight envelope is at least $c_2 f_{Q,2}/(C_0 c_+)$, while the actual integer choice $\lceil f_{Q,2} \rceil$ has arbitrary-start risk at most

$$\frac{2^{17} + (3 + 2^{18})/c_-}{\sqrt{M}}.$$

Thus the least feasible integer weight ratio is $\Theta(f_{Q,2})$ for sufficiently large fixed target multipliers in both experiments. Corollary G.5's constant-versus-logarithmic distinction at $\beta = 1/M$ consequently holds also for arbitrary initialization. The previously established stationary constant 544 remains sharper than the common envelope; it is not invalidated by the extension.

I Binary fixed-depth causal finite evaluators

The one-block real-table family and the causal finite-evaluator family are different. The binary frontier transfers to the latter only after proving a nonlinear Markov concentration bound and an exact finite-label upper inclusion.

Use the temporal-causality and bounded-primitive conditions of Section 2, now with $K = 2$, fixed radius $r \geq 3$ and output budget $D \geq 8$. The capacity must admit one global head, width 16, supplied exact radius-three triple-bit tables, and affine coefficient and row-sum budgets at least two. The main prototype uses row-sum-one projections. Computing the triple bits instead as $\mathrm{ReLU}(u + v + w - 2)$ is an optional different local implementation requiring row-sum budget three. Public masks must allow density $c \leq 1/3$. Historical tables and states cannot use the future boundary label; that label enters only the appended query and decoder. Evaluation remains the fixed-architecture signed finite-evaluation procedure at target $h + 3$.

Theorem I.1 (Binary bounded causal frontier)**.** *Put*

$$H_2(M, \beta, Q) = \max\{M^{-1/2}, \mathcal{W}(Q, M \max\{\sqrt{\beta}, M^{-1}\})\}.$$

For this specified fixed-depth causal finite-evaluator family,

$$c_{\mathrm{arch},2} H_2 \leq \mathcal{R}^{(2)}_{\mathrm{causal,stat}} \leq \mathcal{R}^{(2)}_{\mathrm{causal,all}} \leq \min\{1, 262147\, H_2\}.$$

The proof combines a stationary lower, an exact finite prototype and the arbitrary-start upper. Constants depend on the fixed architecture, not on M, β, Q.

Proof. For the lower, retain the binary source and active reference chain of Proposition G.1. Section B bounds the ideal terminal output's change under any late single-symbol replacement by Γ/M, already including local preprocessing. The reference chain's two-step coupling has expected disagreement count at most 16. Telescoping single-symbol replacements along two coupled continuations therefore bounds the difference of their conditional output expectations by $34\Gamma/M$. For a j-step active suffix and each frozen branch, the Doob and single-transition derivative arguments give

$$\mathrm{Var}_\nu f \leq (34\Gamma)^2 j/M^2, \qquad |(\mathbb{E}_\nu f)'| \leq 34\Gamma j/M.$$

This concerns the expectation of the nonlinear ideal output, not a ratio of expectations or a derivative of finite rounding.

Write $H_b = 2^b$ for the finite boundary alphabet and $\tau_B = 34\Gamma\sqrt{32DH_b}$. On a simultaneous event of probability at least $31/32$, deviations are at most $\tau_B\sqrt{j}/M$. With $J_* = \lfloor \min\{(4e_0)^{-1}, M^2/(4Q^2)\} \rfloor$, the scaled tube has bounded cross-section. A mesh gives at most

$$A_B(1 + Qj/M), \qquad A_B = H_b(2 + 34\Gamma)\big(\lceil 2\tau_B + 6 \rceil + 1\big)^D$$

complete good labels, including the certified finite error at most $1/(8Q)$ and final grid floor. The informative-terminal-A Bayes calculation and pathwise domination from Section G then yield the same truncated logarithmic lower, with constant $(524288A_B)^{-1}$ and $z = Q$. The fixed-architecture IID curve argument and full-history testing supply separate $1/Q$ and $M^{-1/2}$ floors. The binary all-regime completion proves $c_{\mathrm{arch},2}H_2$. No growing-weight uniform IID strengthening is needed for this fixed architecture.

For upper membership, supply the eight exact local triple bits at each historical row. Set the synthetic initial state to zero. A single head has zero query and key projections, values those bits, and masks

$$\mathcal{K}_1 = \{1\}, \quad \mathcal{K}_2 = \{1, 2\}, \quad \mathcal{K}_t = \{3, \dots, t\} \ (3 \le t \le M + 2), \quad \mathcal{K}_{M+3} = \{3, \dots, M + 2\}.$$

These masks are public, causal, boundary-independent and have density at least $1/3$. Width 16 holds the old eight feature slots and eight output slots. Projections and identity padding are exact; historical states do not use the terminal pair.

Crucially, the unchanged shifted Taylor evaluator assigns the *same positive integer* W_* to every zero-logit edge. It need not equal one. For fine denominator $D_f = 2^P$ and count C_{abc}, the exact accumulated denominator and numerator are

$$Z = MW_*, \qquad N_{abc} = W_* D_f C_{abc}, \qquad O_{abc} = \lfloor D_f C_{abc}/M \rfloor .$$

Since D_f/Q is an integer,

$$\lfloor O_{abc}/(D_f/Q) \rfloor = \lfloor QC_{abc}/M \rfloor .$$

Thus the final bank is exactly the uniform histogram bank at $\kappa = 1$ on every history, not merely close to it. Zero counts remain zero, which is necessary for the binary purity argument. The boundary is used only for post-grid block selection. Theorem H.3's upper now applies at $z = Q$ for every initialization, although the encoder implementations are different. Since $\mathcal{W}(Q, Me_0) \ge 1/Q$, the claimed envelope follows. □

The finite-cost composition also remains distinct. Charge the complete signed finite encoder, local-table and mask generation, all-token arrays, every Taylor/edge scan, compact converted records and full arithmetic words from Section P. Add bank conversion and all-token pointwise decoding. The complete addressed triple-bank ledger at $\kappa = 1$ is a conservative overcharge for its decoder suffix; using that bound does *not* execute singleton heads inside the dense causal encoder. Add simultaneous phase costs, rather than taking minima across implementations. Optional PURE unrolling applies to the composed bitwork and has much larger depth.

At fixed depth the necessary and sufficient binary grid order is $Z_{*,2}$ in (43), for both initializations. In particular $\beta = 1/M$ permits $Q \asymp \sqrt{M}$, whereas $\beta \le M^{-2}$ requires $Q \asymp \sqrt{M}\log M$. The larger-alphabet bounded causal theorem retains the β_0 cutoff. These distinctions do not depend on an online estimate of the mixing promise.

J Actual bounded-primitive depth as a third resource

Increasing a pre-cut representation's sensitivity need not require growing attention weights. Repeated bounded affine/ReLU transformations can redistribute grid cells toward small histogram coordinates. The resulting depth question is different from the one-block weight/grid frontier: it concerns actual sequential primitive stages before the same information cut. The construction below uses only zero attention logits, so its attention-weight ratio is one.

Scalar companding—a monotone transform, uniform quantization and inverse reconstruction—is established. Adler, Tang and Polyanskiy analyze probability-vector companders under KL loss, including power transforms and the square-root case [1, Sections I–III, Theorem 5]. The obligations here are instead actual-terminal Markov TV risk, a bounded-primitive dyadic realization, directional finite arithmetic and a matching depth-order lower. Likewise, depth versus *coefficient* precision is an established ReLU-network tradeoff [7, Proposition 4.1]; our final-summary grid is a different resource, and internal coefficient precision and memory remain paid.

J.1 Uniform family and theorem

Fix the alphabet, width, head count, output dimension D, local radius r, number b_0 of boundary-symbol slots, fan-ins, coefficient/operator bounds, state ranges, mask density and number of primitive affine/ReLU operations per stage. Only the number $d \geq 1$ of stages may grow. Attention-free positionwise stages are allowed; a syntax requiring a head at every stage may instead execute paid zero-output heads. A stage may not contain an uncharged growing circuit.

The initial synthetic terminal state is a function *only* of public data and the frozen finite boundary field S. Every source symbol used by that initial query, including any terminal local window, must be included in the b_0 slots before branch counting. Historical local tables occur strictly before the synthetic terminal index $T = M + 3$. Every frozen branch obeys the primitive and ideal-range contracts on every formal history, even if its boundary symbols are inconsistent with that history. The cut and decoder access restrictions are unchanged.

For lower bounds, a finite evaluator must have a certified uniform pre-cut error at most E/Q, with E independent of d, M, Q. We first prove such a certificate for the same signed Taylor/affine evaluator used earlier. Arbitrary discontinuous rounded programs without that certificate are not included.

Theorem J.1 (Depth at the statistical grid)**.** *Within a uniform bounded-primitive family having sufficient fixed capacity for the complete K^3 histogram and the coordinatewise construction below:*

1. *For every fixed $K \geq 2$, every $\beta \in [0, 1]$ and every initialization, a fully paid finite encoder with zero attention logits, fixed width and $d = O(\log \log M)$ attains risk $O_K(M^{-1/2})$ at a dyadic $Q \asymp \sqrt{M}$.*
2. *For fixed $K \geq 3$, $\beta = 0$ and a fixed band $c_- \sqrt{M} \leq Q \leq c_+ \sqrt{M}$ with $0 < c_- \leq c_+ < \infty$, attaining any fixed risk bound $C_{\mathrm{risk}}/\sqrt{M}$ requires $d = \Omega(\log \log M)$, even under stationary initialization. This holds for the uniformly certified growing-depth finite evaluator as well as the ideal family.*

Thus the least sufficient primitive depth is $\Theta(\log \log M)$ at this operating point and sufficiently large fixed target-risk multiplier. This is not an all-Q three-resource frontier, a necessary depth claim for every positive β, or a statement about the depth of a Boolean/PURE expansion.

J.2 Dyadic compander geometry

Use $L \geq 0$ for the number of compander iterations and write $\chi = 2^L$ for its bottom-segment gain. This symbol is deliberately distinct from the attention-weight ratio κ. On $[0,1]$ define

$$T(x) = \min\{1, 4x\} = 4x - \mathrm{ReLU}(4x-1), \qquad g_0(x) = x, \quad g_{L+1}(x) = \tfrac{1}{2} g_L(T(x)) + \mathrm{ReLU}(x - \tfrac{1}{4}). \tag{53}$$

A sequential realization sets $u_0 = x, v_0 = 0$ and, for $0 \leq j < L$, performs

$$u_{j+1} = T(u_j), \qquad v_{j+1} = v_j + 2^{-j}\,\mathrm{ReLU}(u_j - \tfrac{1}{4}), \qquad g_L(x) = v_L + 2^{-L} u_L. \tag{54}$$

The old u_j is retained until both updates finish. Every iteration has a fixed number of bounded-fan-in affine/ReLU primitives. Coefficients have magnitude at most four; $u \in [0,1]$, $v, g_L \in [0, 3/2]$, and affine temporaries have a uniform bound, for example four. The dyadics 2^{-j} have actual description length $O(j+1)$, which is charged.

Induction gives a strictly increasing concave piecewise-linear function with slope 2^j on $[4^{-j-1}, 4^{-j}]$, $0 \leq j < L$, and slope χ on $[0, 4^{-L}]$. Its inverse slopes obey $2^{-j} \leq 2\sqrt{x}$ outside the bottom segment and equal $1/\chi$ on that segment. Integrating, including across breakpoints, yields

$$x - y \leq (2\sqrt{x} + \chi^{-1})\,[g_L(x) - g_L(y)], \qquad 0 \leq y \leq x \leq 1. \tag{55}$$

This inequality is a proof device; the program never evaluates a square root.

For the generic dense-causal comparison witness, acquire radius-three one-hot triple features by paid finite local tables. The predecessor-copy construction is an alternative acquisition schedule in the specialized ledger; its hard singleton heads are not additional dense-family layers. A zero-logit head pools the complete K^3 histogram at every row over that row's own causal prefix, using the density-1/3 masks from Section I. No terminal counts are copied backward into historical states. Apply $g_L/2$ to every coordinate, then make the downward Q-grid cut. The decoder receives only this bank, the exact terminal pair and public metadata. It inverts g_L on twice the represented coordinates, selects the terminal pair's block, normalizes if nonzero, returns uniform otherwise, and performs ordered simplex rounding.

J.3 Downward arithmetic and zero purity

A particularly transparent implementation uses

$$P = h + L + 4, \qquad D_f = 2^P = 16\chi Q.$$

For an exact histogram count A and row length m, put $X = \lfloor D_f A/m \rfloor$, so $\widehat{x} = X/D_f \leq x = A/m$. Identical positive finite softmax weights cancel exactly, just as in Section I. In integers, initialize $U_0 = X, V_0 = 0$ and compute

$$R_j = \max\{0, U_j - D_f/4\}, \qquad V_{j+1} = V_j + \left\lfloor R_j / 2^j \right\rfloor, \qquad U_{j+1} = \min\{D_f, 4U_j\}, \tag{56}$$
$$W = V_L + \left\lfloor U_L / 2^L \right\rfloor.$$

All T updates are exact on the fine lattice. Each positive dyadic-term floor loses less than $1/D_f$. Since g_L is χ-Lipschitz,

$$0 \leq g_L(x) - W/D_f \leq \frac{\chi + L + 1}{D_f} \leq \frac{3\chi}{D_f}. \tag{57}$$

For $x = 0$, every rail and output is exactly zero: there is no positive-offset leakage. Emit $Z = \lfloor QW/(2D_f) \rfloor$. Then

$$0 \le g_L(x) - 2Z/Q < 3/Q.$$

Consequently $y = g_L^{-1}(2Z/Q)$ is defined and lies below x, with

$$0 \le x - y \le \frac{6\sqrt{x} + 3/\chi}{Q}. \tag{58}$$

The one-sided inequality, not merely a symmetric error estimate, is what makes the zero-bank and small-mass analysis simple.

J.4 A uniform extension of the finite evaluator

The preceding minimal-P schedule is useful, but inclusion in the comparison family must also hold for the same generic evaluator used by the lower. One cannot substitute a growing d into the old fixed constant C_{arch}.

Fixed *per-stage* bounds in the affine/ReLU/Taylor argument give computable rational constants $\alpha, \zeta \ge 0$, independent of d, M, h, such that the numerical discrepancy satisfies

$$\delta_{i+1} \le \alpha\delta_i + \zeta(P+1)2^{-P} \quad \text{whenever } \delta_i \le 1, \qquad \delta_0 \le 2^{-P}.$$

The score neighborhood, positive shift and Taylor index offset are uniform because the primitive bounds are uniform. Enlarge these constants to cover the fixed readout as one additional stage. Choose integer $A_* \ge \max\{2, \alpha + \zeta + 1\}$ and then one integer $c_* \ge 4$ satisfying $A_*(c_* + 5)2^{1-c_*} \le 1$; finite search finds it. With

$$t = d + 2, \qquad P = 4h + c_* t, \tag{59}$$

the scalar recurrence gives an error bound $A_*^t(P+1)2^{-P}$. Indeed,

$$A_*^t(P+1)2^{-P+h} \le A_*^t(c_* t + 5)2^{-c_* t} \le (c_* + 5)\,[A_* 2^{1-c_*}]^t \le 1.$$

Here $(4h + c_* t + 1)2^{-3h} \le c_* t + 5$ and $c_* t + 5 \le (c_* + 5)2^t$. Applying the same estimate to each earlier stage gives $\delta_i \le 2^{-h} \le 1$, closing rather than assuming the neighborhood bootstrap. The pre-cut error is therefore at most $1/Q$, uniformly in depth and every formal history.

For the compander choose the maximum of (59) and $h + L + 4$; increasing P preserves the estimate. Exact local one-hot inputs and zero logits again yield the exact downward fine histogram. On positive rails, T remains exact; only the designated dyadic products round downward. A separately floored final division by two adds at most $1/D_f$ in $g_L/2$ space, so the generic evaluator obeys

$$0 \le g_L(x) - 2F_{\mathrm{fin}}(x) \le \frac{\chi + L + 3}{D_f} \le \frac{5\chi}{D_f}.$$

After the final Q floor, the total loss is less than $(2 + 5/16)/Q < 3/Q$. Hence (58) and exact zero purity hold for this *same* uniformly certified evaluator. Unused zero-value heads produce zero exactly. This is a sufficient $P = 4h + O(d + 1)$ certificate, not a minimal working-precision theorem.

J.5 Normalization and arbitrary-start occupation

Let the selected true histogram coordinates be $x_c = C_c/M$, whose mass is $r = J/M$. For $J \geq 1$, the removed mass $e = \sum_c (x_c - y_c)$ satisfies

$$e \leq 6\sqrt{Kr}/Q + 3K/(\chi Q).$$

If $e < r$, the two-term normalization triangle inequality gives $\mathrm{TV}(y/\sum y, x/r) \leq e/r$. If $e = r$, the decoded bank is zero and any simplex fallback has distance at most $1 = e/r$. Thus, including this branch,

$$\mathrm{TV}(\widehat{p}_{\mathrm{raw}}, C/J) \leq \frac{6\sqrt{K(K+1)M}}{Q\sqrt{J+K}} + \min\left\{1, \frac{3K(K+1)M}{\chi Q(J+K)}\right\}. \tag{60}$$

Adding the add-one difference $K/(J+K)$ and final rounding $(K-1)/Q$ also covers $J = 0$.

The universal occupation bound $\mathbb{P}(J < w) \leq \min\{1, K^4 w/(M+1)\}$ for integer $w \geq 1$ already holds for every initialization. Besides the reciprocal bound expressed by Ψ, it gives

$$\mathbb{E}(J+K)^{-1/2} \leq \frac{2K^2}{\sqrt{M+1}}. \tag{61}$$

To check the fractional form, for $u > K$ the event $J + K < u$ equals $J < \lceil u - K \rceil$, whose positive integer threshold is at most u. For $u \leq K$ it is empty. With $a_0 = K^4/(M+1)$, layer cake therefore gives

$$\mathbb{E}(J+K)^{-1/2} \leq \int_0^\infty \min\{1, a_0/v^2\}\, dv = 2\sqrt{a_0}.$$

Using (60), (61) and the inherited ORIGINAL add-one risk yields

$$\begin{aligned}\mathcal{R} \leq \min\Bigg\{1,\, 2^{13} H_{\mathrm{stat}} + \Psi\left(\frac{K^5}{M+1}\right) + \frac{12K^2\sqrt{K(K+1)} + K - 1}{Q} \\ + \Psi\left(\frac{3K^5(K+1)}{\chi Q}\right)\Bigg\}, \qquad H_{\mathrm{stat}} \leq \frac{K^{5/2}}{\sqrt{M+1}}.\end{aligned} \tag{62}$$

Here $\Psi(x) = x(1 + \ln(1/x))$ on $0 < x < 1$, $\Psi(0) = 0$ and $\Psi(x) = 1$ for $x \geq 1$, as earlier. This step inherits the same statistical input as the triple bank; it is not an independent new proof of that input.

Set

$$m = \mathrm{bitlength}(M+1), \qquad h = \max\{4, \lceil m/2 \rceil\}, \qquad L = \lceil \log_2 m \rceil = \mathrm{bitlength}(m-1).$$

These choices require only integer bitlengths. Then $\sqrt{M+1} \leq Q \leq 2\sqrt{M+1}$ and $m \leq \chi < 2m$. The elementary Ψ estimates used for the triple bank imply $\Psi(C/(mQ)) \leq 3C/Q$ and $\Psi(K^5/(M+1)) \leq 2K^{5/2}/\sqrt{M+1}$. Equation (62) is therefore $O_K(M^{-1/2})$. The complete primitive depth is $d = O(L+1)$, counting every affine/ReLU composition. Neither β nor a transcendental oracle is used.

J.6 A depth-independent rescaled lower tube

Let the uniform per-stage bounds be those of Section B: at most H heads, values bounded by V, score and value Lipschitz bounds S_*, V_L, row-update Lipschitz bound F_L, local-table diameter Δ and terminal projection bound G. Put

$$C_{\mathrm{att}} = e^{2B}/c_{\mathrm{mask}}, \quad a = F_L(1 + 2HVC_{\mathrm{att}}S_*), \quad b = F_L H C_{\mathrm{att}}(V_L + 2VS_*), \quad w_0 = 1 + \ln 2.$$

Increase a if needed to cover attention-free stages. The nonnegative matrix recurrence from Section B, applied only to late source changes, gives

$$\|\Delta_i F\|_\infty \le \gamma_d/M, \qquad \gamma_d = \max\{1, rG\Delta[(a+b)^d - a^d]w_0^{d-1}\} \le C_*^d \tag{63}$$

for $i \ge T/2$, where the fixed constant $C_* = \max\{2, \max(1, rG\Delta)\max(1,(a+b)w_0)\}$ suffices. Because S is frozen, the changed source coin affects at most r historical initial states, all strictly before T, and does not change the synthetic initial state. Thus the estimate uses off-diagonal entries only; no diagonal a^d terminal-input contribution has been silently discarded.

Write $\gamma = \gamma_d$, $z = \gamma Q$, $H_b = 3^{b_0}$ and $\tau = \sqrt{5DH_b}$. On an IID active suffix of length j in the late half, each frozen branch and coordinate satisfies

$$|(\mathbb{E}_p F)'| \le \gamma j/M, \qquad \mathrm{Var}_p(F) \le \gamma^2 j/(2M^2).$$

A union bound gives simultaneous deviations at most $\tau\gamma\sqrt{j}/M$ with probability at least $9/10$. If $z\sqrt{j}/M \le 1/2$, a parameter mesh with spacing $M/(zj)$ and the certified E/Q arithmetic error give at most

$$A_{\mathrm{box}}(1 + zj/M), \qquad A_{\mathrm{box}} = 2H_b N_{\mathrm{box}}^D, \quad N_{\mathrm{box}} = \lceil 2\tau + 2E + 4\rceil + 1 \tag{64}$$

complete good messages. The actual boundary label is selected after sampling the unconditional product suffix. All constants in (64) are independent of d and γ. In particular the old depth-dependent global square-sum integral is not being absorbed into a constant.

For $0 \le \beta \le 1/16$, use the three-symbol stationary witness of Section A. When $1 \le z \le M/4$, restrict to

$$2 \le j \le J_* = \left\lfloor \min\{(4\beta_0)^{-1}, M^2/(4z^2)\} \right\rfloor.$$

Every such suffix is late, has probability at least $1/(32M)$ independently of the common prior parameter, and satisfies the tube condition. The complete-message Bayes lemma and harmonic integration give

$$\mathcal{R} \ge \frac{c'_*}{z} \ln\left(1 + \frac{1}{8}\min\left\{\frac{z}{M\beta_0}, \frac{M}{z}\right\}\right), \qquad c'_* = \frac{1}{8192 A_{\mathrm{box}}}. \tag{65}$$

For $z > M/4$, the displayed right side is at most $c'_* M/(8z^2) < 2c'_*/M$, which is dominated by the full-history stationary IID lower $1/(128\sqrt{M+2})$. Thus (65) holds for every $z \ge 1$; the large-z case does not posit nonexistent rare windows. For the growing-depth finite evaluator above, take $E = 1$.

Finally set $\beta = 0$ and $c_-\sqrt{M} \le Q \le c_+\sqrt{M}$. If $\gamma \ge M^{1/4}$, then already $\gamma \gtrsim \ln M$. Otherwise,

$$z \ge c_-\sqrt{M}, \qquad M/z \ge M^{1/4}/c_+,$$

so the logarithm in (65) is at least $(\ln M)/8$ for all sufficiently large M. A risk bound $C_{\mathrm{risk}}/\sqrt{M}$ forces $\gamma \ge [c'_*/(8c_+C_{\mathrm{risk}})]\ln M$. Together with (63), this gives

$$d \ge \frac{\ln\ln M + O(1)}{\ln C_*}.$$

The constants depend only on the fixed primitive budgets, band and risk multiplier. This proves the necessary half of Theorem J.1; the preceding construction proves sufficiency. The lower sources are stationary, hence also admissible in the arbitrary-start minimax problem.

J.7 Paid depth and arithmetic ledgers

There are two complete implementation choices. They should not be combined coordinatewise into an artificial smaller budget.

For the generic growing-depth finite evaluator, let U count supplied numerical records across *all* stages and F count all public mask/metadata bits; neither is depth-independent. With B_p the maximum supplied description length and $n = M + 3$, a guarded word can have

$$w = O_{\text{fixed}}\big(P + \log(n+1) + \log(d+1) + \log(P+1) + \log(U+F+2)\big).$$

Acquisition pays $UB_p + F$ bits, conversion pays $O(U(B_p+P)^2)$ work, and converted records occupy $O(UP)$ bits. Fixed arithmetic microcode has a safe $O(w^4)$ charge. Online sequential work is

$$O_{\text{fixed}}\big(dn^2(P+1)w^2 + (U+F)w + w^4\big),$$

including all edges, Taylor steps, primitive updates, local input acquisition and tape rewinds. A conservative phase-sum peak is

$$O_{\text{fixed}}\big(UB_p + UP + F + n + nw + w^4 + B_p + P\big).$$

Only numerical arrays roll in $O(nw)$ space; total parameter and mask storage still grows with the stages. Supplied-data generators add their actual work and scratch. The inverse/normalization decoder and breakpoint generator are additional costs, not already absorbed by this encoder ledger. At the encoder-to-decoder handoff, explicitly convert the complete all-token bank to the decoder format, paying $O(nK^3(w+b))$ bit writes and the simultaneous input/output bank volume $O(nK^3(w+b))$. Add the decoder's program, parameters, working arrays and generation scratch to all other retained data at that phase; an encoder-only peak is not the composed peak.

For the specialized attention-free compander stages, put $\ell = \lceil \log_2(n+K+2) \rceil$ and $b = 128(h + \ell + L + 1)$. Persistent scalars use b bits, full products use $2b$, and guarded temporaries at most $4b+4$. This b bounds the specialized evaluator and its decoder, not the generic FA encoder: the latter retains its full w-bit words, which may be larger when the constant in (59) is large. A conservative fixed-loop budget is

$$\begin{aligned} T_0 &= O\big([n^2K^6 + n(L+1)K^6 + (L+1)^2]b^2\big), \\ S_0 &= O\big(n^2 + nK^3b + (L+1)K^6b + b^2\big). \end{aligned} \tag{66}$$

These charges include dense zero entries, paid local features, all-token histograms and companders, complete output banks, decoder scans, parameter generation, masks, controllers, copies, clears and scratch. Stage parameters cost $O((L+1)K^6b)$ bits. If dummy dense heads are actually run at every stage, replace the first time term by $n^2(L+1)K^6$ and charge any duplicated mask storage.

The inverse is finite exact arithmetic: generate the $L+2$ endpoints $0, 4^{-L}, \dots, 4^{-1}, 1$, evaluate their dyadic g_L values, scan every interval with a fixed tie rule, and solve its affine formula. The generation cost $O((L+1)^2b^2)$ is included in (66). Inverse outputs have a common dyadic denominator dividing $2^{h+2L+O(1)}$. Normalization, positive fallback and final ordered rounding then use exact integers; no square-root, logarithm or exponential oracle is present.

An addressed evaluator stores $O(T_0b)$ instruction bits, uses $O(T_0S_0b)$ forward work and $O(T_0b + S_0)$ peak, and pays $O((T_0+S_0)b^2)$ offline work. A separate PURE expansion with $T = T_0 + b$, $S' = S_0 + b$ has depth $O(T)$, width $O(S')$, sparse addressed coefficient bits $O(bTS')$, dense coefficient bits $O(bT(S')^2)$ and dense work $O(b^2T(S')^2)$. Phase peaks add the chosen program or coefficients, masks, input, output and rolling storage. At the statistical operating point $P = O(\log M)$ and $b = O(\log M + \log K)$: the gain in primitive depth does not make internal precision or total memory constant.

K The mixing-dependent depth threshold

The preceding beta-free depth result is a special case of a sharper statement. The same finite dyadic compander exploits a mixing promise through its public gain budget, without estimating that promise or changing its attention weights. Define

$$e_K = \begin{cases} \max\{\sqrt{\beta}, M^{-1}\}, & K = 2, \\ \max\{\beta, M^{-1}\}, & K \geq 3, \end{cases} \quad f_K = 1 + \ln\left(1 + \frac{Q}{Me_K}\right), \quad G(z,t) = 1 + \ln(1 + z/t).$$

Theorem K.1 (Mixing-sensitive primitive depth)**.** *Fix the uniform primitive budgets of Section J, including capacity for its full histogram and rolling compander, and a nonempty legal band $c_-\sqrt{M} \leq Q \leq c_+\sqrt{M}$. For every fixed positive risk multiplier, statistical-order risk requires*

$$1 + d \geq c_{\mathrm{nec}}(1 + \ln f_K).$$

For a sufficiently large fixed multiplier it is attainable with $1 + d \leq C_{\mathrm{suff}}(1 + \ln f_K)$. Both statements hold for stationary and arbitrary initialization. Constants depend on the fixed alphabet, primitive capacities and band; necessity also depends on the requested multiplier. The conclusion is about actual primitive depth, not Boolean depth, internal bit precision or an all-resolution minimax surface.

K.1 The same compander with a sharper occupation analysis

Write $\chi = 2^L$, $z = \chi Q$. We first prove for the unchanged finite program of Section J that

$$R_{\mathrm{all}} \leq \min\{1, C_K[M^{-1/2} + Q^{-1} + G(z, Me_K)/z]\}. \tag{67}$$

For $K \geq 3$, retain its normalization terms

$$A_J = \frac{6\sqrt{K(K+1)M}}{Q\sqrt{J+K}}, \qquad B_J = \frac{3K(K+1)M}{z(J+K)}.$$

Equation (61) gives $\mathbb{E}A_J \leq 12K^2\sqrt{K(K+1)}/Q$. Add-one risk and its empirical correction cost $O_K(M^{-1/2})$, including $J = 0$, and final rounding costs $(K-1)/Q$. For $\beta \geq 1/M$, apply the exact half-suffix mixture and stationary occupation bound of Section D to $\mathbb{E}\min(1, B_J)$, not to a different predictor. With $\epsilon_0 = (1-\beta)^{\lfloor M/4 \rfloor}$ the bound is

$$\frac{6K^3(K+1)}{z}\left[17 + \ln\left(1 + \frac{160z}{K(K+1)M\beta}\right)\right] + \epsilon_0\Psi\left(\frac{6K^5(K+1)}{z}\right).$$

The elementary residual estimate following (28) bounds these two terms by $[144K^3(K+1) + 30K^5(K+1)]G(z, M\beta)/z$. At $\beta = 1$ the residual is zero. For $\beta < 1/M$, universal occupation gives $\Psi(3K^5(K+1)/z) \leq 3K^5(K+1)G(z,1)/z$. This proves (67) for larger alphabets and its weaker beta-cutoff version for $K = 2$.

The sharper binary cutoff requires a new purity argument; replacing β by $\sqrt{\beta}$ in the preceding calculation would be invalid. Put $u(x) = \min\{\chi x, \sqrt{x}\}$. On $x \leq 4^{-L}$, $g_L(x) = \chi x$. On $4^{-j-1} \leq x \leq 4^{-j}$,

$$g_L(x) = 2^j x + 2^{-j-1} - 2^{-L-1}, \qquad 0 \leq j < L.$$

Writing $x = t4^{-j}$ with $1/4 \leq t \leq 1$, the intercept after scaling lies between $1/4$ and $1/2$. The inequalities $t + 1/4 \geq \sqrt{t}$ and $t + 1/2 \leq (3/2)\sqrt{t}$ give

$$u(x) \leq g_L(x) \leq \tfrac{3}{2}u(x), \qquad u(cy) \geq \sqrt{c}\,u(y) \quad (c \geq 1,\ cy \leq 1). \tag{68}$$

For $L = 0$ the same statement is immediate. The composed scalar finite map in (56) is nondecreasing and downward, including the coarse cut; it is not necessary that every subtractive affine primitive be coordinatewise monotone. Both binary coordinates use the same code and precision. If their selected bank is positive, a largest true coordinate x_0 has largest label $Z_0 \geq 1$. Put $y_0 = g_L^{-1}(2Z_0/Q)$. Then

$$g_L(x_0) < g_L(y_0) + 3/Q \leq \tfrac{5}{2} g_L(y_0) \leq \tfrac{15}{4} u(y_0),$$

so (68) gives $x_0 < 16y_0$. If p is either designated empirical exit fraction and v its decoded normalized value, then for $p \leq 1/2$, $v \leq 16p/(1-p) \leq 32p$, hence $|v - p| \leq 32p$; for $p > 1/2$ unit loss suffices. Intersecting with (60) gives, on positive banks,

$$|v - p| \leq A_J + \min\{32p, B_J\}, \qquad A_J = \frac{6\sqrt{6M}}{Q\sqrt{J+2}}, \quad B_J = \frac{18M}{z(J+2)}.$$

For the true exit weights $q = (a, b, e, t_1)$ from Section H, Lipschitz truncation yields

$$|v - q| \leq A_J + 32 \min\{q, A_0/(J+2)\} + 33|p - q|, \qquad A_0 = 18M/(32z). \tag{69}$$

The orientation and true q are proof devices, not inputs. The empirical error on $J > 0$ is bounded by the add-one comparator plus $\mathbb{E}[2/(J+2)]$, both $O(M^{-1/2})$.

If both labels are zero, $u(x_i) < 3/Q$, so $x_i \leq 3/z + 9/Q^2$ and $J/M \leq 6/z + 18/Q^2$. Taking an integer ceiling in (25) gives

$$\mathbb{P}(\text{zero bank}) \leq 16(6/z + 18/Q^2 + 1/M).$$

This includes $J = 0$, and does not assert that the program tests $J = 0$ exactly. For full support and $\beta M^2 \geq 1$, insert A_0 into (50). After multiplication by 32 its reciprocal part is at most

$$\frac{300096}{z}\left[1 + \ln\left(1 + \frac{16z}{9M\sqrt{\beta}}\right)\right] \leq \frac{600192}{z} G(z, M\sqrt{\beta}),$$

and its remaining term is $128M^{-1/2}$. For singleton stationary support, the absorbing bound proved above gives instead $4608/z + 8704/M$, including transient initial states. Positive-β support classification exhausts these cases. For $0 \leq \beta < M^{-2}$, use the beta-free general occupation argument, with $Me_2 = 1$; equality belongs to the strong branch. Together with $\mathbb{E}A_J \leq 48\sqrt{6}/Q$ and output rounding, these estimates prove (67) for $K = 2$; $C_2 = 2^{23}$ is a safe constant.

K.2 A binary tube uniform in depth

Let $\Gamma_d \leq C^d$ be the late-coordinate influence from (63), with fixed C, and put $z = \Gamma_d Q$. The three-symbol tube already gives, when $\beta \leq 1/16$ and $z \leq M/4$,

$$R_{\text{stat}} \geq \frac{c}{z} \ln\left(1 + \frac{1}{8}\min\{z/(Me_K), M/z\}\right).$$

Here and below c is independent of depth. For binary sources the following argument proves the same formula when $\beta \leq 1/256$. Use the binary witness and reference active chain of Proposition G.1, with $e = e_2$ and $J_* = \lfloor \min\{(4e)^{-1}, M^2/(4z^2)\} \rfloor$. Then $J_* \geq 4$, $J_* \leq M/4$, and $z\sqrt{J_*}/M \leq 1/2$. The reference law at suffix length j is dominated pointwise by the stationary actual path law with factor $1/(32M)$; every random suffix symbol is late. Freeze a boundary branch and an ideal nonlinear output f_s. The common $1/8$ atom of the two-step active chain couples two continuations with

expected number of differing symbols at most 16. Telescope replacements on complete formal histories. Each costs Γ_d/M, already including local preprocessing, so the conditional continuation range is at most $34\Gamma_d/M$. The all-formal-history contract permits the intermediate hybrid words. The Doob variance and the derivative obtained by inserting the signed transition derivative at each step are therefore

$$\mathrm{Var}(f_s) \le 34^2\Gamma_d^2 j/M^2, \qquad |(\mathbb{E}f_s)'| \le 34\Gamma_d j/M.$$

Only the ideal output is differentiated; the finite floors need no smoothness. With $H_b = 2^{b_0}$, take

$$\tau = 34\sqrt{32DH_b}, \quad N_b = \lceil 2\tau + 2E + 4\rceil + 1, \quad A_b = 36H_bN_b^D.$$

Chebyshev over all DH_b branches gives probability at least $31/32$ of simultaneous deviation at most $\tau\Gamma_d\sqrt{j}/M$. A mesh of spacing $M/(34zj)$ has at most $36(1+zj/M)$ centers. After the fixed E/Q evaluation error and the floor, the complete good alphabet has at most $A_b(1+zj/M)$ labels. Its cross-section constant contains neither d nor Γ_d. The reference terminal state is informative A with probability at least $1/8$, so good-and-informative probability is at least $3/32$. Under the same uniform prior on $[1/4, 3/4]$, neighborhoods of radius $|I|/(64R_j)$ around the $R_j = A_b(1+zj/M)$ predictions cover prior mass at most $1/32$. Integrating outside them gives reference loss at least $1/(2048R_j)$. Pathwise domination and disjoint suffix events yield

$$R_{\mathrm{stat}} \ge \frac{1}{65536A_bM}\sum_{j=4}^{J_*}(1+zj/M)^{-1} \ge \frac{1}{524288A_bz}\ln\left(1+\frac{1}{8}\min\{z/(Me_2), M/z\}\right).$$

For the last inequality, put $u = z/M \le 1/4$, shift $j = k+4$, use $1+u(k+4) \le 2(1+uk)$ and integrate the decreasing summand; $J_* - 3 \ge J_*/4$ handles even $J_* = 4$. This proof averages one common prior and all seeds, never a falsely uniform endpoint-conditioned prior.

K.3 Uniform inversion of the fixed-band bounds

Choose the public dyadic gain $f_K \le \chi < 2f_K$. Since $G(\chi Q, Me_K) \le f_K + \ln\chi$, its ratio to χ is at most two. Equation (67) proves sufficiency with $L = O(1+\ln f_K)$ and actual $d = O(L+1)$. For necessity put $x = Q/(Me_K)$, $f = 1+\ln(1+x)$, $\Gamma = \Gamma_d$. Choose fixed $F \ge 2(1+\ln 8)$ large enough that

$$\frac{e^{t-1}-1}{8c_+^2 t} \ge e^{t/2} - 1 \quad (t \ge F).$$

If β is above the tube's allowed threshold, $e_K \ge 1/16$, so $f \le 1+2c_+$ and $\Gamma \ge 1$ suffices. If $\Gamma Q > M/4$, then $\Gamma > \sqrt{M}/(4c_+)$ whereas $f \le (1/8 + c_+)\sqrt{M}$. If $f \le F$ use $\Gamma \ge 1$, and if $\Gamma \ge f$ there is nothing to prove. In the remaining case $1 \le \Gamma < f$, the two logarithm branches in the tube are each at least $f/2$: the first uses $\ln(1+\Gamma x/8) \ge f - 1 - \ln 8$, and the second uses $M/(\Gamma Q) \ge x/(\Gamma c_+^2) > x/(fc_+^2)$. A risk bound $C_{\mathrm{risk}}/\sqrt{M}$ therefore forces $\Gamma \ge cf/(2c_+C_{\mathrm{risk}})$. Thus $\Gamma \ge c_*f$ in every finite-domain case. Taking logarithms in $\Gamma \le C^d$, and using $1+d \ge 1$ on bounded f, proves the necessary statement. Stationary inclusion transfers it to arbitrary starts.

At $\beta = 0$ both alphabets need $\Theta(1+\log\log M)$ depth. At $\beta = 1/M$, binary sources need only constant depth, while every fixed $K \ge 3$ still needs the double logarithm. This is not a global square-root mixing assertion. The public promise can index the gain budget; an implemented selection from a rational promise pays its description and rational comparisons (binary square-root comparisons can be squared). The uniform evaluator uses $P \ge \max\{4h + c_{\mathrm{FA}}(d+2), h+L+4\}$. Its full encoder word is not bounded by a decoder-only word formula. All costs and conversion/coexistence charges in Section J.7 remain in force.

L Packing more information through a fixed coordinate bank

A grid floor established for a fixed prefix, a one-block local ratio, or a fixed architecture need not survive optimization over growing-depth nonlinear encoders. This section isolates the issue on IID sources before returning to the general Markov target. Here Q is an internal cut resolution. Decoders may output arbitrary real parameters in the information lower bounds; the constructive decoder produces a paid finite dyadic parameter, not necessarily a multiple of $1/Q$.

In the IID sections we use $M \geq 1$ and dyadic $Q \geq 2$, overriding the earlier global defaults; specific propositions may impose further restrictions. Every transition row is Bernoulli(p), with an arbitrary initial pair. The M generated symbols are IID and independent of that pair; their next-symbol target is p. The pair kernel has $P^2(s,\cdot) = \mathrm{Bern}(p)^{\otimes 2}$ for every s, so this is a legal witness for every minorization promise, including $\beta = 1$. Exactly $J \geq 2$ coordinates in $[0,1]$ are downward quantized at $Q = 2^h$, $h \geq 1$. Only a fixed finite boundary label and source-independent metadata survive alongside them. For the depth lower, that boundary consists specifically of at most b bits at fixed public positions (conditional on a public independent seed). An arbitrary nonlocal finite-valued statistic is not admitted by that lower.

Theorem L.1 (IID capacity and constructive partial folding)**.** *With boundary alphabet size at most H, every procedure has*

$$R \geq \max\left\{\frac{3}{64\sqrt{M}}, \frac{1}{16H(Q+1)^J}\right\}.$$

For every integer $0 \leq r \leq (J-1)h$, a causal zero-logit bounded-primitive encoder of fixed width $O(J)$ and depth $d \leq d_0 + 2r$ has a fully paid finite evaluator satisfying

$$R \leq \frac{1}{2\sqrt{M}} + \frac{2}{Q2^r}.$$

Coefficients have magnitude at most four, retained states lie in $[0,1]$, and the finite pre-cut error from its ideal encoder is at most $1/Q$ on every formal history. At $r = (J-1)h$ the optimal IID risk in this sufficiently deep interface class is $\Theta_{J,H}(\max\{M^{-1/2}, Q^{-J}\})$.

Proof. For the alphabet lower fix all independent seeds. There are at most $R_* = H(Q+1)^J$ decoder outputs. Under a uniform prior on $[1/4, 3/4]$, neighborhoods of radius $1/(8R_*)$ about them cover at most half the prior, leaving expected absolute error at least $1/(16R_*)$, regardless of how samples choose labels. Average the seeds. For the statistical lower fix the initial pair to $(0,0)$ and take $p_\pm = 1/2 \pm 1/(16\sqrt{M})$. Their gap is $1/(8\sqrt{M})$, one-bit chi-square is at most $1/(12M)$, and product chi-square is at most $e^{1/12} - 1 \leq 1/11$. Thus sample-law TV is less than $1/4$. Integrating the minimum of the two densities and using the triangle inequality gives average loss at least $3/(64\sqrt{M})$. These witnesses need not coincide; their maximum is valid.

For the upper define the tent

$$T(u) = 2\min(u, 1-u) = 2u - 4\,\mathrm{ReLU}(u - 1/2), \quad 0 \leq u \leq 1.$$

Induction gives, on $[v/2^e, (v+1)/2^e]$,

$$T^e(u) = \begin{cases} 2^e u - v, & v \text{ even}, \\ v + 1 - 2^e u, & v \text{ odd}. \end{cases}$$

Neighboring formulas agree at endpoints. If $r = 0$ retain only time zero. Otherwise let $m = \lceil r/h \rceil$, $t_i = ih$ for $0 \leq i < m$, and $t_m = r$. There are $m + 1 \leq J$ retained rails $y_i = T^{t_i}(x)$, each gap $e_i = t_{i+1} - t_i$ lies in $[1, h]$, and unused rails are zero.

Here is an exact decoder, including orientation and endpoints. Start with feasible interval $C = [0, 1]$ and affine $y_i = a_i x + b_i$, initially $a_0 = 1, b_0 = 0$. Given label $k_i < Q$, intersect C with the inverse of the closed interval $[k_i/Q, (k_i + 1)/Q]$, sorting its endpoints when $a_i < 0$. For $k_i = Q$, impose $y_i = 1$: since $a_i \neq 0$, a valid fiber is a singleton and can be returned immediately. Empty intersections decode to zero on invalid inputs. A singleton formed by any other intersection can also be returned immediately. For a remaining nonsingleton interval and next gap $e \leq h$, its coarse bin is contained in the closed branch of T^e with index

$$v = \left\lfloor \frac{k_i}{2^{h-e}} \right\rfloor .$$

It cannot cross a branch boundary internally because 2^e divides Q. Compose the appropriate affine formula; continuity validates shared endpoints. Inductively $|a_i| = 2^{t_i}$. Thus the final interval has length at most $1/(Q2^r)$; its lower endpoint z satisfies

$$z \leq x \leq z + (Q2^r)^{-1}.$$

All endpoints lie on the fixed dyadic grid of denominator 2^{h+r}. The decoder uses $O(J)$ signed integer operations on $O(h + r)$-bit quantities, not a table enumerating Q^J cells. Ordinary unsigned base-Q digit reading would be wrong because the orientations alternate.

At history row $t \geq 3$, a zero-score head pools the bits at positions $3, \ldots, t$; the synthetic terminal pools positions $3, \ldots, M + 2$. Rows one and two use public singleton self fallback. A common mask density $1/4$ suffices. Each history row computes its own causal prefix mean, not a terminal mean copied backward. Retain the chosen rails while advancing one working rail through r tents. One tent is precisely two bounded-size stages, a ReLU followed by an affine update retaining the old value. Signed preactivation lies in $[-1/2, 1/2]$, coefficients are bounded by four and width is $O(J)$. Any syntax-required dummy heads and their masks are executed and paid.

Choose fractional precision $P \geq h + r$, $D_f = 2^P$. For a row with count A and m bits compute $U_0 = \lfloor D_f A/m \rfloor$, then perform

$$U \longleftarrow 2 \min\{U, D_f - U\}.$$

These tent updates are exact integers, including zero, the midpoint and the endpoint. Their labels are the exact ideal labels of $\widehat{x} = U_0/D_f$. The decoder therefore gives $0 \leq x - z \leq (Q2^r)^{-1} + 2^{-P} \leq 2/(Q2^r)$. Each intermediate ideal/finite rail differs by at most $2^r 2^{-P} \leq 1/Q$. Identical positive Taylor weights cancel in the zero-logit mean, and the signed full-product affine evaluator performs these integer tent updates exactly. Thus the same generic finite family includes the construction; one may additionally pay the sufficient $P \geq 4h + c_{\mathrm{FA}}(d + 2)$ from (59). No one-sided tent-output error is asserted: the decoder recovers the initial fine scalar. Finally $\mathbb{E}|A/M - p| \leq 1/(2\sqrt{M})$ proves the risk bound for every initial pair. □

The tent iterations themselves are established constructions [10, Section 3.3]. Multi-coordinate representation of a high-resolution scalar and geometric reconstruction also precede this experiment [11, Sections 2.1–2.4 of the cited version]. The statement here concerns the explicit statistical, depth and finite-cut contracts, not priority for folding or output-coordinate packing.

L.1 An influence-reveal lower valid at growing depth

The next argument replaces a depth-dependent global variance constant by a reveal selected from the actual deterministic influences.

Theorem L.2 (Summed-influence floor)**.** *For every real $Q \geq 1$, observe $N = M+2$ IID Bernoulli(p) bits with $p \in [1/4, 3/4]$. Conditional on each source-independent seed, let the exact boundary consist of at most b bits at public fixed positions. For every frozen branch s and output coordinate $j \leq D$, assume on all formal histories*

$$|f_{s,j}(x) - f_{s,j}(x^{(i)})| \leq c_i^{s,j}, \qquad \sum_i c_i^{s,j} \leq B, \quad B \geq 1.$$

The actual pre-cut output a satisfies $\max_j |a_j(x) - f_{S(x),j}(x)| \leq E/Q$, with fixed D, b, E and deterministic public B. The complete cut is the coordinatewise downward Q grid, boundary and independent metadata. Then every unrestricted decoder obeys

$$\sup_p \mathbb{E}_p|\widehat{p} - p| \geq c(D, b, E)/(QB).$$

For the uniformly bounded causal primitive family and its certified finite approximants this implies $R \geq c/(QC^d)$, with one fixed $C \geq 2$ independent of depth.

Proof. Set

$$F = D2^b, \quad C_0 = 100F, \quad L_0 = F + b, \quad m = \lceil C_0 Q^2 B^2 \rceil,$$
$$P_0 = 128\sqrt{L_0(C_0+1)}, \quad K_0 = \lceil 2E + 6 \rceil + 1, \quad R_0 = 8K_0^D.$$

If $N < m$, grant the full history and use the two-point calculation above with N observations. Its risk is at least $1/(128\sqrt{N})$ and hence $[128\sqrt{C_0+1}QB]^{-1}$, because $m \leq (C_0+1)Q^2B^2$. Otherwise use one uniform prior on $I = [1/4, 3/4]$. Fix and grant the encoder seed. For every (s, j) choose its m largest influence bounds, breaking ties publicly. Let U be the union of all these indices and all boundary positions. It is independent of p and the data. Grant $Y = X_U$ to the decoder; with $n = |U|$,

$$m \leq n \leq Fm + b \leq L_0 m \leq L_0(C_0+1)Q^2B^2.$$

This is a proof-only oracle, not an upper-construction port. For each unrevealed coordinate, $c_i^{s,j} \leq B/m$, while their sum is at most B. Given $Y = y, p$, the remaining bits are still independent. Efron–Stein gives

$$\mathrm{Var}(f_{s,j} \mid Y = y) \leq \tfrac{1}{2}\sum_{i \notin U} (c_i^{s,j})^2 \leq B^2/(2m).$$

Chebyshev and a union over all F branches give probability at least .995 of deviations at most $1/Q$, uniformly for every y, p. The conditional mean $h_{s,j}(p)$ is a finite product-measure polynomial. Differentiating one free factor at a time yields

$$h'_{s,j}(p) = \sum_{i \notin U} \mathbb{E}_p[f_{s,j}(X_i = 1) - f_{s,j}(X_i = 0) \mid Y = y],$$

so $|h'_{s,j}| \leq B$. No posterior is differentiated and no branch-consistency condition is imposed on the free bits.

If y contains t ones, its posterior density on I is proportional to $p^t(1-p)^{n-t}$. Put $z = t/n$ and $I_y = I \cap [z - 3/\sqrt{n}, z + 3/\sqrt{n}]$. Hoeffding gives joint probability of $p \notin I_Y$ at most $2e^{-18}$. Markov therefore yields a set $\mathcal{T}$ of revealed patterns of probability at least $1 - 200e^{-18} > .99$ whose posterior mass of I_y is at least .99. For $y \in \mathcal{T}$, the posterior mode p_0 is z clamped to I. With $\ell(p) = t \ln p + (n-t)\ln(1-p)$,

$$|\ell'(p_0)| \leq 16\sqrt{n}, \qquad |\ell''(p)| \leq 16n \quad (p \in I).$$

The first follows from distance at most $3/\sqrt{n}$ to the interval and $p_0(1-p_0) \geq 3/16$. One direction from p_0 remains in I for length $1/4$. Over an inward segment of length $1/(64\sqrt{n})$, Taylor's theorem loses at most $1/4 + 1/512 < \ln 2$. Integrating it gives posterior density at most $128\sqrt{n} \leq P_0 QB$. This treats boundary modes, including truncated extreme counts, without a normal approximation.

For $y \in \mathcal{T}$, its actual boundary branch is now fixed. Mesh I_y at spacing at most $1/(QB)$ including endpoints. At most $2 + 6QB/\sqrt{n} \leq 8$ centers suffice. The mean-curve error, random tube, finite error and floor imply that every good integer label is within $E + 3$ of a center's scaled mean. Each coordinate has at most K_0 possibilities, so the entire good alphabet has at most R_0 messages, independent of p. It may depend on y and the seed; the good event need not be observable from the message. The joint posterior good probability is at least .98. Fix decoder randomness. Neighborhoods of radius $(8R_0P_0QB)^{-1}$ around its R_0 predictions have posterior mass at most $1/4$. Subtracting this mass leaves conditional Bayes risk at least $(16R_0P_0QB)^{-1}$. Average y, then both seeds, and use that the original experiment is less informative than the oracle experiment. A valid constant is

$$c = \min\{(128\sqrt{C_0 + 1})^{-1}, (32R_0P_0)^{-1}\}.$$

The branch $N < m$ is selected by public deterministic quantities, so its different two-point prior creates no seed-dependent minimax interchange.

For the causal application use the actual nonnegative perturbation majorant $H = aI + b_{\mathrm{att}}A$ from (16). Since A is row stochastic, H^d has row sum $(a + b_{\mathrm{att}})^d$. A source bit affects at most ρ historical local states of diameter Δ. The frozen synthetic initial state does not change. Therefore one can take

$$c_i = G\Delta \sum_{u=i}^{\min(N,i+\rho-1)} (H^d)_{Tu}, \qquad \sum_i c_i \leq \rho G\Delta (a + b_{\mathrm{att}})^d.$$

This sums actual matrix entries, not the logarithmic pointwise envelope or its factorial square-sum constant. Increase a for positionwise stages and use the fixed density-adjusted attention constants. Then $B_d = \max\{1, \rho G\Delta(a + b_{\mathrm{att}})^d\} \leq C^d$ for $C = \max\{2, \max(1, \rho G\Delta)\max(1, a + b_{\mathrm{att}})\}$. The finite approximation contract, not Lipschitzness of internal floors, supplies admission. This proves the depth claim. □

L.2 The all-dyadic statistical-order sandwich

Combine the two theorems for the IID arbitrary-initialization experiment. Its statistical lower uses a fixed initial pair; its influence lower uses a stationary product initial pair so that all $M + 2$ bits are IID. Both are allowed, and their witnesses need not coincide. Every admitted encoder satisfies

$$R \geq \max\left\{\frac{3}{64\sqrt{M}}, \frac{1}{16H(Q+1)^J}, \frac{c}{QC^d}\right\}. \tag{70}$$

Thus risk at most $A/\sqrt{M}$ requires

$$A \geq 3/64, \quad Q^J \geq \frac{\sqrt{M}}{16HA(3/2)^J}, \quad d \geq \max\left\{0, \frac{\ln(c\sqrt{M}/(AQ))}{\ln C}\right\}.$$

Conversely, for any fixed $\kappa_0 > 0$, if $Q^J \geq \kappa_0\sqrt{M}$, choose

$$r_* = \min\{(J-1)h, \max\{0, \lceil \log_2(\sqrt{M}/Q)\rceil\}\}.$$

Then $Q2^{r_*} \geq \min\{\sqrt{M}, Q^J\}$, so depth $d_0 + 2r_*$ gives risk at most $[1/2 + 2\max\{1, \kappa_0^{-1}\}]/\sqrt{M}$, and $d_0 + 2r_* \leq d_0 + 2 + 2\log_{2,+}(\sqrt{M}/Q)$. Necessary and sufficient constants are different. At $M \asymp Q^{2J}$ the depth order is $\Theta(\log Q)$, but no matching exponential base for risk versus depth follows. IID upper guarantees do not transfer to general Markov sources by inclusion; the IID lower obstructions do.

L.3 The full cost of the folded representation

Let $n = M + 3$, $d = O(1 + r)$ and $w = C_J(P + \log(n + 1) + \log(d + 1) + 1)$, with full $2w$-bit products. Decoder integers of $O(h + r)$ bits fit because $P \geq h + r$. A deliberately conservative implementation has

$$T_0 = O_J(dn^2(P + 1)w^2 + (d + n)w^2), \qquad S_0 = O_J(dn^2 + dJ^2w + nJw + w^4).$$

These budgets include every token and stage, dense masks including excluded edges, dense zero coefficient slots, optional dummy heads, source and local acquisition, counts and divisions, coefficient and positional generation, retained-time selection, labels, signed interval decoding, output and clearing/isolation. The $P+1$ factor pays generic Taylor evaluation; bespoke uniform pooling can omit it. Label storage is $J(h+1)$ bits because endpoint label Q is allowed, plus the fixed boundary; this is not the full internal precision or memory. A bounded-arity addressed tape realization stores $O(T_0 w)$ instruction/address bits, scans the S_0-bit tape per instruction, and has safe online work $O(T_0 S_0 w)$, peak $O(T_0 w + S_0)$ plus generator scratch, and offline work $O((T_0 + S_0)w^2)$. No noncut count, mean, work array or other token output remains accessible to decoding. A PURE Boolean/ReLU unrolling has its actual much larger depth and gate-slot costs; it is not a fixed-width $d_0 + 2r$ Boolean circuit.

M Robust IID packing through a noisy finite cut

The preceding IID capacity results concern a clean final grid. They do not by themselves give robustness to a fault immediately *before* that grid. This section first exhibits a vulnerability of the particular mean-and-fold representation, then gives changed continuous representations that withstand a bounded pre-grid fault. The upper bounds remain IID results: here every transition row is Bernoulli(p), so the M generated bits are independent, the initial pair is arbitrary, and the actual next-symbol target is p. Absolute parameter error is binary TV loss. An IID lower witness is also a lower witness for larger Markov classes; an IID upper bound is not a Markov upper bound.

Throughout this section $J \geq 2$ denotes the number of exposed scalar coordinates, *not* the source alphabet size K or a context occupation count. We fix J and a noise multiplier $c \geq 0$, and write $Q = 2^h$. Constants may depend on J, c and fixed primitive budgets. Increasing J changes width, label count, description lengths and constants; none of the estimates below is dimension-uniform. The source alphabet in these upper constructions is fixed at $K = 2$.

Geometric scalar-to-vector encoding, stretching/folding and nearest-curve reconstruction are established ideas. Analog tube-packing work treats these mechanisms under an AWGN formulation [9]; two-coordinate low-precision curve representations and nearest-curve decoding also appear in Uss et al. [11]. Their Gaussian or distributional noise assumptions are not the bounded adversarial channel studied here. The obligations addressed below are the stated statistical experiment, bounded-size sequential primitives, a complete finite cut and fully charged arithmetic. No novelty claim for curve packing, folding or reconstruction follows.

M.1 Fault location and a reflection collision

For the robust upper bounds the channel acts on the *actual* finite pre-cut bank $a \in [0,1]^J$. An adversary chooses e with $\|e\|_\infty \le c/Q$, possibly depending on the entire history and using randomness; the received labels are

$$q_j = Q^{-1} \lfloor Q\, \mathsf{C}(a_j + e_j) \rfloor, \qquad \mathsf{C}(u) = \min\{1, \max\{0, u\}\}. \tag{71}$$

Clipping is part of this declared channel; equivalently one can require the disturbed coordinates to remain in the cube. After the cut only these labels, the declared finite boundary field and source-independent public metadata/seeds survive. Counts, history, unquantized rails, duplicate outputs, residual work arrays and timing channels do not survive.

This is different from a canonical *post-grid* corruption that must map the grid $\{0, 1/Q, \dots, 1\}$ to itself. In that latter model every coordinate perturbation of magnitude strictly less than $1/Q$ is the identity. A fault on a finer state followed by coarse quantization need not be the identity, even below half a coarse cell. Moving the fault boundary is a change of experiment, not a contradiction between bounds.

Write

$$\mathsf{T}(u) = 2u - 4\,\mathrm{ReLU}(u - \tfrac{1}{2}), \qquad 0 \le u \le 1.$$

The following result concerns only the specific clean folded encoder, not every encoder with J outputs.

Proposition M.1 (First-orientation erasure)**.** *Let $M = 2^m$, $h \ge 1$, $P \ge \max\{m, Jh\}$ and $D_f = 2^P$. The encoder starts from the fine mean $\lfloor D_f A/M \rfloor / D_f = A/M =: x$, where A counts the generated ones, and retains*

$$(x, \mathsf{F}(x), \dots, \mathsf{F}^{J-1}(x)), \qquad \mathsf{F} = \mathsf{T}^h.$$

One source-independent map of these fine-grid rails, applied before their downward Q-quantization, with displacement at most δ in ℓ_∞, gives, for every decoder and $0 < \delta \le 1/2$,

$$\max\{R(p_-), R(p_+)\} \ge \frac{\delta}{2}\, w_r\, [1 - 2e^{-M\delta^2/2}]_+, \qquad p_\pm = \tfrac{1}{2} \pm \tfrac{\delta}{2}. \tag{72}$$

Here $r = 0, w_0 = 1$ if no history-dependent side label survives. If the exact terminal pair and fixed-initial-pair flags survive, one may take $r = 2$ and $w_2 = p_+p_- = 1/4 - \delta^2/4$, provided $M \ge 2$. More generally, for side information consisting of functions of fixed generated-bit positions, choose an even $r \le M$ covering those positions and set $w_r = (p_+p_-)^{r/2}$. In particular, when $M\delta^2 \ge 2\ln 4$, the no-side lower is $\delta/4$ and the terminal-pair lower is at least $3\delta/64$.

Proof. Reset the first rail to $1/2$ whenever $|x - 1/2| \le \delta$, and otherwise leave it unchanged; leave every other rail unchanged. This is a common map depending only on its input and public δ, not on p. It is legal on the entire fine grid, because $D_f/2$ is integral, and moves at most δ. A non-dyadic δ merely specifies a finite lookup condition; no bounded-depth adversary implementation is asserted.

Without side information, couple a generated history at p_+ with its complement at p_-, keeping the initial pair fixed. Their exact means satisfy $x' = 1 - x$. The identity $\mathsf{T}(1-x) = \mathsf{T}(x)$ implies equality of all retained rails after the first. On $E = \{|x - 1/2| \le \delta\}$ both first rails are reset to $1/2$, so their *complete* quantized observations agree, including endpoint labels.

Complementing every generated bit would change a retained terminal pair. Instead pin the r fixed positions to the *same balanced binary pattern* under both sources, and complement only the

remaining bits. This event has probability w_r under each source; its pinned sum is $r/2$, so again the two full counts sum to M. Every allowed side bit is identical. For the terminal pair use its pattern 01. Initial bits are fixed and excluded from A. This constructs a common subprobability coupling of the unconditional observations, not a conditional-risk claim that discards the cost of the pinned event.

Conditionally on that pattern, $A = r/2 + B$ with $B \sim \mathrm{Bin}(M - r, p_+)$. The conditional mean of A/M is at distance at most $\delta/2$ from $1/2$. For $M > r$, Hoeffding's inequality therefore gives

$$\mathbb{P}(E^c \mid \text{pattern}) \le 2\exp\left\{-\frac{M^2\delta^2}{2(M-r)}\right\} \le 2e^{-M\delta^2/2}.$$

If $M = r$, the mean is exactly $1/2$ deterministically. Thus identical complete observations have coupled mass at least $w_r[1 - 2e^{-M\delta^2/2}]_+$. Couple decoder seeds identically on this mass. For any decoded value z, $|z - p_+| + |z - p_-| \ge \delta$. Integrating, discarding losses elsewhere, and dividing by two proves (72), also for randomized decoders. Finally $w_2 \ge 3/16$ and the stated sample condition makes the bracket at least $1/2$. □

For a concrete subcell separation fix $J \ge 2$, let $M = Q^4$, take $P \ge \max\{4h, Jh\}$ and $\delta = 1/(4Q)$. All comparisons are dyadic and exact, $\delta < 1/(2Q)$, and $M\delta^2 = Q^2/16 \to \infty$. The terminal-side attacked risk is $\Omega(Q^{-1})$, whereas the clean folded decoder has risk at most

$$\frac{1}{2\sqrt{M}} + \frac{2}{Q^J} \le \frac{5}{2Q^2}.$$

Only the first orientation rail is disturbed; the deeper rails already have exact reflection symmetry. This does not establish fragility with an arbitrary fixed-cardinality history label: such a label could contain an orientation bit. Nor does it cover a non-dyadic mean whose two rounded complementary values need not sum to one, an internal arithmetic fault, or a redesigned/protected encoder.

M.2 The common separated grid path

We next describe the geometry shared by two different continuous parameterizations. For an even dyadic $N = 2^k \ge 2$, define an ordered axis-aligned path $\mathcal{C}_{j,N}$ recursively. The one-dimensional path is $[0, 1]$. At level j, put a copy of $\mathcal{C}_{j-1,N}$ in each plane $x_j = i/N$, $0 \le i < N$, in forward order for even i and reversed order for odd i. Follow each copy by an outer-axis connector of length $1/N$ to plane $(i + 1)/N$. There is no copy in plane 1.

Lemma M.2 (Separated path geometry)**.** *The path is simple, has endpoints* 0 *and the last coordinate unit vector, and has an ordered list of E_j edges with*

$$E_1 = 1, \quad E_j = NE_{j-1} + N = O_j(N^{j-1}).$$

Its vertices lie on the N-grid, every edge has length at least $1/N$, incident consecutive edges are perpendicular, and nonincident edges have Euclidean distance at least $1/N$. If two points on the path have distance $\Delta < 1/N$, their connecting subpath has length at most $\sqrt{2}\Delta$.

Proof. Induct on j. Different copy planes have separation at least $1/N$. Nonadjacent connector intervals have that separation in the outer coordinate; adjacent connectors are anchored at the two opposite lower-dimensional endpoints, whose distance is one. A connector and a nonadjacent copy are likewise separated in the outer coordinate. In either adjacent plane its anchor is an endpoint of

the lower path. A grid vertex not belonging to an axis-aligned grid edge is at least $1/N$ from that edge; by inductive simplicity, the endpoint belongs only to its incident lower edge. Thus all other lower edges remain at least $1/N$ away, and that sole incident edge is perpendicular to the connector. Within a copy use induction. This also proves no repeated vertices or nonlocal contacts. Even N makes the final connector end at the original lower endpoint in plane 1, giving the stated parent endpoints. Edge counts and minimum lengths follow directly from the recursion.

For two points at distance below $1/N$, containing edges must be equal or incident. On one edge the intervening length is Δ. On perpendicular incident edges it is $a+b \le \sqrt{2}\sqrt{a^2+b^2} = \sqrt{2}\Delta$. The simple ordered path has no alternative itinerary between them. □

The actual cut has only $(Q+1)^J$ coordinate labels, multiplied by the explicitly allowed boundary alphabet. The finite-alphabet lower in Theorem L.1 applies irrespective of the geometric representation.

M.3 A dyadic robust encoder, including its finite lattice

Define a bounded extended first tent by

$$\mathsf{T}_{\mathrm{ext}}(u) = 2u - 4\,\mathrm{ReLU}(u - \tfrac{1}{2}) + 4\,\mathrm{ReLU}(u-1).$$

On $[0, 1+1/(4N)]$ its output lies in $[0,1]$; subsequent tents are ordinary T. Let $P_1(t) = t$. For $j = 2, \dots, J$, put

$$L_j = N^{j-1}, \qquad \eta_j = (2L_j+2)^{-1},$$

and define

$$\begin{aligned} v &= \mathsf{T}^k(t), & A_\triangle &= \mathsf{T}^{k+1}(t), \\ B_\triangle &= \mathsf{T}^k(\mathsf{T}_{\mathrm{ext}}(t+1/(4N))), & & \\ z_j &= \min\{1 - A_\triangle, L_j\,\mathrm{ReLU}(\tfrac{1}{2} - B_\triangle)\}, & s_j &= A_\triangle/2 + z_j, \\ S_j(t) &= t - s_j/N, & u_j(t) &= \mathsf{C}(v + v/L_j - 1/(2L_j)), \\ P_j(t) &= (P_{j-1}(u_j(t)), S_j(t)). & & \end{aligned} \tag{73}$$

The factor L_j is not a primitive coefficient. Starting with $b = \mathrm{ReLU}(1/2 - B_\triangle)$, perform $(j-1)k$ stages $b \leftarrow \min\{1, 2b\}$. Taking the minimum with $1 - A_\triangle \le 1$ makes this saturated computation exactly equivalent to multiplication by L_j. Minima use $\min(f,g) = f - \mathrm{ReLU}(f-g)$, and clipping uses two ReLUs. All scalar coefficients have magnitude at most four and all intermediates have a fixed bounded range. Sequentially compute a level, retain S_j, and recurse on u_j: this uses $O(J)$ rails and $O(J^2k + J)$ actual fixed-size primitive stages. No multiplication of variable rails, discrete branch oracle or increasing intra-stage circuit is used.

Theorem M.3 (Fixed-dimensional robust IID rate)**.** *Fix $J \ge 2, c \ge 0$. Choose integer $a \ge 1$ with $2^a \ge 8\sqrt{J}(c+1)$, and let $h \ge a+1$, $N = 2^{h-a}$. There is a causal finite encoder exposing exactly J coordinates, with width $O(J)$, bounded scalar states, coefficients at most four and primitive depth $O_J(1+h)$, such that every attack in* (71) *satisfies*

$$\mathbb{E}|\widehat{p} - p| \le \frac{1}{2\sqrt{M}} + \frac{1 + 2\sqrt{2J}(c+1)2^{(J-1)a} + (J-1)2^{Ja}}{Q^J}. \tag{74}$$

A rational nearest-segment decoder and its $O_J(N^{J-1})$ arithmetic operations are paid. Thus for fixed J, c the robust IID minimax order is $\max\{M^{-1/2}, Q^{-J}\}$ on this range of Q, once this depth and paid resources are allowed. No logarithmic-time decoder assertion is made.

Proof. We verify the formula, its inverse stability, and its finite evaluation.

Phase calculation. Write $t = (i + r)/N$, $0 \le i < N$, $0 \le r \le 1$, with adjacent endpoints identified by continuity. If tri is the period-one triangular wave of height one and slopes ± 2, tent iteration gives $A_\triangle = \mathrm{tri}(r)$ and $B_\triangle = \mathrm{tri}(r + 1/4)$. On $0 \le r \le 1/2$ the positive part in (73) vanishes and $s_j = r$. On the second half, $A_\triangle = 2(1 - r)$ and $\mathrm{ReLU}(1/2 - B_\triangle) = \min(A_\triangle, 1 - A_\triangle)$. Substitution yields

$$s_j = \min\{r, (2L_j + 1)(1 - r)\}.$$

The two expressions cross at $r = 1 - \eta_j$. Therefore

$$S_j((i + r)/N) = \begin{cases} i/N, & r \le 1 - \eta_j, \\ i/N + (r - 1 + \eta_j)/(N\eta_j), & r \ge 1 - \eta_j. \end{cases} \tag{75}$$

The clipped u_j traverses $[0, 1]$ with alternating orientation on $r \in [\eta_j, 1 - \eta_j]$ and is fixed at its endpoints on the two remaining portions. Its active derivative has magnitude $N(1 + 1/L_j)$. Thus (73) traces exactly $\mathcal{C}_{j,N}$: initial clipping gives a possible vertex pause, whereas final clipping coincides with the moving connector. This proves the asserted geometry without assuming a staircase oracle.

Speeds and vertex fibers. Every moving edge has speed at least N^{j-1}, by induction on copies and by connector speed $1/\eta_j = 2N^{j-1} + 2$. A global coordinate slope bound is

$$V_1 = 1, \quad V_j \le \max\{N(1 + 1/L_j)V_{j-1}, 2N^{j-1} + 2\} \le 3 \cdot 2^{j-2}N^{j-1} \quad (j \ge 2).$$

Let λ_j bound the full preimage interval of any vertex. An initial clip contributes η_j/N; an inherited endpoint pause contributes at most λ_{j-1}/N, with interior inherited pauses no larger. A reversed copy may move its inherited initial pause to the end, but its final clip is a moving connector, not a second new pause. Nonzero connectors separate consecutive copies. Consequently

$$\lambda_1 = 0, \qquad \lambda_j \le \eta_j/N + \lambda_{j-1}/N \le (j - 1)/(2N^j).$$

For two image points at distance $\Delta < 1/N$, Lemma M.2 bounds moving path length by $\sqrt{2}\Delta$. At most two whole vertex fibers need be added, including endpoint preimage choices. Indeed an incident-edge local subpath cannot reach either far endpoint: that endpoint is at least $1/N$ from the opposite edge. The same-edge case is no worse. Thus any preimages satisfy

$$|t - t'| \le \frac{\sqrt{2}\Delta}{N^{J-1}} + \frac{J - 1}{N^J}. \tag{76}$$

This also covers $\Delta = 0$ and the global endpoints.

Noise and decoding. For a finite input scalar t, exact evaluation gives $w = P_J(t)$. Attack, clipping and the final grid give $\|q - w\|_2 \le r_0 := \sqrt{J}(c + 1)/Q$. Enumerate the E_J segments, project q onto each by clamping its varying coordinate, compare squared rational distances, and break ties by public edge order. A nearest point w' satisfies $\|w' - w\|_2 \le 2r_0 < 1/N$ by the choice of a. Choose any actual preimage t' on its selected moving affine piece. Equation (76) gives the pointwise bound

$$|t' - t| \le \frac{2\sqrt{2J}(c + 1)}{QN^{J-1}} + \frac{J - 1}{N^J}. \tag{77}$$

The edge table stores rational geometric and parameter endpoints. Build it recursively by applying the inverse oriented affine map u_j to each child interval and append the connector interval $[(i +$

$1-\eta_j)/N,(i+1)/N]$. Stationary intervals need not be decoded: an incident moving piece supplies an endpoint preimage. There are E_J positive-length moving intervals; their endpoint numerators and denominators have $O_J(h)$ bits, because only a fixed number J of factors N and $2L_j+2$ occur. Projection, squared-distance comparison by cross multiplication and inverse interpolation therefore require no square root or real oracle. List generation, storage and scanning cost $O_J(E_J)$ bounded-word arithmetic operations and $O_J(E_Jh)$ bits.

Exact uniform precision. Set

$$\nu_J=\left\lceil \log_2(3\cdot 2^{J-2})\right\rceil+1,\quad p_0=Jh+\nu_J,\quad G=\tfrac{1}{2}(J-1)(J-2)k,\quad P=p_0+G.$$

A direct P-bit mean would not in general leave enough exact divisibility after every recursive division. Instead scale input bits by 2^{-G}, pool by causal zero-logit attention, then apply G bounded doublings. Identical positive finite Taylor weights cancel in the attention ratio. At a row with exact empirical mean x, the actual scalar is exactly

$$2^G\,2^{-P}\left\lfloor 2^{P-G}x\right\rfloor=2^{-p_0}\left\lfloor 2^{p_0}x\right\rfloor=:t.$$

The ideal scaling and doublings cancel and return x. Every doubling remains in $[0,1]$.

Suppose an incoming recursive scalar has fractional exponent at most e, initially $e=p_0$. Since $e\ge k+2$, the shift $1/(4N)$ fits on that grid. Completed tent chains give exponents at most $e-k$ for v and $e-k-1$ for $A_\triangle,B_\triangle$. Their half-biases, saturated doublings and minima preserve these bounds. Hence s_j has exponent at most $e-k$, so s_j/N and S_j have exponent at most e. The divided term v/L_j has exponent at most $e+(j-2)k$; its bias exponent $(j-1)k+1$ also fits because $e\ge k+2$. Thus u_j uses at most $(j-2)k$ additional bits. Summing over descending levels gives exactly the guard allowance G. Intermediate ReLUs fit before the completed-tent divisibility improvement is used. All stages after the initial attention floor are exact on the single P grid under signed full-product affine/single-floor evaluation; partial products are not sequentially truncated.

Use the causal masks $\{3,\dots,t\}$ at history rows $t\ge 3$, all M generated positions at the synthetic terminal, and public singleton fallbacks at rows $1,2$. Their common density is at least $1/4$. Historical rows compute their own prefixes, never a final mean copied backwards. The arbitrary initial pair is ignored. The scalar error is $|t-x|\le 2^{-p_0}\le Q^{-J}$, and the global coordinate slope bound gives $\|P_J(t)-P_J(x)\|_\infty\le V_J2^{-p_0}\le 1/Q$. Thus the same encoder has a direct uniform pre-cut certificate on every row and formal history. No corresponding $1/Q$ claim about every intermediate rail is needed.

Finally $\mathbb{E}|x-p|\le 1/(2\sqrt{M})$. Add the scalar rounding error to (77) and substitute $N=Q/2^a$ to obtain (74). Every recovery estimate was pointwise, so adaptive or randomized attacks are included. The lower order follows from zero-attack admissibility and Theorem L.1's independent statistical and finite-alphabet floors. □

Corollary M.4 (Two-coordinate exact-lattice version)**.** *For $J=2$, choose $2^a\ge 8(c+1)$ and $h\ge a+1$. The same two-coordinate formula has a uniform exact evaluator with only $P=2h$ fractional bits, plus count/address guards, and*

$$\mathbb{E}|\widehat{p}-p|\le\frac{1}{2\sqrt{M}}+\frac{1+4(c+1)2^a+2^{2a}}{Q^2}.$$

Proof. Here $L_2=N$ and no recursive guard bits are needed. Pool $t=2^{-2h}\left\lfloor 2^{2h}x\right\rfloor$. With exponent $P=2h$, the completed tent outputs v and $(A_\triangle,B_\triangle)$ have exponents at most $P-k$ and $P-k-1$,

respectively. Hence s_2 has exponent at most $P-k$, and both s_2/N and v/N fit on the original P grid. The shifted first tent is evaluated as displayed in (73), not by halving an already full-precision input. Since $P \geq k+2$, all biases fit. Thus every subsequent listed stage is exact. The planar path has N horizontal edges and N vertical connectors, with $\eta = (2N+2)^{-1}$. For a horizontal edge at height i/N, set $f = X$ if i is even and $f = 1-X$ otherwise; a preimage is

$$t' = \frac{i+\eta+(1-2\eta)f}{N}.$$

For its following connector a preimage is $t' = (i+1-\eta)/N + \eta(Y - i/N)$. These formulas include endpoints. Enumerating $2N$ edges costs $O(N)$ rational operations and $O(Nh)$ bits if the full table is retained. The planar version of (77) is $4(c+1)/(QN) + 1/N^2$; adding the Q^{-2} mean error proves the claim. The smaller separation choice uses the planar norm directly rather than the conservative general-J choice of a. □

M.4 Removing pauses for the statistical-order depth sandwich

The additive N^{-J} inverse error in (76) is harmless for the fixed-J rate at $N \asymp Q$. It is not harmless when N is selected from a shallower statistical target and Q approaches $\sqrt{M}$. A shifted parameterization eliminates that error. Its coefficients are rational, generally not dyadic; the finite arithmetic extension is proved explicitly below.

For $k \geq 1$, write $V_k(u) = \mathsf{T}^{k-1}(\mathsf{T}_{\mathrm{ext}}(u))$. At level j, retain $A_\triangle, B_\triangle, z_j, s_j, S_j$ from (73), but replace its folded input by

$$\widetilde{u}_j(t) = \mathsf{C}\left(\frac{V_k(t+\eta_j/(2N)) - \eta_j/2}{1-\eta_j}\right), \qquad \widetilde{P}_1(t) = t, \quad \widetilde{P}_j(t) = (\widetilde{P}_{j-1}(\widetilde{u}_j(t)), S_j(t)). \tag{78}$$

The first shifted tent must be $\mathsf{T}_{\mathrm{ext}}$, including at $t=1$; ordinary tents are valid only after its output returns to $[0,1]$. Both shifts lie in its stated domain. The new coefficients have $O_J(\log N + 1)$-bit rational numerator/denominator descriptions, magnitude at most four, and $1/(1-\eta_j) \leq 6/5$. The same saturated-doubling implementation and sequential recursion give bounded intermediates, width $O(J)$ and depth $a_J + b_J \log_2 N$, for fixed syntax-dependent constants.

Lemma M.5 (Finite rational-data conversion)**.** *The growing-depth certificate of Section J extends to finite rational parameter descriptions a/b, $b>0$, provided their actual description lengths and conversion costs are paid. If the signed numerator and positive denominator have total length at most B_{par}, conversion to the working grid costs $O((B_{\mathrm{par}}+P)^2)$ bitwork and $O(B_{\mathrm{par}}+P)$ scratch. With fixed per-stage bounds a sufficient common working precision remains*

$$P = 4h + c_{\mathrm{ar}}(d+2), \tag{79}$$

for a fixed computable c_{ar}, and the actual pre-cut discrepancy is at most $1/Q$ on every formal history.

Proof. Compute the signed integer quotient $\left\lfloor 2^P a/b \right\rfloor$ by exact long division, including the negative-floor correction when the signed quotient has nonzero remainder. The converted parameter $2^{-P}\left\lfloor 2^P a/b \right\rfloor$ differs from a/b by less than 2^{-P}. Long division gives the stated work and scratch bounds. This replaces only the supplied-data conversion; subsequent full signed affine products, their single output floor, ReLUs and finite normalized attention are unchanged. For example, bounded fan-in gives a coefficient-conversion error bounded by a fixed multiple of 2^{-P} times the fixed state

bound. Thus the one-step affine/Taylor/normalization estimate used in Section J depends on the same conversion bound, not on dyadic ideal coefficients.

Consequently fixed per-stage constants α, ζ give $\delta_{i+1} \le \alpha\delta_i + \zeta(P+1)2^{-P}$ whenever $\delta_i \le 1$, and $\delta_0 \le 2^{-P}$. Enlarge the fixed constants to cover readout and choose $A_{\mathrm{ar}} \ge \max\{2, \alpha+\zeta+1\}$. Choose integer $c_{\mathrm{ar}} \ge 4$ by finite search so $A_{\mathrm{ar}}(c_{\mathrm{ar}}+5)2^{1-c_{\mathrm{ar}}} \le 1$. For $t = d+2$, the same scalar induction gives

$$\delta_i \le A_{\mathrm{ar}}^t (P+1)2^{-P} \le 2^{-h},$$

because

$$A_{\mathrm{ar}}^t (P+1)2^{-P+h} \le (c_{\mathrm{ar}}+5)[A_{\mathrm{ar}}2^{1-c_{\mathrm{ar}}}]^t \le 1.$$

The estimate holds at every earlier stage, closing the $\delta_i \le 1$ bootstrap. Initial mean rounding, coefficient conversion and all later arithmetic are included. Append exact working-lattice clipping of final finite coordinates to the cube, with fixed identity clips on the ideal outputs; this cannot worsen the error. The fixed overhead is counted in d. This is an explicitly enlarged finite-description class, not an assertion that the rational snake stages are lattice exact or that the earlier dyadic input specification already included them. □

Theorem M.6 (Robust statistical-order capacity/depth sandwich)**.** *Fix $J \ge 2, c \ge 0$, a fixed positional boundary budget of at most b bits, and the uniform bounded-primitive family of Theorem L.2, with enough fixed width for* (78)*. Boundary positions are public and source-independent, historical processing is causal, synthetic initialization uses only public data and the declared frozen boundary branch, and all ideal bounds hold on every formal history. Use the rational finite-description extension of Lemma M.5, or another admitted evaluator with a fixed E/Q certificate. Let $H = 2^b$. There are fixed $C \ge 2$ and $\gamma = \gamma(J, b, E) > 0$ such that*

$$\mathcal{R}_{\mathrm{rob}} \ge \max\left\{\frac{3}{64\sqrt{M}}, \frac{1}{16H(Q+1)^J}, \frac{\gamma}{QC^d}\right\}. \tag{80}$$

For the constructive bounds take $E = 1$ and put

$$B_{\mathrm{sep}} = 4\sqrt{J}(c+2), \qquad K_{\mathrm{rec}} = 2\sqrt{2J}(c+2).$$

For every dyadic $N \ge 2$ with $N \le Q/B_{\mathrm{sep}}$, there is an admitted encoder with

$$\mathcal{R}_{\mathrm{rob}} \le \frac{1}{2\sqrt{M}} + \frac{K_{\mathrm{rec}}}{QN^{J-1}}, \qquad d \le a_J + b_J \log_2 N. \tag{81}$$

There is also a constant-depth scalar-mean encoder, padded to exactly J coordinates, with risk at most $1/(2\sqrt{M}) + (c+2)/Q$ for every dyadic $Q \ge 2$.

In particular, set $\kappa_\star = (2B_{\mathrm{sep}})^{J-1}$ and $A_\star = 1/2 + \max\{c+2, K_{\mathrm{rec}}\}$. For every *$M \ge 1$ and dyadic $Q \ge 2$ with $Q^J \ge \kappa_\star\sqrt{M}$, the constructions attain*

$$\mathcal{R}_{\mathrm{rob}} \le A_\star/\sqrt{M}, \qquad d \le A_J + B_J \log_{2,+}(\sqrt{M}/Q). \tag{82}$$

Conversely, risk at most $A/\sqrt{M}$ necessarily implies

$$\begin{aligned} A \ge 3/64, \qquad Q^J &\ge \frac{\sqrt{M}}{16HA(3/2)^J}, \\ d &\ge \max\left\{0, \frac{\ln(\gamma\sqrt{M}/(AQ))}{\ln C}\right\}. \end{aligned} \tag{83}$$

Necessary and sufficient constants are distinct. This is not a same-risk-multiplier if-and-only-if frontier or a sharp exponential-base depth theorem.

Proof. **No-pause phase and inverse.** The staircase still satisfies (75). On row i, the shifted triangular wave and clip give exactly

$$\widetilde{u}_j((i+r)/N) = \begin{cases} r/(1-\eta_j), & i \text{ even},\ r \le 1-\eta_j, \\ 1-r/(1-\eta_j), & i \text{ odd},\ r \le 1-\eta_j, \\ 1, & i \text{ even},\ r \ge 1-\eta_j, \\ 0, & i \text{ odd},\ r \ge 1-\eta_j. \end{cases}$$

For an even row the wave starts at $\eta_j/2$; subtracting this and dividing by $1-\eta_j$ gives the first expression. The odd row is its reversal. The triangle turns at phase $1-\eta_j/2$, but throughout that turn it remains beyond the same clipping threshold, so the endpoint was already reached at $1-\eta_j$. The extended first tent verifies the final row and $t=1$ as well. Adjacent row formulas agree.

Thus a whole child copy occupies $[i/N, (i+1-\eta_j)/N]$ and its connector the rest of the row. Movement starts immediately. Where the folded input is fixed, the staircase moves strictly; where the staircase is fixed, the folded input moves. Induction proves that (78) traces the same simple edge list without stationary intervals and is injective, including at vertices and global endpoints. Copy speeds are multiplied by $N/(1-\eta_j) \ge N$ and connector speed is $1/\eta_j \ge N^{j-1}$. Lemma M.2 therefore yields the improved local inverse

$$|t-t'| \le \frac{\sqrt{2}\,\|\widetilde{P}_J(t)-\widetilde{P}_J(t')\|_2}{N^{J-1}}, \qquad \|\widetilde{P}_J(t)-\widetilde{P}_J(t')\|_2 < 1/N. \tag{84}$$

There is no vertex-fiber term: vertices have zero duration, and equal image points have equal parameters.

Finite cut and decoder. Pool each row's own causal mean with the same public masks as above and apply the bounded primitives (78); the ideal terminal scalar is the *exact* mean $x = A/M$. Lemma M.5 gives a finite bank within $1/Q$ of $\widetilde{P}_J(x)$ coordinatewise, uniformly over all rows/formal histories. Attack, clipping and the final floor give

$$\|q - \widetilde{P}_J(x)\|_2 \le \sqrt{J}(c+2)/Q =: r_0.$$

Enumerated nearest-segment projection produces w' at distance at most $2r_0 \le 1/(2N) < 1/N$ from the true curve point. Apply (84) and invert the selected affine piece to obtain $|\widehat{x}-x| \le K_{\mathrm{rec}}/(QN^{J-1})$ pointwise. Add $\mathbb{E}|x-p| \le 1/(2\sqrt{M})$. No additional initial-mean rounding term is added: that error is already included in the uniform finite-arithmetic tube.

For completeness the exact table construction sends a child interval $[\alpha,\beta]$ to

$$\left[\frac{i+(1-\eta_j)\alpha}{N}, \frac{i+(1-\eta_j)\beta}{N}\right]$$

in an even row, and to

$$\left[\frac{i+(1-\eta_j)(1-\beta)}{N}, \frac{i+(1-\eta_j)(1-\alpha)}{N}\right]$$

in an odd row. Append its connector interval $[(i+1-\eta_j)/N, (i+1)/N]$. These positive-length intervals tile $[0,1]$ without gaps. Each endpoint has $O_J(\log N)$ rational bits; projection of a Q-grid observation, squared-distance comparison and inversion use $O_J(\log N + h)$ bits. The $E_J = O_J(N^{J-1})$ entries, their generation and their scan are paid, not replaced by an arbitrary-real nearest-point oracle.

For the scalar fallback retain the causal mean and $J-1$ zeros, use the same generic finite certificate, and decode the first received coordinate. Finite error, fault and final floor total at most $(c+2)/Q$. This construction is available even when no separated snake fits, including small dyadic Q.

Capacity and depth algebra. If $Q \geq \sqrt{M}$, use the scalar fallback. Otherwise let N be the least power of two at least $(\sqrt{M}/Q)^{1/(J-1)} > 1$. Then

$$2 \leq N < 2(\sqrt{M}/Q)^{1/(J-1)} \leq Q/B_{\mathrm{sep}}$$

under $Q^J \geq (2B_{\mathrm{sep}})^{J-1}\sqrt{M}$, and $QN^{J-1} \geq \sqrt{M}$. Equation (81) proves the risk bound and $\log_2 N < 1 + (J-1)^{-1}\log_2(\sqrt{M}/Q)$ gives the claimed depth after fixed constants are enlarged. The sufficient capacity constant is deliberately large; this argument does not assert sufficiency for every positive capacity constant at the same risk multiplier.

For necessity the adversarial supremum includes the zero attack. Theorems L.1 and L.2 apply to this same bounded ideal family and fixed E/Q finite tube, after its fixed bounds are enlarged once to include the new primitives. The statistical witness fixes the initial pair, whereas the summed-influence witness uses its stationary product initial pair; arbitrary-initialization minimax contains both. The fixed-position boundary restriction is essential for the latter witness and cannot be replaced by an arbitrary nonlocal H-valued history statistic. These independent witnesses give (80). Finally $Q+1 \leq 3Q/2$ for $Q \geq 2$, and rearranging each term gives (83). Rational ideal descriptions do not invalidate that lower: it uses only fixed primitive bounds and the certified finite tube, not finite-map Lipschitzness or equality of ideal and finite floor labels. □

Consequently robust statistical order in this common family requires $Q^J = \Omega(\sqrt{M})$, and requires logarithmic primitive depth when $\sqrt{M}/Q$ diverges. A sufficiently large fixed capacity ratio suffices with $O_J(1+\log_+(\sqrt{M}/Q))$ depth. This extends neither the fixed-depth $1/Q$ obstruction to arbitrary depths nor the IID upper bounds to unknown Markov rows. The shifted rational construction and the exact dyadic plateau construction are different implementations with different arithmetic certificates.

M.5 Fully paid work, storage and isolation

We record conservative complete ledgers; they are existence bounds, not optimal running-time or working-precision claims. Primitive depth counts bounded-size attention/affine/ReLU stages. It is not the depth of a Boolean/PURE expansion or the number of bit operations.

For the exact dyadic construction let $n = M+3$, $d = O_J(1+h)$, use its stated P, and take

$$w = C_J\big(P + \log(n+1) + \log(d+1) + 1\big)$$

with sufficient guards for signed full products, counts, addresses and rational decoder integers. With $E_J = O_J(N^{J-1})$, safe arithmetic bitwork and storage bounds are

$$\begin{aligned} T_0 &= O_J\big(dn^2(P+1)w^2 + (d+n+E_J)w^2\big), \\ S_0 &= O_J\big(dn^2 + dJ^2w + nJw + E_Jw + w^4\big). \end{aligned} \tag{85}$$

For the specialized two-coordinate case use $E_2 = 2N$ and $P = 2h$; this gives the corresponding $O(N)$ decoder and table charges. The $(P+1)$ factor safely pays generic Taylor evaluation although bespoke uniform pooling can omit it. All tokens/stages, every dense mask entry including excluded edges,

dense affine/QKV zero slots, optional dummy heads, coefficient/bias descriptions and generation, scaled input acquisition, causal prefix means and divisions, rolling row arrays, full products, counters, segment generation/table, output labels and clearing are included. Network primitive depth is logarithmic in Q; table generation and decoding work are polynomial in N.

For the rational no-pause construction, use $E_J = 1$ for the scalar fallback, $B_{\mathrm{par}} = O_J(\log N + 1)$ (constant for that fallback), and the generic precision (79). In a fully expanded sequential public layout let

$$U = O_J(dn^2), \qquad F = O_J(dn^2 + n\log(n+1))$$

count supplied scalar records and public mask/metadata bits, including zero and excluded slots. Set

$$w = C_J\big[P + B_{\mathrm{par}} + h + \log(n+1) + \log(d+1) + \log(P+1) + \log(U+F+2) + 1\big].$$

Acquisition pays $UB_{\mathrm{par}} + F$ actual bits; conversion pays $O(U(B_{\mathrm{par}} + P)^2)$ work. Exact rational coefficient/table generation costs at most $O_J((U+F+E_J)w^2)$ bitwork when compact rational records and single-bit masks are streamed with $O_J(w)$ reusable generator scratch. Recursive edge tables retain $O_J(E_J w)$ bits. This does not assume a free or retained $(U+F)w$-padded generator array. A safe ledger including these phases is

$$\begin{aligned} T_0 &= O_J(dn^2(P+1)w^2 + (U+F+E_J+n+d)w^2 + w^4), \\ S_0 &= O_J(UB_{\mathrm{par}} + UP + F + nw + E_J w + w^4 + B_{\mathrm{par}} + P). \end{aligned} \tag{86}$$

The storage bound is a safe phase-sum charge for supplied/converted data, masks, rolling activations, rational tables and conversion/microcode scratch, not rolling activations alone. It includes all source acquisition, repeated sequential tape movement, Taylor operations, bounded primitives, rational projections/comparisons/inversions, final labels and access isolation.

For either complete ledger, an explicitly non-unit-cost addressed unrolling can retain $O(T_0 w)$ instruction/address bits and scan an S_0-bit data tape per instruction. Conservative online bitwork is $O(T_0 S_0 w)$, peak storage is $O(T_0 w + S_0)$ plus the actual offline phase scratch, and offline generation costs $O((T_0 + S_0)w^2)$. A separate Boolean/ReLU PURE compilation pays its actual larger bit-operation depth, width and held/dense gate slots; the displayed logarithmic primitive-depth statements do not describe that compilation. No arbitrary-length unit-cost arithmetic, giant real constant, logarithmic-time nearest-curve oracle or uncharged table is granted.

Before decoder access every noncut source-dependent history, count, work and token array is disabled or cleared, with clearing work charged. The J closed-grid labels require $J(h+1)$ bits, not Jh bits, because label Q is allowed; the declared boundary adds at most b bits in the depth theorem. Public programs do not depend on p. The much larger working precision, count guards, parameter descriptions and program storage remain paid. These results establish neither internal-fault robustness, training reachability, hardware behavior, optimal total cost, nor a dimension-uniform or full-Markov robust upper theorem.

N Robust packing for the actual terminal Markov row

The IID upper in the previous sections is not a general prediction theorem. A terminal Markov row may have very few earlier visits, and robust decoding can create positive mass in coordinates whose true count is zero. The construction below combines a complete triple histogram, the compander, separated-curve packing, and a different post-cut normalization. Its proof explicitly controls that false mass.

Fix $K \geq 2$, packing multiplicity $R \geq 2$, fault constant $c \geq 0$ and $M \geq 64$. The unknown homogeneous second-order table and its initial pair law are arbitrary. For each triple let $x_{abc} = C_{abc}/M$. Exactly $D = RK^3$ coordinates in $[0,1]$ cross the cut after coordinatewise real perturbations bounded by c/Q, clipping, and downward Q-quantization. Perturbations can be correlated, history dependent and randomized. Only the terminal pair, two fixed-position symbols, survives exactly; any query or mode is its function or public. The decoder emits an exact finite rational simplex vector. It is not constrained to the internal Q grid. This differs from the final output-rounding step of the earlier concrete implementations, not from their information lower bounds, which already allowed arbitrary real decoders.

Choose integer $a \geq 1$ with $2^a \geq 8\sqrt{R}(c+2)$, let $Q = 2^h$ with $h \geq a+1$, and set $N_g = Q/2^a$. Define

$$B = 2\sqrt{2R}(c+2)2^{(R-1)a} + (R-1)2^{Ra}, \quad \varepsilon = 2B/Q^R, \quad \chi = 2^L,$$
$$\Delta_0 = 4\varepsilon^2 + 2\varepsilon/\chi.$$

Constants involving radicals are analysis constants; executable budget choices may replace them by explicit integer upper bounds.

Theorem N.1 (Robust general-Markov upper)**.** *There is a causal bounded-primitive encoder of fixed width $O_R(K^3)$ and depth $O_R(1+L+\log Q)$, with bounded states and coefficients of magnitude at most four, whose fully paid finite implementation and labels-only rational decoder satisfy, for every table, initial law and allowed attack,*

$$R_{\mathrm{all}} \leq \min\{1, 2^{13}\mathcal{H}_{\mathrm{orig}} + 8K^{5/2}\varepsilon + \Psi(K^5\Delta_0)\}. \tag{87}$$

Here $\mathcal{H}_{\mathrm{orig}}$ is the add-one scale in (22); without a declared mixing promise take $\beta = 0$, so $\mathcal{H}_{\mathrm{orig}} \leq K^{5/2}/\sqrt{M+1}$. With $m = \mathrm{bitlength}(M+1)$, $L = \lceil \log_2 m \rceil$ and $Q^R \geq 2B\sqrt{M}$, the risk is $O_{K,R,c}(M^{-1/2})$. At the smallest sufficient dyadic scale $Q = \Theta_{R,c}(M^{1/(2R)})$, primitive depth is $O_{K,R,c}(\log M)$.

Proof. Acquire the two preceding symbols through a paid radius-three causal local table, not a sparse attention head mislabeled as dense. A triple indicator is $\mathrm{ReLU}(e_a(X_{t-2}) + e_b(X_{t-1}) + e_c(X_t) - 2)$. A zero-logit statistical head pools the K^3 indicators over each row's own observed prefix; rows one and two use public fallback, and the synthetic terminal pools all M triples but is never itself a triple. Statistical masks have fixed positive density. Boundary symbols are acquired by a paid scan at the fixed terminal positions. Apply (54) coordinatewise and put $t = g_L(x)/2 \in [0, 3/4]$. Feed each such scalar into the dyadic plateau snake of Theorem M.3, with dimension R and grid parameter N_g, and run all K^3 copies in parallel. Its bounded-stage realization and the compander give the stated width and depth. Every optional later dummy head is paid.

Use the uniform finite evaluator on the *entire composition*. Fixed per-stage bounds give one constant c_{FA} independent of M, Q, d such that $P \geq 4h + c_{\mathrm{FA}}(d+2)$ ensures pre-cut discrepancy at most $1/Q$ on every row and formal history. Increase P to cover all dyadic coefficient exponents $O_R(h)+L+O(1)$. This is the uniform recurrence (59), not substitution of growing depth into a fixed-architecture constant. There is no separate, hidden re-pooling or exact-mean floor after companding. Before the attack append the exact working-lattice operation $\mathcal{C}(z) = \mathrm{ReLU}(z) - \mathrm{ReLU}(z-1)$ to every exposed coordinate. The corresponding ideal clips are identities. Count their fixed stage overhead in d before choosing P above, and charge their all-token rails, coefficients, writes and clearing in the ledger below. Nonexpansiveness preserves the stated certificate and gives an actual bank in $[0,1]^D$. After the attack, nonexpansive clipping and the final floor put the received vector

within Euclidean distance $\sqrt{R}(c+2)/Q$ of the ideal snake point. Its nearest-segment decoder returns $\widehat{t}$ with

$$|\widehat{t}-t| \le \frac{2\sqrt{2R}(c+2)}{QN_g^{R-1}} + \frac{R-1}{N_g^R} = B/Q^R.$$

The local inverse is pointwise for every input, not an IID-mean statement. The nearest projected point is within $2\sqrt{R}(c+2)/Q < 1/N_g$ of the true point, as required. Clip $2\widehat{t}$ to $[0, g_L(1)]$ and call the result $\widehat{z}$. Then $|\widehat{z} - g_L(x)| \le \varepsilon$. Set $\widehat{x} = g_L^{-1}(\widehat{z}) \in [0,1]$.

Unlike the downward decoder, this reconstruction may exceed x, including at $x = 0$. If $\widehat{x} \le x$, (55) bounds the error directly. Otherwise write $e = \widehat{x} - x > 0$ and apply that inequality with upper endpoint $x + e$:

$$e \le 2\sqrt{x+e}\,\varepsilon + \varepsilon/\chi \le 2\sqrt{x}\,\varepsilon + 2\sqrt{e}\,\varepsilon + \varepsilon/\chi.$$

Since $2\sqrt{e}\,\varepsilon \le e/2 + 2\varepsilon^2$, both signs satisfy

$$|\widehat{x} - x| \le 4\sqrt{x}\,\varepsilon + 4\varepsilon^2 + 2\varepsilon/\chi. \tag{88}$$

This covers zero coordinates, endpoints, $L = 0$ and breakpoints. The leakage terms cannot be discarded by citing zero purity from the downward-only implementation.

Select the block $S = Y_M$ using only the exact terminal pair and output

$$\widehat{q}_b = \frac{\widehat{x}_{Sb} + 1/M}{\sum_c \widehat{x}_{Sc} + K/M}.$$

Its denominator is positive even for a true empty row. It does not receive J, counts or the clean mass. For nonnegative vectors u, v of positive masses A, B, splitting through v/A gives $\mathrm{TV}(u/A, v/B) \le \|u - v\|_1/A$. With $r = J/M$, Cauchy–Schwarz and (88) give

$$\sum_b |\widehat{x}_{Sb} - x_{Sb}| \le 4\varepsilon\sqrt{Kr} + K\Delta_0,$$

$$\mathrm{TV}(\widehat{q}, q) \le \frac{4\varepsilon\sqrt{KM}}{\sqrt{J+K}} + \min\left\{1, \frac{MK\Delta_0}{J+K}\right\}.$$

The true mass is an analysis denominator only. Universal actual-terminal occupation (25), its exact integer layer cake (26), and (61) give

$$\mathbb{E}\,\mathrm{TV}(\widehat{q}, q) \le 8K^{5/2}\varepsilon + \Psi(K^5\Delta_0).$$

No conditional independence of successors at the terminal row is assumed. Add the inherited full-history add-one risk to obtain (87).

At $\varepsilon \le M^{-1/2}$ and $\chi \ge m$, the leakage is at most $4/M + 2/(\chi\sqrt{M})$. The capped function Ψ is increasing, concave, subadditive and satisfies $\Psi(u) \le 2\sqrt{u}$. Thus the quadratic contribution is $O_K(M^{-1/2})$. For fixed $A = 2K^5$, if $A/(\chi\sqrt{M}) \ge 1$ its capped contribution is at most $A/\sqrt{M}$; otherwise

$$\Psi\left(\frac{A}{\chi\sqrt{M}}\right) \le \frac{A[1 + \ln\chi + (\ln M)/2]}{\chi\sqrt{M}} = O_K(M^{-1/2}),$$

since $\ln\chi \le \chi$ and $\ln M \le \chi \ln 2$. This proves the operating point without a mixing assumption. □

N.1 Using only the packing depth needed by the sample size

The plateau snake's additive N_g^{-R} error is inconvenient when Q is near $\sqrt{M}$. The rational no-pause snake removes it.

Corollary N.2 (Partial-capacity Markov upper)**.** *Put $B_0 = 4\sqrt{R}(c+2)$ and $C_{\mathrm{cap}} = (2B_0)^{R-1}$. For every dyadic $Q \geq 2$ satisfying*

$$Q^R \geq C_{\mathrm{cap}}\sqrt{M},$$

the same general-Markov experiment admits risk $O_{K,R,c}(M^{-1/2})$ at fixed width and depth

$$d = O_{K,R,c}(1 + \log\log(M+2) + \log_+(\sqrt{M}/Q)).$$

This construction uses the explicit rational-description extension in Lemma M.5; it is not silently included in a dyadic-only data contract.

Proof. Use $L = \lceil \log_2 m \rceil$ as above. If $Q \geq \sqrt{M}$, transmit $t = g_L(x)/2$ itself in one coordinate and pad the other $R-1$ with zeros. Complete finite evaluation, fault and floor give compander-space error at most $2(c+2)/Q$; no snake is needed. If $Q < \sqrt{M}$, let $u = (\sqrt{M}/Q)^{1/(R-1)} > 1$ and choose the least dyadic $N \geq u$. It satisfies $u \leq N < 2u$ and is executable by the least integer $k \geq 1$ with $Q^2 2^{2k(R-1)} \geq M$. The capacity assumption rearranges to $2u \leq Q/B_0$, hence the no-pause geometry is admissible. Apply Theorem M.6's curve to t after companding, using its pointwise inverse, not its IID statistical upper. Uniform evaluation of the entire composition and the channel give

$$|\widehat{t} - t| \leq \frac{2\sqrt{2R}(c+2)}{QN^{R-1}}, \qquad |\widehat{z} - g_L(x)| \leq \frac{4\sqrt{2R}(c+2)}{QN^{R-1}} \leq \frac{A_0}{\sqrt{M}},$$

where $A_0 = \max\{2(c+2), 4\sqrt{2R}(c+2)\}$. There is no plateau term or extra mean-floor error. The argument of (88) and (87) holds for this fixed multiple of $M^{-1/2}$ too, including false positive mass. Finally $\log_2 N < 1 + \log_{2,+}(\sqrt{M}/Q)/(R-1)$, and adding the compander stages proves the stated depth. Coefficient conversion uses signed rational division with paid description lengths; bounded saturated doublings replace any apparent coefficient of size N. □

N.2 Finite rational outputs and the unresolved dimension gap

The nearest-segment decoder enumerates $E_R = O_R(N_g^{R-1})$ moving edges in the plateau construction, or $O_R(N^{R-1})$ in the no-pause construction (one in scalar fallback). It projects by coordinate clamp, compares squared rational distances by full cross products, breaks ties publicly and inverts the chosen affine parameterization. Segment data have $O_R(h)$-bit rational descriptions. The inverse compander enumerates $L+2$ dyadic endpoints and scans its $L+1$ affine intervals; generation costs $O((L+1)^2 b^2)$ at decoder word size b. Each $\widehat{x}$ has $O_R(h+L)$ rational numerator/denominator length. Summing K such fractions, adding $1/M$ and normalizing produces exact finite simplex coordinates with $O_{K,R}(h + L + \log(M+1))$-bit descriptions. A common denominator may be a product; no K-uniform bitlength is asserted. No square-root or logarithm oracle is used. An optional separate output grid of resolution Q_{out} adds at most $(K-1)/Q_{\mathrm{out}}$ loss and its actual computation/storage cost; setting $Q_{\mathrm{out}} = Q$ would generally lose the packed statistical-order advantage.

For explicit accounting let $U = O_{K,R}(d+n)$ count supplied scalar/position/local records including dense zeros, $F = O_{K,R}(dn^2)$ mask bits and $B_p = O_{K,R}(h + L + \log(n+1))$ maximum description length. Use distinct words

$$w = C_{K,R}[P + \log(n+1) + \log(d+1) + \log(P+1) + \log(U+F+2) + 1],$$

$$b = C'_{K,R}[h + L + \log(n+1) + 1].$$

A conservative complete instruction budget and phase-sum data peak are

$$\begin{aligned} T_0 &= O_{K,R}(dn^2(P+1)w^2 + (U+F)w + w^4 + U(B_p+P)^2 \\ &\qquad + nK^3[E_R + (L+1)^2 + 1]b^2 + nK^3R(w+b)), \\ S_0 &= O_{K,R}(UB_p + UP + F + nK^3w + w^4 + B_p + P \\ &\qquad + K^3E_Rb + nK^3b + (L+1)b + b^2). \end{aligned}$$

Both old w-bit and new b-bit banks coexist during conversion; $O(nK^3R(w+b))$ explicitly pays reading and writing them, even for a larger-than-necessary P. Then the old banks are cleared or isolated with paid writes. The n factor conservatively pays all-token decoding rather than mixing a terminal-only cheap phase into a different ledger. Counts, local tables, dense zero slots, excluded mask edges, full products, original descriptions, conversion scratch, edge/breakpoint generation, exact normalization, boundary acquisition, output serialization and generator scratch are charged. With $v = C\max(w, b, B_p)$, a scanning tape implementation has online work $O(T_0S_0v)$, instruction/address storage $O(T_0v)$, peak $O(T_0v + S_0)$ plus actual generator scratch and offline work $O((T_0+S_0)v^2)$. PURE expansion is a separate much larger bit-depth construction.

The IID witness yields the universal alphabet lower only with the *whole* cut dimension $D = RK^3$:

$$R_{\text{all}} \gtrsim \max\{M^{-1/2}, [H(Q+1)^D]^{-1}\}.$$

It does not prove that the sufficient $Q^R \gtrsim \sqrt{M}$ is necessary. In the fixed-positional-boundary, bounded-influence, uniformly certified family, Theorem L.2 also gives $R \geq c/(QC^d)$ by choosing zero attack. At the named scale $Q = \Theta(M^{1/(2R)})$, statistical order forces $d = \Omega(\log M)$ and matches the upper order. Away from that scale neither the D versus R capacity gap nor the additive compander-depth term is resolved. These are fixed-K, R results, not dimension-uniform minimax optimality or a training theorem.

O Independent fault amplitude and effective precision

The preceding robust constructions hold for noise of size c/Q with fixed c. An independent amplitude cannot be handled by allowing a hidden constant to grow with $Q\delta$. We now rederive the channel bounds uniformly. This section's upper is IID-only. The general-Markov theorem above remains a c/Q theorem; no independent-amplitude Markov upper is asserted here.

Keep the IID experiment with $M \geq 1$, $J \geq 2$ exposed coordinates, fixed positional-bit boundary budget b, and uniform primitive budgets. The actual finite bank $a \in [0,1]^J$ approximates its ideal bank f within E/Q on every frozen-branch formal history, with fixed $E \geq 1$. One real pre-cut perturbation e, $\|e\|_\infty \leq \delta$, precedes clipping and the Q floor:

$$q_j = Q^{-1}\lfloor Q\,\mathrm{clip}(a_j + e_j)\rfloor, \quad Q = 2^h \geq 2, \quad 0 \leq \delta \leq 1/4.$$

The attacker can observe the history, bank and public seed, and randomize, but cannot change the exact side label. Real perturbed values need not lie on the finite evaluator's fine grid. Arithmetic elsewhere is reliable. This is neither the canonical post-grid fault model nor an internal repeated-noise model. Put

$$s = M^{-1/2}, \quad \tau = \max\{Q^{-1}, \delta\}, \quad Q_{\text{eff}} = \tau^{-1} = \min\{Q, 1/\delta\},$$

with $1/0 = \infty$. Thus $0 < \tau \leq 1/2$. The minimax risk takes the supremum over allowed attacks as well as sources and initial laws.

Theorem O.1 (Effective precision at fixed dimension)**.** *Let $H = 2^b$ and $c_{\mathrm{cap}} = [16H(3/2)^J]^{-1}$. With the primitive influence constant $C \geq 2$, there is $\gamma = \gamma(J, b, E) > 0$, independent of M, Q, d, δ, such that*

$$R \geq \max\{3s/64, c_{\mathrm{cap}}\tau^J, \gamma\tau/C^d\}. \tag{89}$$

For a finite upper construction supply an integer $m \geq 1$ certifying $\tau \leq t = 2^{-m} < 2\tau$. At depth $O_J(1 + \log Q_{\mathrm{eff}})$, risk $O_J(s + \tau^J)$ is attainable. More precisely, choose integer $D_0 \geq 12\sqrt{J}$ and set $\kappa_J = 2^J(2D_0)^{J-1}$. If

$$Q_{\mathrm{eff}}^J \geq \kappa_J\sqrt{M},$$

then a fixed risk multiplier $A_J = 1/2 + \max\{3, 6\sqrt{2J}\}$ is achievable at depth $O_J(1 + \log_+(\tau\sqrt{M}))$. Conversely, risk at most $A/\sqrt{M}$ requires

$$A \geq 3/64, \quad Q_{\mathrm{eff}}^J \geq (c_{\mathrm{cap}}/A)\sqrt{M}, \quad d \geq \max\{0, \ln(\gamma\tau\sqrt{M}/A)/\ln C\}.$$

Necessary and sufficient constants are distinct. At fixed J and fixed $\delta > 0$, the positive floor $c_{\mathrm{cap}}\delta^J$ persists regardless of transmission precision or depth.

Proof. Zero attack gives the alphabet lower $[16H(Q+1)^J]^{-1} \geq c_{\mathrm{cap}}Q^{-J}$. For $\delta > 0$, apply the same deterministic coordinate attack for every source and history:

$$A_\delta(a) = \delta\lfloor a/\delta \rfloor.$$

It stays in $[0, 1]$ and moves down by less than δ, including endpoints. Its coordinate alphabet has at most $\lfloor 1/\delta \rfloor + 1$ values; final flooring can only merge them. The same prior-covering proof therefore gives

$$R \geq \frac{1}{16H(\lfloor 1/\delta \rfloor + 1)^J} \geq \frac{\delta^J}{16H(5/4)^J} \geq c_{\mathrm{cap}}\delta^J.$$

The attack is independent of the unknown parameter, not chosen differently for two testing hypotheses. Fixing and averaging public seeds is legitimate and does not add a source-dependent port. For the depth term, when $\delta \leq 1/Q$, zero attack invokes Theorem L.2 at resolution Q. When $\delta \geq 1/Q > 0$, use the coarse attack but grant its full indices $k_j = \lfloor a_j/\delta \rfloor$ to the decoder. This oracle determines the received labels $\lfloor Q\delta k_j \rfloor/Q$ and is more informative. The influence theorem permits real resolution $Q' = 1/\delta \geq 4$ and applies to the *original* bank a because its approximation error is $E/Q \leq E/Q'$. Hence its coarse labels obey the lower $\gamma\delta/C^d$. No Lipschitz property of the discontinuous attacked bank is used. Exact real simulation is a mathematical lower-bound grant, not a constructive finite-computation claim. For this influence argument use a stationary product initial pair so all $M + 2$ bits are IID. For the separate statistical bound choose zero attack and a fixed initial pair, leaving M informative bits. The minimax includes both; taking their maximum proves (89).

For the upper take the no-pause curve P_J proved in Section M, with any dyadic $N \geq 2$ such that $N \leq 1/(D_0 t)$. Its stage count is $a_J + b_J \log_2 N$, with constants independent of noise; it has nonincident-edge separation $1/N$ and local inverse bound

$$\|P_J(u) - P_J(v)\|_2 < 1/N \implies |u - v| \leq \frac{\sqrt{2}\|P_J(u) - P_J(v)\|_2}{N^{J-1}}.$$

Apply its uniformly certified rational finite evaluator to the actual IID mean x. The channel gives $\|q - P_J(x)\|_2 \leq \sqrt{J}(\delta + 2/Q) \leq 3\sqrt{J}\tau$. The nearest projected curve point is thus within

$6\sqrt{J}\tau \le 6\sqrt{J}t \le 1/(2N) < 1/N$ of $P_J(x)$. Inverting the selected rational edge yields, pointwise under every allowed attack,

$$|\widehat{x} - x| \le \frac{6\sqrt{2J}\tau}{N^{J-1}}.$$

Add $\mathbb{E}|x - p| \le s/2$. The scalar fallback transmits the mean and $J-1$ zero rails and gives $R \le s/2 + \delta + 2/Q \le s/2 + 3\tau$.

For full packing, if $D_0 t \le 1/2$, choose the largest dyadic $N \le 1/(D_0 t)$; then $N > 1/(2D_0 t)$ and the error is at most $6\sqrt{2J}(2D_0)^{J-1}2^J\tau^J$. If $D_0 t > 1/2$, use scalar fallback and $\tau > 1/(4D_0)$, which implies $3\tau \le 3(4D_0)^{J-1}\tau^J$. This covers coarse cases and equality at $N = 2$. For statistical selection the capacity premise implies $t^{-J} > (2D_0)^{J-1}\sqrt{M}$. If $t\sqrt{M} \le 1$, use scalar fallback. Otherwise choose the least dyadic $N \ge (t\sqrt{M})^{1/(J-1)}$. It is at least two and

$$N < 2(t\sqrt{M})^{1/(J-1)} \le 1/(D_0 t),$$

so it is admissible and has approximation error at most $6\sqrt{2J}/\sqrt{M}$. Its logarithm is at most a fixed constant plus $\log_{2,+}(\tau\sqrt{M})/(J-1)$. The choices can be made by exact comparisons $t^2 M \le 1$ and $2^{2k(J-1)} \ge t^2 M$, avoiding numerical square roots. Rearranging (89) proves necessity. □

The scale certificate exists with $1 \le m \le h$, but extraction from an unspecified arbitrary real δ is not an algorithm. The constructive theorem uses the supplied finite integer only. For rational δ its certificate can instead be computed with at most $h+1$ exact comparisons, paying the rational input's bitlength. This public noise-budget information carries no source information. No such certificate is needed for the lower bound.

For clarity about uniform arithmetic, convert each bounded rational coefficient a/b by the signed quotient $\lfloor 2^P a/b \rfloor$, with error below 2^{-P}, work $O((B_{\mathrm{par}} + P)^2)$ and scratch $O(B_{\mathrm{par}} + P)$. The fixed-stage affine/Taylor recurrence therefore remains independent of N, Q, M, δ. Taking $P = 4h + c_{\mathrm{FA}}(d+2)$ closes the same bootstrap and gives total pre-cut error $1/Q$; final lattice clipping cannot enlarge it. All history rows compute their own causal means. No exact-lattice assertion is made for rational snake stages.

One complete conservative ledger sets $n = M + 3$, $U = O_J(dn^2)$ supplied scalar slots and $F = O_J(dn^2 + n\log(n+1))$ mask/metadata bits, including zero and excluded slots. Let $B_{\mathrm{par}} = O_J(1 + \log N)$, $E_J = O_J(N^{J-1})$ and use $E_J = 1$ for scalar fallback. If only m is supplied, its input length is $L_{\mathrm{noise}} = O(1 + \log(h+1))$; if rational δ is supplied add its full description length. A safe common guard word is

$$w = C_J[P + B_{\mathrm{par}} + h + L_{\mathrm{noise}} + \log(n+1) + \log(d+1) + \log(P+1) + \log(U + F + 2) + 1].$$

With compact streamed public records, rational edge tables and all generator/selection work paid,

$$T_0 = O_J\big(dn^2(P+1)w^2 + (U + F + E_J + n + d + L_{\mathrm{noise}})w^2 + w^4\big),$$
$$S_0 = O_J\big(UB_{\mathrm{par}} + UP + F + nw + E_J w + w^4 + B_{\mathrm{par}} + P + L_{\mathrm{noise}}\big).$$

These include every token, masks, zero slots, full products, parameter conversion, scale selection, exact rational projection/comparison/inverse, output and isolation. Scanning-tape online work is $O(T_0 S_0 w)$, instruction storage $O(T_0 w)$, peak $O(T_0 w + S_0)$ plus offline scratch, and offline work $O((T_0 + S_0)w^2)$. Transmitted labels cost $J(h+1)$ bits plus at most b positional bits. Primitive depth is not this bit-machine depth or that of a PURE expansion.

Consequently consistency at fixed J requires both $Q \to \infty$ and $\delta \to 0$. Growing dimension is only a possible escape outside this fixed-dimension theorem, not a proved sufficiency claim. IID lower

witnesses transfer to larger source classes; the upper does not. Real pre-grid attacks, fine-lattice attacks on a particular folded encoder, and canonical post-grid noise are separate experiments and their impossibility statements are not interchangeable.

P Finite resource interpretation

The theorems distinguish information at the cut from the cost of producing it. None is a total-memory or runtime optimality claim. The fixed-depth and one-block ledgers below complement the genuinely depth-dependent ledgers in Section J.7. They illustrate what is actually paid, not an optimal implementation claim.

For the affine/ReLU evaluator, let U be the actual number of supplied numerical records, B_{param} their maximum description length, and F the explicit masks and other public-data bits. With $n = M + 3$, a full signed online word has

$$w = O_{\mathrm{arch}}(P + \log n + \log(P + 1)),$$

augmented by $\log(U + F + 2)$ when needed for addresses. Converted public parameter records occupy compact $P + O_{\mathrm{arch}}(1)$ slots, not w-bit workspace slots. Their total is $O_{\mathrm{arch}}(UP)$; padding them to w bits is a separate format with $O(Uw)$ storage and writes. Original descriptions remain a separate UB_{param} charge.

One fixed-control multitape schedule scans position-ordered projected arrays, streams row-major masks and biases, and rewinds key/value tapes for every query row. All edges, including masked ones, are checked. Exact products, integer Taylor steps and division give online bitwork

$$O_{\mathrm{arch}}(n^2(P + 1)w^2),$$

plus supplied-parameter conversion $O(U(B_{\mathrm{param}} + P)^2)$. A safe retained-data and working-memory bound is

$$O_{\mathrm{arch}}(UB_{\mathrm{param}} + UP + F + n + nw + w^4 + B_{\mathrm{param}} + P).$$

These are phase sums including code, masks, input, both rolling arrays, output, controllers and scratch. A generator for supplied data adds its actual cost; arbitrary tables are not presumed free to generate. The compressed prototype also pays the finite log-constant generator, $O(p_0^3 \log^2(p_0 + 1))$ work, and the full downstream decoder implementation. Shared buffers reduce peaks only under an explicit lifetime schedule.

For the actual triple-bank program, let $\ell = \lceil \log_2(n + K + 2) \rceil$, $k = \mathrm{bitlength}(\kappa)$ and $b = 64(h + \ell + k + 1)$. Persistent words have b bits; products retain their full $2b$ bits and guarded temporaries use at most $4b + 4$. A conservative dense Boolean schedule has

$$T_0 = O(n^2 K^6 b^2), \qquad S = O(n^2 + nK^3 b + K^6 b + b^2). \tag{90}$$

It pays the two singleton heads, all local gates, zero-inclusive dense maps, all-token K^3 banks, denominator and numerator accumulations, block selection, output normalization, copies, clears, masks, controllers and addresses. The cache consists of the actual integers $1, \kappa$. There is no numerical attention matrix hidden in the space bound.

An addressed sequential evaluator storing $O(T_0 b)$ instruction bits can scan the S-bit tape for each instruction, costing $O(T_0 S b)$ forward bitwork and $O(T_0 b + S)$ peak. Its offline generator pays $O((T_0 + S)b^2)$ work. Alternatively, a pure Boolean/ReLU expansion with $T = T_0 + b$, $S' = S + b$

has depth $O(T)$, width $O(S')$, $O(TS')$ gates and identity holds, $O(bT(S')^2)$ dense coefficient bits, and $O(b^2T(S')^2)$ dense work. The stored parameters, input, public constants, rolling layers and output must all be included in each phase peak. These are distinct complete implementations, not coordinates to combine into an artificial best-of-both estimate.

At $\kappa = \text{bitlength}(M+1)$, the weight bitlength is $O(\log\log M)$, but h and accumulation length are still $O(\log M)$. Effective score range $\ln\kappa$ is neither floating-point exponent precision nor the total bits used by the model. Likewise fixed ideal attention depth does not imply fixed depth after arithmetic unrolling.

Q Dependencies and remaining limitations

The earlier full-history add-one/selector estimates and their event-radius inputs are stated in Appendix D. The original compressed-head inversion identities and stationary singular-operator occupation estimate are also identified there as inherited inputs. The present draft preserves their mathematical interfaces and the earlier proofs of the information-cut, binary-return and depth conclusions. Its accompanying source manifest binds the complete original proof bodies, subsequent proofs, and review records by exact versions and hashes. It does not substitute a review label for a proof or claim that finite diagnostics establish a universal theorem.

The newly integrated arguments are internal to this text: mixing-sensitive compander analysis and binary depth-uniform coupling, deterministic influence reveal and posterior control, partial folds and exact finite decoding, the reflection attack, dyadic and rational robust path parameterizations, symmetric inverse/false-mass control at the actual terminal Markov row, and the independent-amplitude effective-precision argument. Where a new theorem uses the inherited add-one risk, that dependency is displayed explicitly. A standalone rederivation of that earlier statistical chain is outside this synthesis; it remains available in the accompanying full dossier.

The general-Markov packing upper has sufficient exponent R, whereas its universal finite-alphabet lower has dimension $D = RK^3$. No sharp dimension-capacity frontier is asserted. The independent-amplitude upper is IID-only. The smaller partial-capacity Markov depth upper does not establish an all-resolution optimal compander term. Constants are not sharp in growing alphabet or dimension. Neither the fixed-family $1/Q$ lower nor a particular folded encoder's vulnerability is universal over redesigned deep representations. Exact positional side bits cannot be replaced by arbitrary nonlocal statistics in the summed-influence lower.

The constructions establish finite representability and fully charged existence ledgers, not convergence of training, practical constants, efficient nearest-segment decoding, minimal working precision, minimum total memory, or an attention-specific advantage over a fully paid direct counter. Their arithmetic certificates are analytic; finite diagnostics and LLM review are not external peer review, formal verification or measured execution of an emitted network. Geometric packing, companding, tents, weighted counting and generic depth–precision ideas are established prior mechanisms. This targeted literature comparison is not a novelty certificate or an exhaustive search.

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